Enterprise architecture

the selection process of an Enterprise Architecture Toolset to support understanding and governing the enterprise

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MASTER'S THESIS

Enterprise Architecture

The selection process of an Enterprise Architecture Toolset to support understanding and governing the enterprise

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s res s es of ad a on e od o s dy o e Science and n ee n a e ec n sce n e s e, ndh o en (/ e). e ad a on oo ace n e e d o n o a on Sys e s, n e se A c ec e as e a ea o ex e se, and as ca ed o a s.

s ec on cs o e ands s a oba eade n co o e e s on se s, n , e ec c s a e s, ed ca d a nos c a n and a en on o n , and o e od c s. s 4,000 e oyees n o e an 0 co n es a e ac e n e a eas o n , cons e e ec on cs, do es c a ances, se cond c o s and ed ca sys e s. e ad a on oo ace a e de a en o o a e .

s an s o o e o a ds o e s'. a eans a e d e en a eas a e o be ab e o coo e a e. a can es n s a n e sa e eso ces n e e d o . o nd o se ces a e s ab e o beco e a s a ed se ce and o ad s e se ces o e d e en b s nesses e se o a c ec es s necessa y.

n o de o a abo a c ec e a o a s s necessa y. An n e se A c ec e oo o des s o a s and s o s e a c ec e od ca . e oo a es a d a o e be een e b s ness and oss b e and s a so ca ab e o s o n e c an es n e a c ec e.

na y, o d e o an y s e so s ca e ed, ees an ee, ees e n a and an B s, a en o e deas, co en s and s o . Bes des e an o an Maa en Le s o s s o and o n s e d. n a c a o d e o ah an, Ma anne, Ma and e es o y ends and a y o e o e and s o d n s ad a on ocess and o o y s dy.

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B Abstract

"The improvement of understanding is for two ends: first, our own increase of knowledge; secondly, to enable us to deliver that knowledge to others." [John Locke (1632 – 1704), British philosopher]

Understanding is an important issue in Enterprise Governance. Decision making in the enterprise can be supported by a better understanding of the structure, processes, and functions of the enterprise, supported by understanding the impact of the decisions. By the introduction of information systems, enterprises are becoming more and more complex to understand. Also customers are becoming more and more demanding and this results in the fact that understanding the enterprise becomes more necessary than ever before.

Enterprise Architecture is a possible solution for the lack of understanding. It is still a young profession and at this time there is limited consensus on definitions and methodology. It will be considered as a meta-architecture, which is a set of architectures. With Enterprise Architecture the different views and architectural descriptions can be compared. In this thesis the IT-architectures are considered as the added value of an Enterprise Architecture. The definition of Enterprise Architecture, which will be used during this thesis, is based on the IEEE definition for architecture. This definition considers systems and systems can be subdivided into information systems, enterprise systems, aspect systems, and subsystems. The enterprise itself can also be considered as a system.

In this thesis, Enterprise Architecture is split into three main aims, mentioned as three approaches. These approaches are:

- **IT-centric:** the most important objective is Business IT alignment;
- **Business process-centric:** the most important objective is to analyze and improve business processes;
- **Governance-centric:** the most important objective is to understand the business better and to govern it better.

Besides these main objectives of the three approaches, Enterprise Architecture can be beneficial for the enterprise in another eight ways. For instance, these three approaches all result in better communication with the stakeholders.

Enterprise Architecture is a complex subject and the Enterprise Architecture framework is a methodology to structure the architecture in a domain and an Enterprise Architecture. Frameworks are used to create a logical structure for classifying and organizing the different architectural descriptions of the enterprise. There is a shift in the USA, where the Clinger-Cohen Act and the Sarbanes Oxley Act, has led to a great increase in the use of Enterprise Architecture and its frameworks. In this thesis two frameworks will be discussed. First the Zachman framework will be described because it is seen as the first framework in the world and still is important nowadays. The second framework is the Integrated Architecture Framework. This framework will be used in the results of this thesis.

Enterprise Architecture tools are developed to support Enterprise Architecture and frameworks in a formal and systematic way. Enterprise Architecture can be supported by a nds of oos. ese oos cons s of a ode n a d and a e os o y o s o e e nfo a on sys e a ca y. e e a e e e a n a oa es n ese oos. ese oos a e o n cose and c ose o ea o e , so n e e e becO e ome nd of oo a s n a e nc ona es of ese e nds of oos. ese a oa es a e:

- **EA top-down approach:** s a oa s s a ed a e o (s a e es and d e s) and o do n o e bo o (e n ca e en a on). s s e exec e's e ;
- **EA bottom-up approach:** s a oa s a s a e bo o (e e n ca e en a on) and oes o e o (e s a e es and d e s). s s e oyee's e ;
- **EA change management approach:** s a oa s a s n e dd e o s o so e o ec s. e oo s a e on y sed o so e o ec s. s s o e ana e s o e oyee's e s ec e.

e second a of s es s be desc b n e oo se ec on ocess o s. n e se A ec e as o s o s n e 'o a ds ome s's a e y. s s a e as e ob ec e n o e coo e a on n e e d of a ec e be een e d e en d s ons of s. e e o e s needs a oo o a a e e bo on co o a e e e as on d s ona e e . e an ose s a be e B s ness a n en and o be ab e o a e be e dec s ons n e e d of . e e o e e oo a s s e e e en s of s bes ad o be se ec ed.

e s s e n e oo se ec on ocess as o e e a e a on s . s on s con a ned e an oos a e e a a ab e and e e e d e en a oa es e e a en n o acco n . e e o e a eas ome oo o e y a oa as se ec ed. n s se ec on ocess a es o n o a on as sen o e oo endo s. e e o an e on s as ed sed o a s o s of e oos. n s ase e an oc as on c ea n d e en odes and e a ons s be een e odes. e odes, e a e o be e e a ed and s o ed e e e of a oo , a e o be n a ony e A a e o . A e e ocess, e e e oo endo s o e s o s e e n ed o e a esen a on and de ons a on of e oo . Based on ese esen a ons and a so based on e es s o e Me s o o as cons de ed as e os o s n oo o s. e e o e Me s as se ec ed o a es case.

e an ose of e es as o nd o e oo s ab e o e c cons s ency, n s, and o e a be een d e en e s of e sa e a ec e. ese as ec s e e ec ed by s n e yd a and e A doc en d n s es case. e e o e As s e e ence a ec es of yd a and A e e e en ed n e n e se A ec e oo . A so a d e as c ea ed. s s e A e and o e A doc en n s es, a e es, b s ness nc es and nc es e e a en. A e e en h ese e e s a o e as c ea ed. s e s based on e A a e o and e n as e e a ed o e A, yd a, and A e s. n e es case Me s d d no s o any e ec ons.

e n od c on of an n e se A ec e o s e ess a o of o o do. s of a a be e es case Me s as o be ed. n s es case a ea se e as o be sed, o see o coo e a on Me s o s. Second e e as o be e e a ed a a y and n ness a on e e oyees o se an n e se A ec e.

Contents

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A o of I o ec s co d be be e ad ed o e en e se e e of n e se A ec e (A). In 2003 a Inc. s a es a , *"IT project failures in industry and Government account for an astounding $ 75 billion in losses each year."* A so se o b ness e ec es e o [Me a o]:

- e an 25% of I o ec s a e ed na o , de ned, o ec e e oa s;
- e an 2% f nda en a y ad anced e b ness s a e c oa s o e en e se.

A o of I o ec s fa d e o a ac of nde s and n . e I de a en acce s e e e en s o e b ness a e e en s de n on a s oo s e. o o e and o e en e ses A s e so on o a n b ness I . A e s o den fy e eo e o a e n o ed, no on y o desc be e e e en s, b a so o ana yse ese e e en s and n e e e n e b oade con ex o e en e se. A e a A e s n ans a n e e e en s n o I . A s no on y abo ana ys s, b s a so a ocess be een e o e o an s a ed nde s and n .

B ness I a n en s no e on y eason o de oy an A. e o A ocesses can be ea a ed n a be e ay o e en e se can be o e med be e . In ese cases A con b es o a be e nde s and n o e en e se and s s a on. e ocesses can be ea an ed n s c a ay a ey be e s o e b ness oa s o e c o e s o e en e se. e ana e en can o e n e en e se, a s ocesses, be e beca se ey nde s and e s a on and e ac o e dec s ons. e o A a c ea s on can be fo a ed, c based on e d e en e s ons o ea y a fo e y es ed n a Babe e con s on. Based on s c ea s on be e dec s ons can be ade.

Be e nde s and n s no eas abe, b e o o n e a o s a es e case fo a ec e. I a o se as o be b e nc a as a des n n s ead. A ec e s ed o co n ca e s dea e con ac o . a ec e e nc a and e con ac o can co n ca e, e n c a y can dec de f ay be b d, e con ac o can dec de e cos ce, e c. A so e con ac o no s a e needs o b d n a e a so f e o se can be b d a a . s exa e s o s a a ec e s sed fo co n ca on and dec s on a n . e o a ec e e e s a be e nde s and n o a e nc a ad n nd. In e case o b d n a o se no nc a e e e a cons c o b d s o se o any nd o a ec e. B en I a ca ons a e o be b , s no a o do a o any nd o a ec e.

e o A e d e en e s, o n ons, n e es s, and conce ns o e s a e o de s n e en e se a e a en n o acco n . e n o a on a as o be ac ed can be seen fo o any e s ec es. In an en e se a o o eo e a e n o ed. ey a a e d e en e s, o n ons, n e es s, and conce ns. ey a oo a e en e se fo a d e en on o e . In o de o co e o a be e nde s and n o e en e se a e d e en e s a a e e e an a e o be a en n o acco n . ese e e an e s on e en e se a e e e nd ed e cen nde enden no nd ed e cen co a b e, b o e e ey e ea o e s abo e sys e an e e co d do a ome. o e a ood s a on abo o d e en e s on e sa e ob ec can ex s s, an anc en nd an a e a s ca ed: "*The blind men and the elephant*" be en [s e ca], [S e e an 2003].

S x b nd en ed n a a e. me day an e e an ca e o a a e, b ey ad no dea a an e e an as. ey dec ded, "*Even though we cannot see it, let us go and feel it anyway.*" A o e en o e beas and o ed e e e an . "*Hey, the elephant is a pillar,*" sa d e s an o o ed s e . "*Oh, no! It is like a rope,*" sa d e second an o abbed e a . "*Oh, no! It is like a thick branch of a tree,*" sa d e d an o ed e n . "*It is like a big hand fan,*" sa d e o an o o ed e ea . "*It is like a huge wall,*" sa d e an o ed e be y. "*It is like a solid pipe,*" sa d e s x an o e b aced e s o e e e an . ey be an o a e and e e y one o e ns s ed a s a e o e an a as . A se an ca e and as ed e , "*What is the matter?*" ey sa d, "*We cannot agree to what the elephant is like.*" ac one o e o e conc s on. e se an ca y ex a ned o e , "*All of you are right. The reason every one of you is telling it differently is because each one of you touched a different part of the animal. So, actually the elephant has all those features you felt.*" "*Oh!*" e e yone sa d. e e as no o e . ey e a y a ey a e e .

Figure 1-1: The blind man and the elephant

e o a o s a e s do no ass e o a e e o e s beca se one a o e s no n. By s a n e d e en e s ec es a o e co e nde s and n o e a ses. s o nde s and e en e se a e d e en e s a e o be a en n o acco n. A d e en e s on o e sa e s o y as e b nd en and e e e an s a es s an s o y, c s na a ed by Mo ana a a ed n Mo a (

- **Understanding:** this is complex analogy and it is answered by answers to why questions.

Information, knowledge and understanding follow a hierarchy. Understanding uses both information and knowledge and knowledge uses only information. Today most enterprises focus on the information and knowledge, but they forget to create understanding. Understanding is to be able to explain the complexities and behaviour of the system and the reason behind it. In section 3.5 the benefits of EA will be discussed and when an EA achieves understanding these benefits could be accomplished. In chapter 4 the frameworks will be discussed and the what, how, and why questions are part of these frameworks. In many cases EA focuses only on the information part, but that is not enough to create understanding.

This thesis is about Enterprise Architecture and especially about the selection process of an Enterprise Architecture Tool. The purpose is to understand and run the enterprise, which is another important thing of this thesis. According to Bernard the main processes of enterprises establish an EA by following seven steps. These steps are [B?? 2003]:

1. **Agree on the Need:** senior management has to agree that a description of the EA is needed, before an EA can be established within the enterprise. The different approaches (for EA see section 3.3) may and could aid in this decision. The management decides the reason for creating an EA and therefore they choose an approach;
2. **Establish an Organizational Structure:** who is going to use, manage and maintain the EA, who will make decisions, who will be responsible for it;
3. **Select a Framework:** a framework consists of processes and templates used for EA. A framework organizes the structures and aspects of EA, provides a place to share, discuss and reuse and organizes cross-functional capability of all elements. It supports the establishment of an EA and they often describe specific approaches to organizing an EA [Riet de et al. 2004 ...];
4. **Select a Tool and Repository:** a tool is needed to store and retrieve the models and information of an EA. This has to make sure that the way that different employees can use EA within the enterprise is set;
5. **Organize the Existing Material:** all the existing models and information that is specified by the framework must be collected;
6. **Begin Using the Enterprise Architecture:** EA is new and management has to regain its use. Many employees have to learn a new way of looking to the management of the enterprise. This is a step that requires a lot of governance and a central approach;
7. **Extend and Maintain the Architecture:** once an EA is established, it has to be kept up to date. If there is a change within the enterprise, the concerned models and information in the database have to be changed. Missing models and information have to be developed.

Before the start of an EA project, the value (for the business) and the governance have to take into consideration, to enhance the chance of success. The assignment of this thesis is focused only on step 4. But to select the tool, also steps 1, 2 and 3 have to be taken into account. These steps are decisive in the process of selecting a tool. After that a tool is selected, the chosen approach, framework, and the people, who have to use the tool. These people have to be able to use the tool for their purposes. This has decided to use the

[illegible] communication [illegible] use. In this section [illegible] different tools and [illegible] been [illegible].

[illegible] contains [illegible] and [illegible] are about [illegible]. [illegible] is an introduction to the tool selection process in which [illegible] will be described. [illegible] and the selection process of [illegible] are about the tool selection process itself. An [illegible] tools [illegible] area [illegible] not [illegible]. [illegible] considered selection process as [illegible]. This tool selection process is based on theory and practical experience. The theory is from the book *Business Information Systems by Data Structures* of [illegible]. The practical experience is from [illegible] tool selection processes [illegible] and a tool selection process, which was executed by [illegible] Student.

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s a con ans conce s and bac o nd n o a on on es s es s s based. n od c on as a ed a a ode s ed o h abo a sys . n sec on 2. s sys and ode be desc bed. se A ec e, s n a n o c s es s, s a co os on o ds en se and a ec e. o s a d n on o se o ds be se a a y en and d n on o A o o n sec on 3. .

6\VWHPV

n 40s an b o o s. L d on Be a a y n od ed e a sys o y and oss As by ex ended . s o y s es s a a den ab e a s o a sys a e a ed o o a s. o nde s and a sys as o be cons de ed n s o a y. s no oss b e o nde s and o e sys so e y by exa n n s so a ed a s A eady n e en cen y an oso e s es ed a o e s o an e s s a s. n nde s and n o e sys , s a s canno be cons de ed n so a on. s o e sys and s a e a ons s a e o be a en n o cons de a on. e no on o o s c e a oac n nde s and n e a ons be een a s o o e s ca ed n an e es a , [Mc].

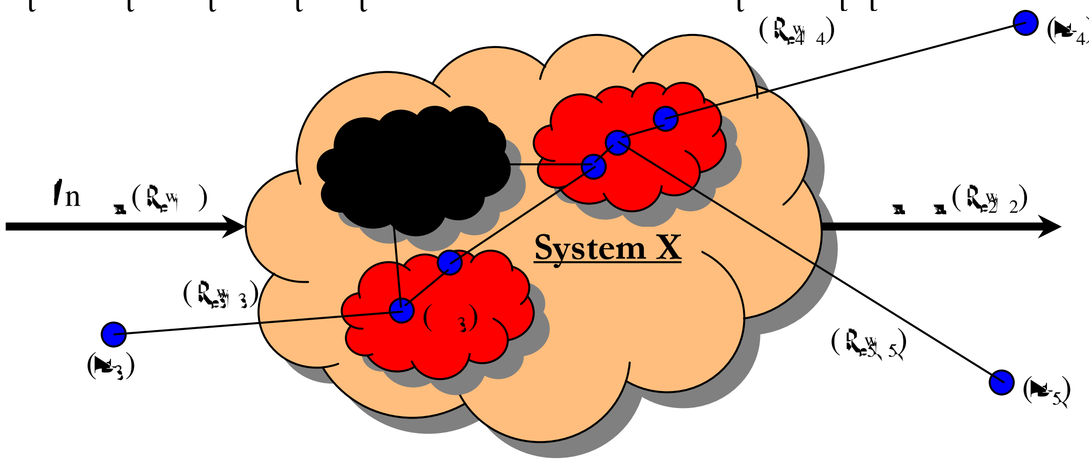

Figure 2-1: A visual representation of System X in its environment

e 2 e esen s a sys and s sys s an o en sys beca se n e ac s s en on en . e n e na e a ons s a e s on e an e ex e na e a ons s and a d s n s es e sys o s en on en . An o en sys , n e ac s s en on en , can e o e by ac n e o e es. n any cases e en on en s o o cons de a on. n ys cs o exa e o es a e ade n dea s a ons o cons de n e n ence o en on en . B n s es s o en sys s be a en n o cons de a on. An o en sys n e ac s s en on en and e sys o e 2 n e ac s s en on en an h and an o a o . Bes des ese a o s e sys n e ac s a so e en s o en es n s en on en . e b c c es e esen o se en s o en es. e a ons s be een e a e esen ed by e b ac es and e e a e so e e a ons s e en s o s de e sys . e b c o d [illegible] [McC...], [...et al. 2004...], [...8]:

1. [illegible]
2. [illegible]
3. [illegible]
4. [illegible]

[illegible]

System: *"A collection of interacting, interrelated or interdependent components forming a complex whole, which is organized to accomplish a specific function or a set of functions."*

[illegible]

System stakeholder: *"A person or group within or outside the system that is involved or affected by the performance of the system."*

[illegible] (...82) [illegible] (S) as: S = < ... > [illegible] [...8]:

- [illegible]
- [illegible]
- [illegible]
- [illegible]

[illegible]

- **Subsystem:** [illegible]
- **Aspect system:** [illegible]

[illegible]

- **State-maintaining system:** [illegible] de e med be a o and a de e med oa . An exa e o s nd o sys e s a so a e sys e . e sa e n e e a e sa e o .

- **Goal-seeking system:** can es ond d e en y o ome o o e d e en e en s n ome o o e d e en en on en s and can es ond d e en y o a a c a e en n an an n en on en od ces a a c a o co e. S sys e s can oose e be a o b ey a e a de e med oa . An exa e o s sys e s a o e e e an a. e can y o e ood n d e en ays n e sa e en on en ;
- **Purposeful system:** can od ce e sa e o co e n d e en ays n e sa e en on en and can od ce d e en o co es n e sa e and d e en en on en s. S sys e s can oose e be a o and e oa . An exa e o s sys e s an en e se o a an be n . ey can oose e ay o ac e e e oa s, b ey a so can an e e oa s.

e os sys e s a e a e ca ab es o e oa see n sys e s and e s a e an a n n sys e s. e oa see n sys e s a e o co se a e ca ab es o e s a e an a n n sys e s [a a eda].

2.1.1 Information Systems

n e so a e a ec e, be desc bed n sec on 2.3, so a e sys e s a e o en cons de ed as sys e s a a e a co abe se o s a es. ese sys e s, a e ca ed ans on sys e s, a e s a e ans ons a d sc e e o en s. ans on sys e s a e s a e an a n n sys e s o oa see n sys e s. A e y o an y e o a sys e s ca ed n o a on sys e . n s sys e e s a e s o ed by a se o n o a on ob ec s. ese sys e s a e o en sed n co e sc ence, nd s a en mee n and e ec ca en mee n [an ee 4].

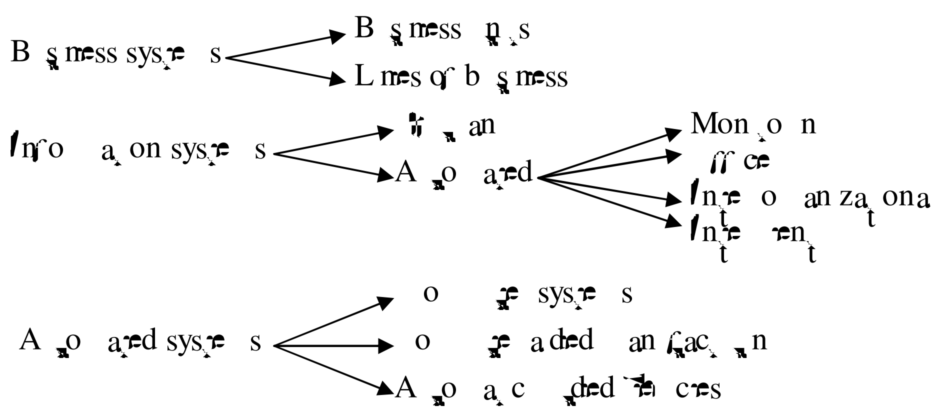

Figure 2-2: A taxonomy of discrete dynamic systems: 'human', 'automated', 'monitoring', 'office', 'inter-organizational' and 'intelligent' are all types of information system [Van Hee1994].

sc e e dyna c sys e s a e a ac e zed by e ac a , a each o en , ey a e n a s a e and e e s a se ence o e en s o n e sys e o ano e s a e. n ac a sys e s a ays n a ce an s a e, b an n n e n be o d e en s a es can ex s . e e s b y es o d sc e e dyna c sys e s a e [an ee 4]:

- B s ness sys e s (s as a ac o y o es a an);

- Information systems (with the automated ones);
- Automated systems (systems that are controlled by an automated information system).

Discrete dynamic systems are a subclass of dynamic systems, but they are very common. A computerized system such as distributed databases, user interfaces, etc., communication systems and decision support systems fall within this class, so do many business systems such as factories, offices and transport enterprises. To consider these systems as discrete systems all kinds of analysis, such as simulation, can be easily executed. Discrete dynamic systems are built of two kinds of components: actors and objects. Actors are the active components in the sense that they consume and produce objects, which are the passive components [Van Hee 94].

Actors can be human beings, machines, devices or networks of other discrete dynamic systems. Actors perform transformations on the state of the discrete dynamic system so that they mostly consume only a subset of the tokens in a state. Objects can be material objects such as tools and cash, or information objects such as insurance policies, diagnoses, advice, designs or financial transfers [Van Hee 94].

In the analysis and design stages of the development process for discrete dynamic systems, the system is seen in terms of models of these systems. A model is in fact another system, but one that is easier to analyze or observe than the original system. But the model has some correspondence to the original system (in certain aspects or areas), so that conclusions drawn from the model are assumed to be valid for the original system as well [Van Hee 94].

The systems which are discussed here, are subject to changes during their lifetime. At any moment in time the system has a certain state, at any later moment this state may have changed. The set of all states that a system may have is called its state space. A system state can be regarded as a snapshot of the system, showing all relevant details. Very often it contains some details which cannot be directly observed. For example, a state may contain information, such as the time at which the current state was reached or the time at which the current state will end. Another approach is to specify directly a set of allowed traces. In practice, it is common to formulate constraints, i.e. so-called rules that together characterize this set. These sets are prefix closed: a prefix of an allowed trace is also allowed. One often distinguishes two types: static and dynamic constraints; static constraints restrict the state space of a system, while dynamic constraints restrict the state transitions that can be made [Van Hee 94].

To define open systems, an open system and system components are defined here in such a way that they form a closed system together. The combination of two open systems is called a composition. An important aspect of system composition is that the concept of communication between systems can be defined. Two systems can communicate if they are components of a compound system. When describing a complex system, it is often decomposed into less complex systems called components. The original system is then a compound system, the composition of the components [Van Hee 94].

In this thesis two different approaches were presented to describe systems. The first approach is to use the general system theory and this approach is mostly used to describe complex systems [illegible] business [illegible]. In the second approach a system is considered as a discrete system and a discrete system is always in a certain state. The number of possible states is its state space. These kinds of systems are mostly used by the IA [illegible] information systems. [illegible] kinds of systems and [illegible] the [illegible] both kinds of systems. To understand and [illegible] systems [illegible] to be considered from different perspectives. In this [illegible] possible perspectives [illegible] different [illegible] considering systems.

To understand a system there are three [illegible] which have to be examined [illegible]:

- **Structure:** The way in which it is divided into the parts of the system and how the parts influence each other (relations). In order to understand the system it [illegible] structure can be understood only if observed in the function of the system. To do this analysis is needed to understand the [illegible], and synthesis [illegible] is needed to understand the [illegible] within the system;
- **Processes:** This is about understanding how the parts of the system respond to the environment. Processes are responsible for [illegible] states of the system and [illegible] the system has;
- **Functions:** To understand the function of a social system the causal-effect relations [illegible] [Sn[illegible]]. A [illegible] good [illegible] is only necessary, not sufficient, for [illegible] good [illegible]. Therefore [illegible] does not explain [illegible] good [illegible]. [illegible] possible to [illegible], hence [illegible], as an object [illegible] observable [illegible] of system behavior. Moreover, since [illegible] sufficient for [illegible] good [illegible], reference to [illegible] environment [illegible] to explain [illegible] good [illegible]. Therefore explanation of a social system [illegible] environment [illegible].

To understand [illegible] structure, processes and functions, it is not possible to understand [illegible] use. For example, a lack of understanding [illegible] between the business and IA. By seeing [illegible] different [illegible], structure, processes, and functions can be better understood. [illegible] IA [illegible] better addressed [illegible] [illegible] 2003].

To study the behavior of a system and its effect on its environment, by experimenting in reality, is often not desirable. That is because [illegible] the costs [illegible], or [illegible] the system and its environment. Another approach is to examine the structure, processes and functions of a system by [illegible] description of reality by means of [illegible] aspects. That can be done [illegible] physical or [illegible] representation. These [illegible] are also known as models. By examining a model from different perspectives, [illegible] discussions and [illegible] possible [illegible], a better understanding and control of the complex reality is achieved. Models are used for [illegible] reasons [illegible]:

- **Design:** used as a blueprint of a system that has to be constructed. In this case it should be easy to [illegible] building blocks of the model into existing [illegible] components;
- **Analysis:** used to analyze a system, for instance performance or [illegible]. If system exists, some [illegible] observable behavior;

- **Specification / documentation:** sed o doc en e nc ona y o e sys e , e. e ode desc bes e be a o o a sys e n an abs ac ay. Mode can be sed as a s ec ca on o e sys e .

(QWHUSULVH

An en e se s a sys e and n e e o s sec on as s a ed a an en e se s an exa e o a os sys e . n s sec on a de n on o an en e se be en. s de n on s based on e de n on o a sys e .

Enterprise: *“A whole of organized people and means intended to create, with the help of processes, products and services where its stakeholders are willing to pay for.”*

e c s o e s o e en e se a e a so s a e o de s o e en e se and ey can d ec y o nd ec y ay o e od c s and se ce c ea ed by e en e se. B o a ays e c s o e a e n o ay. A so o e s a e o de s, e s a e o de s can ay o od c s o se ces. o exa e by n es n n a e en e se od ces oods a a e no so d ye . e de n on o an en e se s e y e e a and as e o e a de n e e a on:

- An en e se does no need o a e a o e;
e o e n en s an en e se oo and s c s o e s a e ay n o by axes;
- A s o s c b s a so an en e se;
- A d son o an en e se can a so be cons de ed as an en e se.

s s o co se an en e se. B a so s od c on sons ('s) a e en e ses o e o n. ey a od ce od c s e e s c s o e s a e n o ay o . e esea c d son o s s an en e se e e s c s o e s a e nd ec y ay n o . s o e s a e ay n o end od c s, b e ce nc des a co ensa on o e esea c and de e o en o e od c .

$UFKLWHFWXUH

e conce o a c ec e s beco n o e and o e o an n en e ses. So e o an easons a e:

- e o e soc e y, nc d n e en e ses, s beco n a d y a o a ed. n a nds o a eas an o s s o ed o e en e aced by n o a on sys e s;
- e o n n o a on sys e s es s n a o d a s o e co ex so beco es o e d c o c ea e sec y ade a e y;
- e c s o e s a e beco n o e de and n . o dea ese obe s a s c a a oac s e ed;
- e des n and ea za on o c an es n an a eady co ex o e. n e ses a e a eady co ex sys e s and s e y d c o a e c an es n an en e se n s c a ay a e en e se can nc on n e ay n e e .

s s c a a oac as been a o e o d o cons c on o cen es. e e s no n as a c ec e'.

In this section [illegible] be defined. There are many definitions [illegible] Business Architecture, Process Architecture, Information Architecture, Application Architecture, Software Architecture, Technical Architecture, and Infrastructure Architecture [illegible].

[illegible] years, [illegible] Electrical and Electronics Engineers, Inc. (IEEE). [illegible] proposed a definition [illegible] broadly accepted. [illegible] the IEEE [illegible] and [illegible] and [illegible] The IEEE Standard 1471-2000: IEEE Recommended Practice for Architectural Description of Software Intensive Systems, [illegible] IEEE 1471 [illegible] [IEEE]:

Architecture: *"the fundamental organization of a system embodied in its components, their relationships to each other and to the environment and the principles guiding its design and evolution."*

[illegible] definition [illegible] explained by [illegible] and explaining [illegible] IEEE 1471 [illegible] In section 2. [illegible] definition of a system and a system [illegible]. The definition of IEEE [illegible]: *"the fundamental organization"* and contains *"environment"*. The fundamental organization [illegible] concepts and [illegible] cannot always be clearly seen [illegible] cannot be described [illegible] *"embodied"*. [illegible] context of the system. A system [illegible] and [illegible] components [illegible] system. [illegible] systems, [illegible] described by characteristics [illegible] section 2. [illegible] characteristics are about [illegible] the system [illegible] the system interests. [illegible] understand a system's [illegible] fundamental characteristics, [illegible] a system [illegible] and [illegible] embedded in [illegible] [A [illegible] 8], [illegible], [illegible] d].

IEEE 1471 [illegible] *"principles guiding its design and evolution"*. The definition of a principle [illegible] used in [illegible] [[illegible]]:

Principle: *"A expression of an idea, a message (culture) or value that comes from corporate vision, strategies, and business drivers, experience or from knowledge of a subject."*

So [illegible] principles [illegible] the system [illegible] design and [illegible] principle [illegible] [[illegible]]:

- [illegible]
- Be [illegible] the decisions [illegible] about the components in the architecture;
- [illegible] the architecture is defined, is consistent and [illegible]

[illegible]

[illegible]

Mission: *"a use or operation for which a system is intended by one or more stakeholders to meet some set of objectives."*

[illegible]

Architectural Description: *"A collection of products to document an architecture."*

[illegible]

Concern: *"Something that interests the system stakeholder because it is important to him or it affects him."*

[illegible]

View: *"A representation of a whole system from the perspective of a related set of concerns"*

[illegible]

Viewpoint: *"A pattern or template from which to construct individual views. A viewpoint establishes the purposes and audience for a view and the techniques or methods employed in constructing a view."*

[illegible]

[illegible]

[illegible]

Model: *"An approximation, representation or idealization of selected aspects of the structure, behaviour, operation or other characteristics of a real-world process, concept or system."* ([illegible] 0. 2 0)

[illegible]

- **Interaction relations:** [illegible]
- **Structural relations:** [illegible]

[illegible]

[illegible]

Rationale: *"the justification behind decisions. It includes the issues that are being resolved, the alternatives being considered and discarded, the criteria used to evaluate alternatives, the assessments of alternatives against criteria, the argumentation, and the decision."*

[illegible]

the system's stakeholders and views, the viewpoints are not system specific. This makes it possible for an architect to reuse viewpoint descriptions [AWG1998].

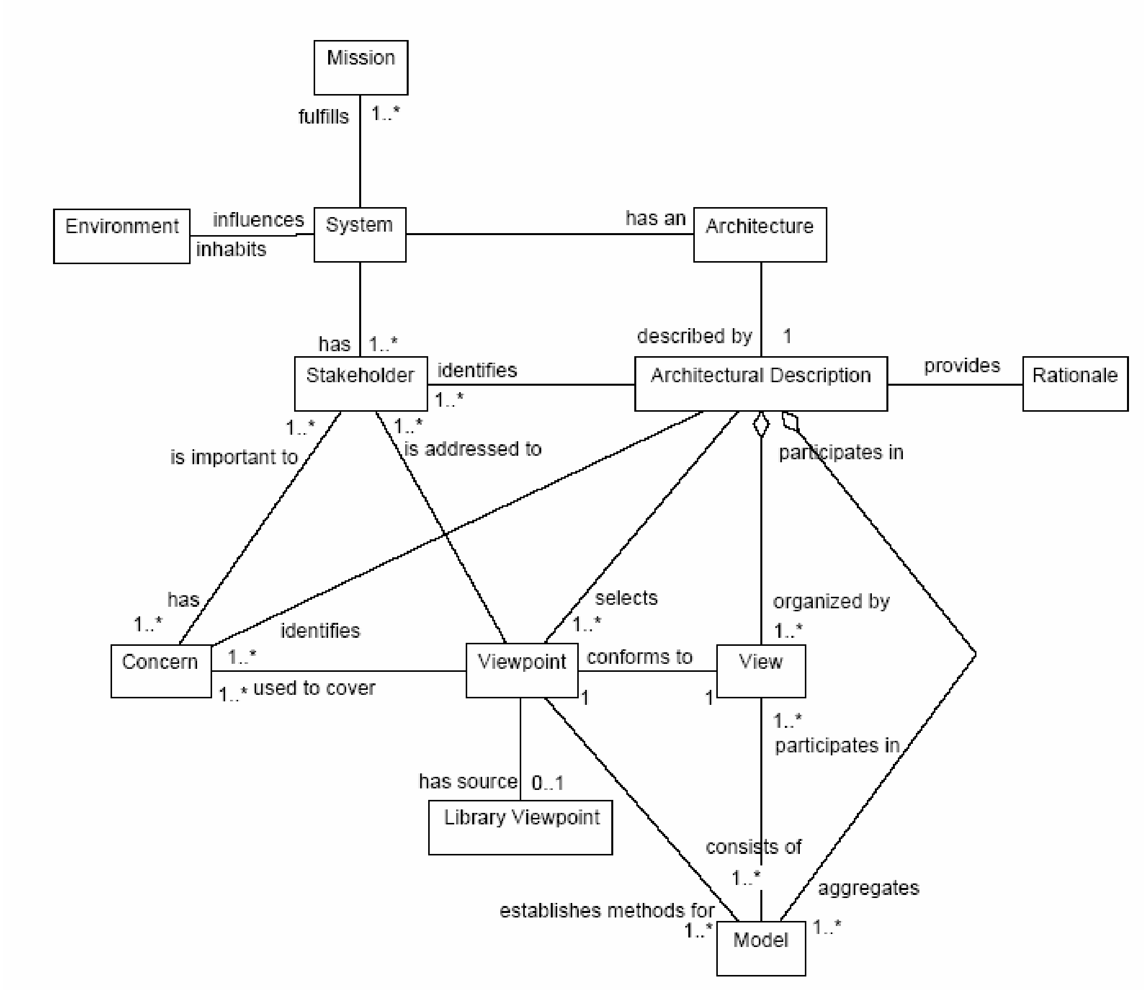

Figure 2-3: IEEE 1471 Conceptual Framework [AWG1998]

Figure 2-3 shows the conceptual framework of the definition in IEEE 1471. This figure also shows the relationship between the different systems and architectures. To be able to understand and make use of those concepts, which are discussed before, there are all these concepts that can be found in Figure 2-3. A system has always one architecture and this architecture can be described by one AD. An AD can consist of many views, models, and viewpoints. The rectangles represent the classes of objects. The lines connecting two objects represent the relations between these objects. The relations which have two roles (one in each direction) and a role can be named. In Figure 2-3 a role's a one to one or more expression is noted. For example: a stakeholder has one or more concerns and a concern is important to one or more stakeholders. The notation in Figure 2-3 is from the Unified Modeling Language (UML) [AWG1998], [UML].

According to the definition in IEEE 1471, an AD should include the following elements [AWG1998]:

- Identification of the system's stakeholders and their concerns judged to be relevant to the architecture. The decisions are made by the owner's of the architecture;

- ef n ons of each e on n as been se ec ed o o an ze e e esen a on of e a c ec e;
- Re esen a on of e a c ec e, o an zed n o e s;
- A eco d of a no n ncons s enc es a on e A 's e ed cons en s;
- Ra onale fo se ec on of e a c ec e.

2.3.1 Evaluation of IEEE 1471

No a s o conc s on on IEEE 4 be desc bed. e fo a on of IEEE 4 s e y s b e, a s beca se e fo a on s an abs ac on fo e exa na on of a de an e of bes ac ces n ese e ods. IEEE 4 does no on y efe o e o e a s c e of ys ca co onen s a a e a sys e , b a so den f es e ocess as ec s of an a c ec e. e ys ca s c e s of en a nda en a as ec of a sys e , s no necessa y. A so does no e e an A o co e e sys e 's bo nda y. e an ad an a e of s def n on s a s c ea and conc se [A], [o s], [e e d e a .]. Bes des a IEEE 4 as e fo o n ad an a es:

- def nes a e e y sys e as an a c ec e, e en f e sys e does no a e an A o f e sys e as no des ned b e o e e ed. e efo e an A can be c ea ed fo e e y ex s n sys e ;
- n s d ec y o e sys e s eo y, so conce s and def n ons can be e sed;
- def nes a a c ec e a es e con ex of e sys e of n e es n o cons de a on;
- def nes a A 's a e s c ed o ee e needs of e e s a e o de s of e sys e , nc d n , b ce a n y no ed o, e b de s of e sys e ;
- e conce ns of e s a e o de s a e os ead y add essed by A 's cons c ed e e s of e sys e , e ac e co e n an den f ed se of sys e conce ns;
- e cond ons on e co e eness, e fo edness, and ana yzab y of e s a e ade ex c o eade s of an A n an a c ec a e on (s as a s a e e ends ne x o e).

IEEE 4 does no e e [e a d]:

- A a c a a c ec e desc on an a e;
- A e ed e o ode ;
- A e ed fo a cons s ency o co e eness c e a.

2.3.2 Software Architecture

No a s ec f c a c ec e be cons de ed and n s es s Sof a e A c ec e s se ec ed as s a on. n s f ed, c s a e a e y ne d sc ne, e e s s no a n fo s anda d. Sof a e A c ec e s a e e des n of a sys e and desc bes e os o an c a ac e s cs of a sys e [o s]. An o an o e of an a c ec e s a can be sed o e fy and a da e o e es of e sys e . Kaz an def nes sof a e a c ec e as [Kaz an e a .]:

"The software architecture of a program or computing system is the structure or structures of the system, which comprise software elements, the externally visible properties of those elements, and the relationships among them."

This defines software architecture as a set of elements, with relations between them. It leads to the following implications [Kazman et al.]:

- **Architecture defines elements:** architecture embodies information about how elements relate to each other. An architecture is foremost an abstraction of a system that suppresses details of elements that do not affect how they use, are used by, relate to or interact with other elements;
- **Systems can and do comprise more than one structure:** also no one structure holds the irrefutable claim to being the architecture;
- **Every software system has an architecture:** because every system can be shown to be composed of elements and relations among them, every system has an architecture. However, that every system has an architecture, does not necessarily follow that the architecture is explicitly described. Unfortunately, an architecture can exist independently of its description or specification, which raises the importance of architecture documentation and architecture reconstruction;
- **The behaviour of each element is part of the architecture:** insofar as that behavior can be observed or discerned from the point of view of another element. This behavior is what allows elements to interact with each other, which is clearly a part of the architecture. This does not mean that the exact behavior and performance of every element must be documented in all circumstances; but to the extent that an element's behavior influences how another element must be written to interact with it or influences the acceptability of the system as a whole, this behavior is part of the software architecture.

The implications are the same as those from the IEEE 1471 definition. But in this definition Software Architecture is only a description of a system's structure. But an architecture has also to identify the process aspects of it. This definition is always suitable for architecture, because it only reflects the description of a system's structure.

Now the most important developments in Software Architecture will be discussed. These will be related to IEEE 1471. In Figure 2.4 is indicated with the numbers where certain developments can be located. The developments are [illegible]:

1. **Architectural frameworks:** frameworks offer standard definitions (the so-called viewpoints), which are ordered in one or more dimensions. Examples of frameworks are Zachman and IAF and these frameworks will be discussed in chapter 4;
2. **Architectural patterns:** these are standard solutions for problems that often occur. These patterns can be seen as standard models, or a kind of models. Architectural patterns describe the collaboration of components at a macro level. In this case the components are seen as black boxes, for which only the input and output is known;
3. **Architectural descriptions:** there has been a lot of research to Architectural Description Languages (ADL's) that can describe the architecture. These languages offer a formal architectural description that can be expressed in interfaces, relations, and functional properties of components;
4. **The evaluation of the architecture:** is about the quality of the architecture. Quality aspects can be seen as specific concerns and therefore evaluation of architecture can be seen as an instrument to measure to which extent the architectural descriptions satisfy the concerns of the different stakeholders. By evaluating the architecture in an early state, the project can be adjusted and other solutions can be examined;

5. **Product-line development:** is aimed to develop the basics of multiple systems at once. The development processes are being separated into domain engineering processes, and application engineering processes, in which the collaboration development is taking place in the first set of processes and the system specific development in the second;
6. **Architectural construction:** extracts the architecture information from existing systems. In terms of figure 2-4 it extracts in the architectural model of the system and transforms it into an architectural description. The internal processes, input and output of the system are used to describe that system.

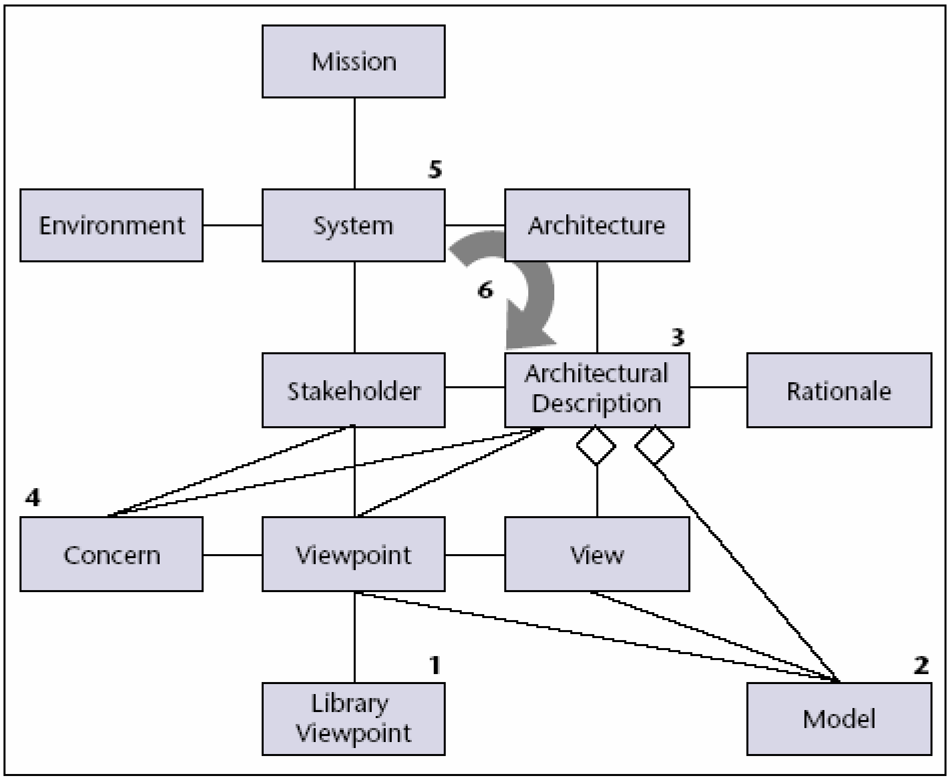

Figure 2-4: Developments in Software Architecture projected in the IEEE 1471 Conceptual Framework [Greefhorst]

(QWHUSULVH $UFKLWHFWXUH

'HILQLWLRQV RI (QWHUSULVH $UFKLWHFWXUH

3.1.1 The new definition of Enterprise Architecture

Enterprise Architecture: *"the fundamental organization of an enterprise embodied in its different architectural descriptions, their relationships to each other and to the environment and the principles guiding its design and evolution."*

Although the name implies another assumption, EA itself is not an architecture. EA is a meta architecture that consists of a set of EA's to all other architectures in there.

An enterprise interacts with its environment and it has the features of the characteristics [illegible] of the [illegible] and/or of an open system in section 2.1.1. In table [illegible] its environment and its environment has influence on the enterprise. Because an enterprise is an open system, the relations between the enterprise and its environment have to be taken into account. These relations are the interactions: the interactions between the enterprise and [illegible] elements in its environment. These relations are described by the different architectures of the enterprise.

The last part of the definition is: *"the principles guiding its design and evolution"*. These principles are the elementary principles of the enterprise. They apply to all architectures in the enterprise. For example: *"Reuse before buy before make"*. Besides these elementary principles there are also principles that are specific for an architecture, these principles only apply to that architecture. In section 2.3 the definition of a principle is given and principles are expressed in rules and guidelines for the enterprise. They also describe what needs to be achieved, the requirements are the way in which they can be achieved. These requirements are the missions (see section 2.3 for the definition of a mission) of the different systems, which are described by the architectures. In section 2.3 every system has a mission, which is the goal of the system. Every system has one or more missions. In section 4.4 will be discussed what the relations are between principles and requirements. The example of a business principle, which was already used, is: *"All employees have to access data, without special software, all over the world."* Two examples of requirements that can be derived from this requirement are:

- An employee can only access data when he/she is authorized;
- An employee has to have access to [illegible] only of the client.

There are more [illegible] characteristics of an open system. The first characteristic is that every system has a goal. The enterprise itself is an open system, so an enterprise has a goal. The elementary goal of a social system is development. Development is the process in which individuals increase their abilities and desires to satisfy their own needs and legitimate desires and those of others. An enterprise has to develop and adapt to its changing environment in order to survive. The way an enterprise wants to develop is written down in its elementary principles. Thus an enterprise is a purposeful system, not a goal-seeking system [Gharajedaghi].

In figure 3 there are four kinds of systems, which are already described in chapter 2. These systems are the enterprise systems, information systems, subsystems, and aspect systems. The arrow between these systems and the entity systems is an ISA relation. It means that [illegible] in the context of this a system is an enterprise, an information system, a subsystem or an aspect system. A system can consist of any number of systems. An enterprise can consist of any number of enterprises. This consists of any number of enterprises such as the food division [illegible]. Because an enterprise consists of any number of systems, the EA is the aggregation of all the A's of those systems.

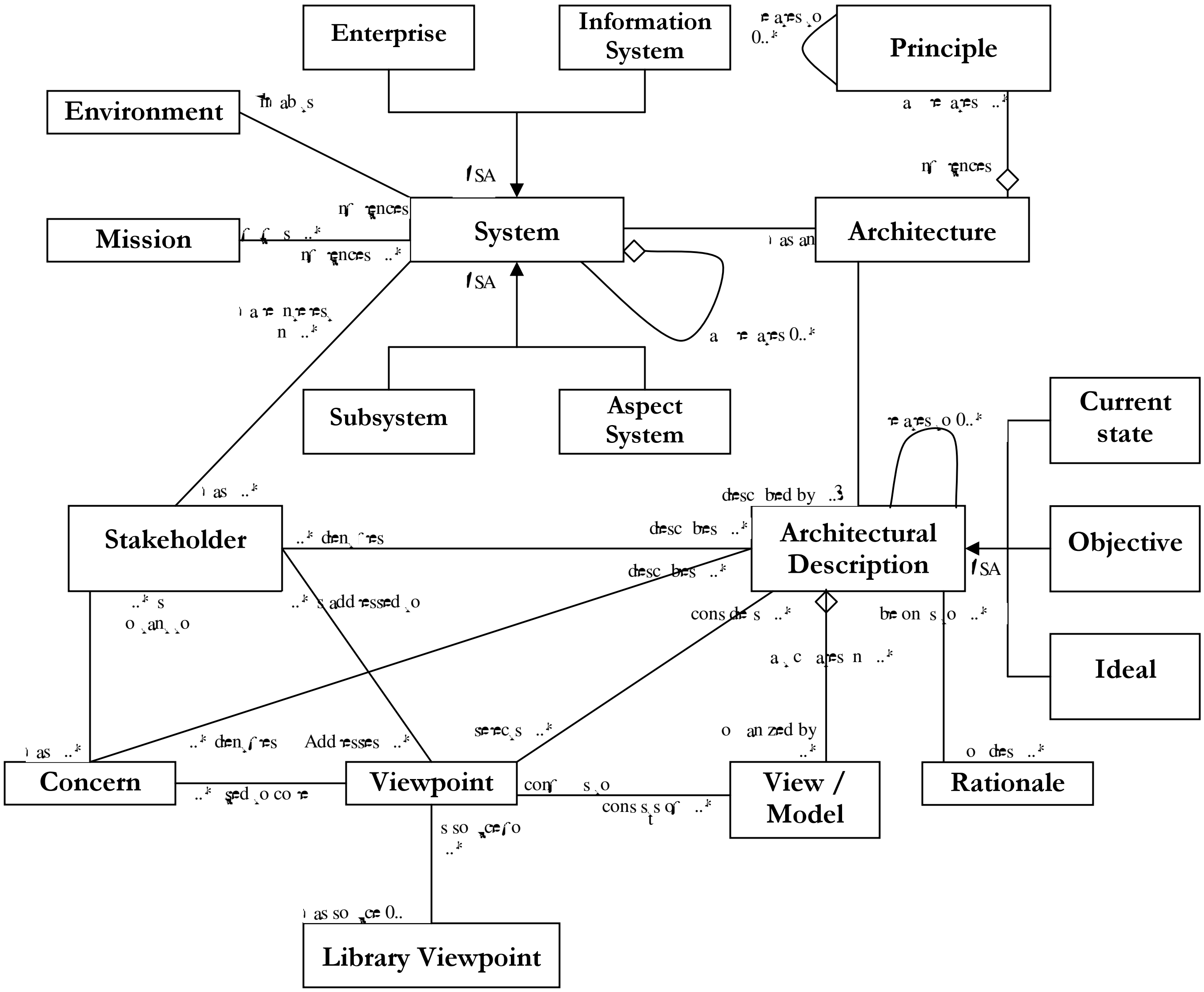

Figure 3-1: The Enterprise Architecture

Regarding to the time dimension, there are three kinds of AD's, which are the current state, the target state, and the ideal state. The AD of the current state is about the existing system at the moment. The AD of the ideal state is about the perfect and [illegible] use. This ideal situation would be realized if there were no technical, organization, and financial or other constraints. [illegible] the other hand, [illegible] situation should [illegible]. An AD of the target state is about the [illegible] that can be realized within a specific time horizon. [illegible] should be [illegible] and [illegible] to [illegible] the cost of the change. The ISA [illegible] will be discussed in section 4.4, also contains these three kinds of AD's. Figure 3-[illegible] is an extended version of Figure 2-3. [illegible] that the [illegible] 4 [illegible] the objects 'Model' and 'View' are merged into one object here. Because a view is exactly one [illegible] of an [illegible], every model can be considered as a view that can be described by one model [illegible] of a view [illegible].

3.1.2 Examples of different architecture descriptions

There are a lot of architecture descriptions that can appear in an EA. Examples are: Business Architecture, Organization Architecture, System Architecture, Application Architecture, Software Architecture, Information Architecture, Governance Architecture, Technology Infrastructure Architecture, etc. All these architectures could be defined in the same way as the definition of IEEE 1471. For example Business Architecture can be defined based on the definition of IEEE 1471 as [illegible]:

Business architecture: *"the fundamental organization of the enterprise embodied in its business domains, their relationships to each other, and to the environment, and the principles guiding its design and evolution."*

Every definition of an architecture can be derived from definition IEEE 1471 by filling in the blanks in the following definition:

… Architecture: *"the fundamental organization of … embodied in its …, their relationships to each other and to the environment and the principles guiding its design and evolution."*

In section 4.4. the six architectures of the IAF [illegible] be defined in this way. Every architecture has its own structures, [illegible], and [illegible]. Section 2.3 describes that an Architecture Description consists of one or more models and examples of these models are: organization charts, data [illegible], [illegible] models, etc. According to the definition of a model in section 2.3, an architecture is also a model because it is also a representation of selected aspects of [illegible], but also [illegible] the characteristics of a real world process, concept or system.

3.1.3 Other definitions of Enterprise Architecture

To get a better understanding of EA a definition of EA was given. To follow that a definition, next some definitions of EA were presented. In this section four of these definitions are presented in comparison with the IEEE definition. In the comparison the strength of the IEEE definition will be discussed.

The first definition is from the Institute for Enterprise Architecture Developments:

"Enterprise Architecture is about understanding all of the different elements that go to make up the enterprise and how those elements interrelate."

"A good definition of ***'enterprise'*** *in this context is any collection of organizations that has a common set of goals/principles and/or single bottom line."*

"A good definition of ***'elements'*** *in this context is all the elements that enclose the areas of People, Processes, Business and Technology."*

(Institute for Enterprise Architecture Developments [IFEAD])

This definition has the same goal as the IEEE definition: to understand the enterprise. But this definition does not take the environment of the enterprise into account. The definition of elements is not specific enough and there is no specification on how to understand the elements. Also there is no reference to the architecture.

The next definition is of the Railroad Retirement Board:

"Enterprise Architecture (EA) is a 'blueprint' of an organization's business processes and the information systems and technology needed to perform those processes efficiently."

(Railroad Retirement Board, Bureau of Information Services [RRB])

This definition already takes the architectures into account: the Process Architecture, the Information Systems Architecture and the Technology Architecture. But it ignores the other architectures and it does not take the environment and the relationships into account.

The next definition is of Maarten Lewis:

"An enterprise architecture consists of:

- *A set of equally important models of the fundamental organization of the enterprise and the interaction with its environment. The models are expressed in a modelling language (often partly a diagram technique) that, often support hierarchical decomposition to deal with components;*
- *Relationships between the models, satisfying rules that guarantee consistency and completeness;*
- *Principles governing its design and evolution.*

The models cover at least the following aspects of an enterprise:

- *Goals and strategy;*
- *Business processes;*
- *Business objects;*
- *Organization structure;*
- *Information systems (hardware and software)."*

This definition is very similar to the first one. The second part is an addition to the first (the actual definition) and that does not belong to the definition of an EA. Every enterprise can decide which models are defined needed to describe the EA. But this definition does not consider the relationships between the architectures. There are only relationships between the models and there are no relationships between the architectures and the environment (the mission of a system). Also in the definition of EA, which was introduced in section 3.1.1, the models do not have to be equally important. In my opinion if you want to achieve a complete description of an EA you need all the models and for that reason they are equally important. But it is impossible to have a complete description of an enterprise because there are so many models and they are continuously changing. EA is used to reach a certain goal and to reach that goal not all the models are needed, not all are equally important. For example, to plan the technical infrastructure an IT architecture is needed, but to plan a business change an IT architecture is already less important. Thus to reach one goal a certain model may be essential and to reach another goal the same model may be less important.

The last definition is of the Global Enterprise Architecture Organisation:

"Enterprise Architecture provides, on various architecture abstraction levels, a coherent set of models, principles, guidelines, and policies, used for the translation, alignment, and evolution of the systems that exist within the scope and context of an Enterprise."

(Global Enterprise Architecture Organisation [GEAO])

This definition consists also of three odels, nc es (n is is s considered a de nes and o c es a e a of the nc es) and the ten on en , b does not con a n the e a ons be een the . In fac , as en oned before, the e a ons s a e the added a e of an EA.

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Besides n a desc on of the ente se, EA has also a ocess as ec n . EA ay s o the ans on f o the c en s a on (the As s a ch ec e) o a o e ffic en and effec e s a on (the To be a ch ec e). In sec on 3. . as ex a ned that a To be a ch ec e co d be s n o an dea s a on and a ea s c s a on. The o e en f o an As s a ch ec e o a ds a To be a ch ec e a e ace n three s e s (see also the a on n F e 3 2). The f s s e s o desc be the As s a on and the second s e s o define the des ed (To be) s a on. The To be s a on can be s n an a e s a e and an dea s a e. The th d s e s the ans o a on ocess f o the c en o the a e s a on, h ch s ca ed n a ans on an [B ne e a .2004], [S].

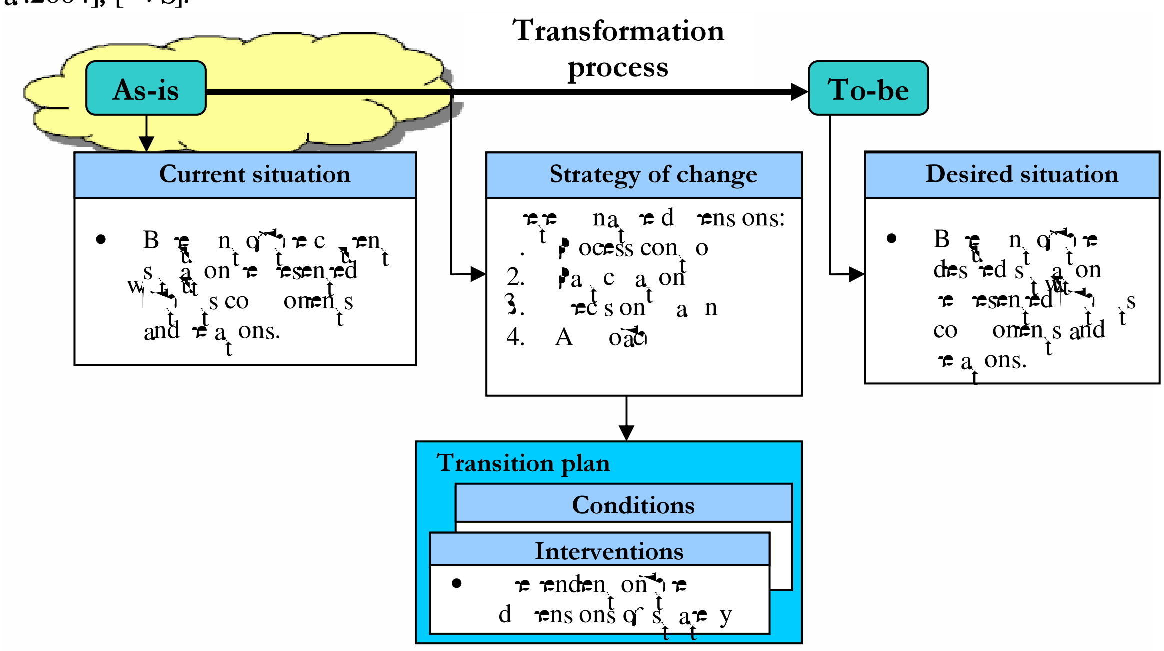


Figure 3-2: The three steps of Enterprise Architecture [Bruijne et al.2004]

The As-is architecture

The As s a ch ec e e esen s the c en s a e of an ente se. The As s a ch ec e s a b e n of the c en s a on n h ch the co onen s and the e a ons be een the a ea . The As s a ch ec e has o be ade f s o no ho of the c en s a on s. This a ch ec e ay e ea nade acies e ncons s encies o effic encies.

The To-be architecture

he o be a ch ec e e esen s he dea and a e s a e o he en e se. sab e n o he dea and a e s a on n ch he co onen s and he e a ons be een he a ea . s a ch ec e e esen s he es s a ons o he en e se. he dea s a on s he s a on ha s o d be ach e ed. B s no oss b e o ach e e h s s a on on a s o no ce. he o e he a e s a on s n od ced and h s s he s a on ha can be ach e ed n he nea e. A b e n o he dea s a on s o d be ade s o a e s e ha he a e s a on no n e e e h he ach e e en o he dea s a on so e he e n he e.

The transformation process

he ans o a on ocess con a ns h ee a s: s a e y o ch an e, n e en on and he ans on an. he ans on an con a ns a he he a c as ec s o he ans o a on ocess and s a oo o e a a e he n e en ons on a se o cond ons and a e ab e. he ans o a on se a es ade by n e en ons and he s a e y o ch an e s he de me o se ec he os o n e en ons. he s a e y o ch an e s based on he c en s a on, des ed s a on and he o de na ed ens ons. hese o d ens ons a e [B me e a .2004]:

- **Process control:** he ans o a on ocess can be o do n o bo o ;
- **Participation:** he ans o a on ocess can be d ec ed o s o he a ch ec s by c ea n an A o can be d ec ed n a n he a ch ec es;
- **Decision-making:** he dec s on a n can be based on consens s o he dec s ons can be ade on he o ;
- **Approach:** he A can be c ea ed and sed based on d e en a oach es, see a so sec on 3.3.

hese o d ens ons a e sed o e a be e nde s and n abo ha a ens and h ch es ons a e ay n a o e n he ans o a on ocess.

7KUHH GLIIHUHQW (QWHUSULVH $UFKLWHFWXUH DSSURDFKHV

he e a e d e en a oach es o A and hey e esen he d e en e s, oa s and se s o A. he o o n desc on o A a es ha n o cons de a on [B 2004]:

> *"… there is no correct definition of the term 'enterprise architecture.' We will continue to use it in BPTrends in both its common senses—as a description of how all the resources of an organization can be structured by the organization's business processes, and as a description of how all of the organization's computer resources are organized."*

he o a on s o s ha he e a e o desc ons o A: one h a b s ness o n o e and one h an o n o e . hese o e s a e he os co on ones. n h s ch a e , he b s ness ocess cen c and he cen c A (nc d n da a, n o a on and n o a on sys e s) be desc bed. A so a h d a oach be e h and s s he o e nance cen c A. he e s no a bes a oach, b an en e se has o se ec an approach has its strength because each approach tries to obtain its goals. The approaches are different and have different objectives because of the differences in uses and ways, have different goals and different views. The different approaches are based on different views, for example tools, and users.

	IT-centric	Business process centric	Governance centric
Objective	Business IT alignment	To manage and improve business processes	To understand the business better, to be able to govern it better.
Model	An overview, for an IT perspective, of all the different IT and business models and resources and how they relate	An overview of all the different business processes and the relations between processes and the interaction between them	An overview of all the domains with interference described from different viewpoints
Decision	IT systems and the architecture	Business processes	Communication between Business and IT; collaboration between all the decisions and analyses
Main actors	IT managers	Business managers	All the managers of interest
Users	IT departments	Business departments	All departments

Table 3-1: An overview of three different EA approaches

In Table 3-1 an overview of the different approaches is given and in the next sections they will be described. To get a good understanding of the different approaches the Enterprise Architecture Pyramid of BPTrends will be introduced. This pyramid shows how all the different elements are related within an organization.

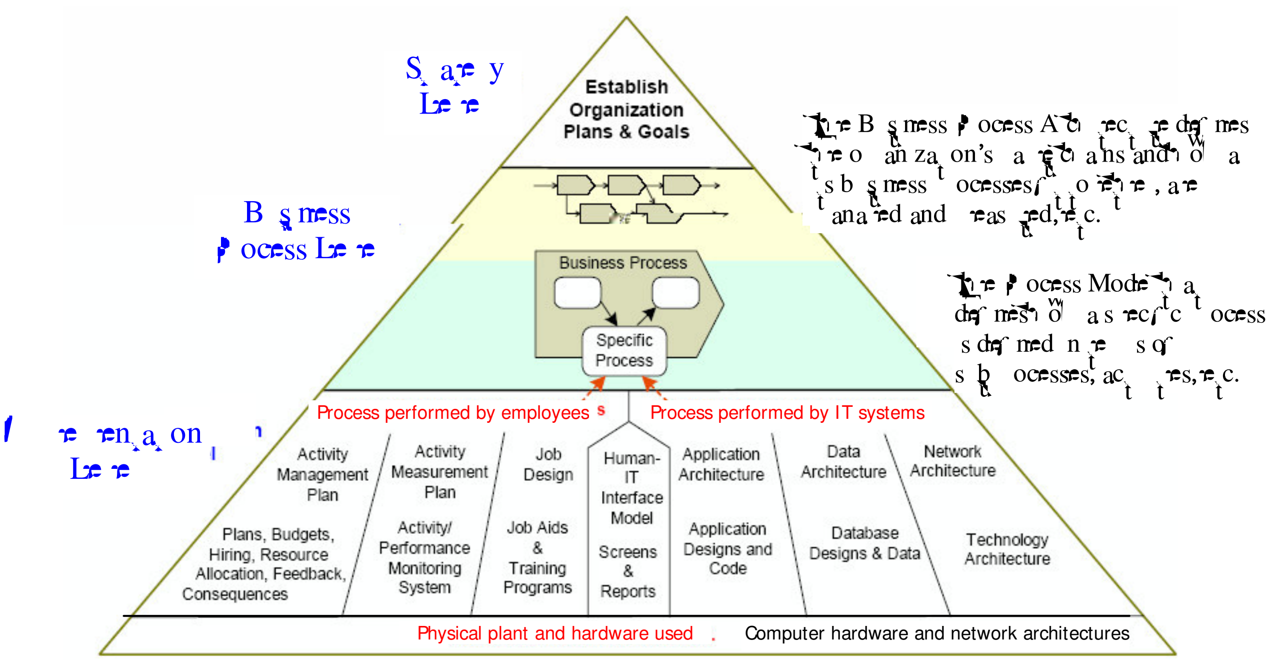
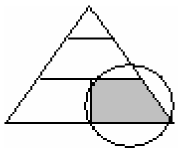


Figure 3-3: The BPTrends Enterprise Architecture Pyramid [BPT2004]

3.3.1 IT-centric



...s a ...oac... a... be considered ...e IT-cen...c a...oac... . The ob...ec...e s ...o c...ea...e B...s...ness IT a...n...en...and EA s...any ...sed ...o ...ffec...eness and ...ff...c...ency. ...EA ...IT ...can ...ea...an...ed n s...a...ay ...a...ac...ee ...s...oas o...a...s ...o e...ff...c...en...and ...e...o...e...s cos...sa...n... . The ...e...on...e ...,...c... be on s...o IT-cen...c EA, s...e...ya...d ...e 3.3. ...s ...es ...o...s...a...e ...oc...s ...IT-cen...c EA ...es n...e bo...o...co ...e, ...c...s...e...e...en...a...on ...e...e...IT sys...e...s. ...n...e...ses, ...c...a...e...sn ...an IT-cen...c EA, ...on y cons...de...e a...eas ...e en...e...se ...a...s necessa...y ...o de...ne...e IT. B...s...ness a...c...ec...e...n...s con...ex...e...s ...o...ode s...e b...s...ness, ...c...a...e ...s...a...y ...s...ocesses and da...a...ode...s, ...a...s...o...e...des...n...IT so...ons [B...2004], [...o...a].

...e IT-cen...c EA s an a...oac...a...s...os...y ...sed n...e IT ...o d and ...s...sed ...o ...e...e...a...n...an a...c...ec...e ...a...o...des an o...e...e...a...e d...e...en...IT ...ode...s and ...eso...ces and ...o...e y ...e a...e. ...n...s a...c...ec...e...e...e s ...oc...s on...e IT sys...e...s and ...e a...c...ec...e. ...e b...s...ness s on y desc...bed ...o...a...do...n...e de...and ...o IT and ...s de...and s cons...de...ed as ...en and ...s s no...d sc...ssed. IT on y ...nces ...e b...s...ness ...en ...e ...ec...no...o...y enab...es ...e...b...s...ness ...ocesses. ...e IT a...c...ec...e ...as ...o n...o...e b...s...ness ...ocesses and ...e EA s a...oo...o do ...a...as ...ood as ...oss b...e. ...s ...e...e...s a ac...co...n...ca...on be...een ...e b...s...ness and ...e IT. ...e n...a...o s ...an IT-cen...c EA a...e ...e IT ...ana...e...s and ...s a...c...ec...e...s...an y ...sed by ...e IT de...a...en... . Mos...en...e...ses and ...eo...e, ...o a...e ...sn EA, a...e ...sn ...o...an IT ...on ...e... . ...a...s beca...se a o...co...an...es s...cons...de...EA as ...e ...es...ons b...y ...e IT de...a...en... .

...e...ode ...an EA n ...e 3.4 cons...s s ...e ...o...aye s ...o...e IA ...a...e...o..., ...c...be d sc...ssed n sec...on 4.4. ...s ...ode can be ado...ed ...n an EA ...an IT-cen...c con...e... . ...n...s, ...e IT s de...n...od...ced EA and ...ey ado...an IT-cen...c a...oac...and ...ey a...e ...s n...e IA ...a...e...o... . ...e ...o...e ...o...aye s ...be d sc...ssed ...e...e and ...ese a...e [...an 2003]:

- **Business:** b...s...ness s...c...es, ...e a...ons, ...as s and ac...es s...o...d be de...ned ...o...e ...e...e ...de...a...o...c...e...e...o...ance ...e...cs can be ...a da...ed and ...e ...ec...no...o...y s...o...nee ds ...den...ed;
- **Information:** ...e ...o...a...on ...aye...cons s...s ...o...s ...o...a...on, ...o...s ...doc...en...a...on, ...e needs ...o...a...on, ...e so...ces ...o...a...on and ...o...a...on ...e...c...an...e ...e...o...d...o...s...de. ...s ac...y de...nes ...ey ...o...a...on ...o...s and ...c...a ac...e...s...cs ...n a b...s...ness a...ea a...a...e...e ...de...a...a can be ...sed ...o access ...e...a...n...es and ...o...e...y a...n...e...n...e o...e a...en...e...se a...c...ec...e. IT a so ...de a desc...on ...o...a...on...o...e...en...and sec...y se...ces ...e...ed ...o...e...o...a...on sys...e...s and ...ec...no...o...y ...as...e [...];
- **Information-Systems:** ...e...e ...de...a...s de...ed ...o...e b...s...ness and ...o...a...on ...aye...o..., ...c...de...nes...e...e...ed so...ons, ...c...e se..., ...nc...ons, ...ea...es and s...anda...ds;
- **Technology Infrastructure:** ...s ...aye de...nes...e ...ec...no...o...y se...ces ...o...ded, no...o...ey a...e ...e...en...ed.

...s ...aye s so...e...es...en...oned as ...e ...a...a ...aye .

[illegible]

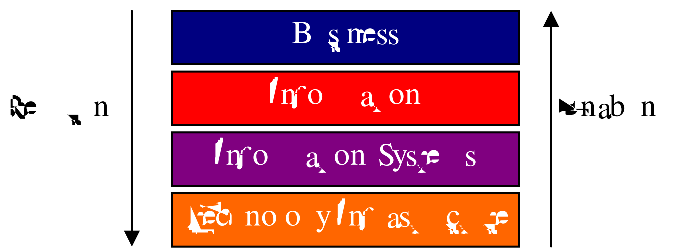
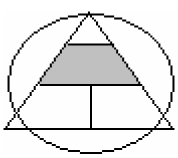


Figure 3-4: The four layers of Enterprise Architecture in an IT-centric approach.

[illegible]

3.3.2 Business process-centric



[illegible]

"… *we first design business processes in a more abstract way, without considering implementation, and then we design the information systems and the organization hand in hand.*"

[illegible]

[illegible]

ocesses) n e con ex . ocesses a e edes ned o se bo an and ca ab es n s a ay a c en s be o e sa s ed, e. by be e and/ o c ea e od c s o se ces. n s a ec e e s a oc s on e b s ness ocesses and s cons de ed o s o ese b s ness ocesses. e b s ness ocesses a e be n o zed e e o A. n e ocess o o z n e e s no co n ca on and s me e cons de ed n e s s a e o B s ness ocess edes n (B). e n a o s o a b s ness ocess cen c A a e b s ness ana e s and s c ea ed and sed by e b s ness de a en s.

e ocesses n e en e se can be d ded n ee nds o ocesses. ese ocesses (no n n d e en na es) a e an ee2002]:

- **Production processes or Primary processes:** od ce e od c s and se ces o e en e se. ese ocesses o de o and dea e c s o e s;
- **Support processes or Secondary processes:** s o e od c on ocesses. An o an a o ese ocesses concen a es on a n a n n e eans o od c on;
- **Managerial processes or Tertiary processes:** d ec and coo d na e e od c on ocesses and e s o ocesses. bec es and econd ons a e o a ed and ese ocesses enco ass e a n enance o con ac s s a o de s.

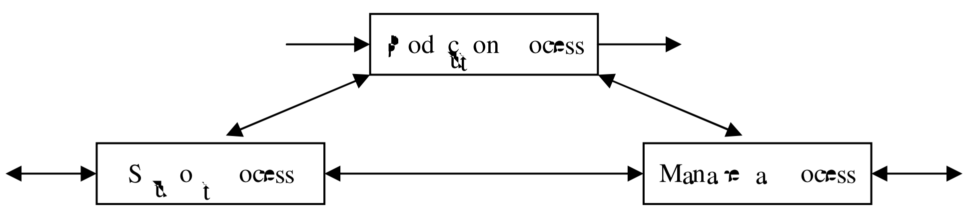


Figure 3-5: The three kinds of processes [VanHee2002]

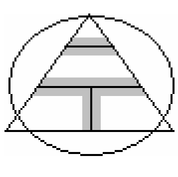
e 3 5 e esen s e ee nds o ocesses e a e a ons s and e e a ons s e o d o s de. e s o and ana e a ocesses a e con n o s (ey ay con a n d sc e e s b ocesses), b e od c on ocesses a e s a y case d en and e e o e d sc e e. n abe 3 2 a e e a ons s o eac ocess o e ocesses and e o d o s de be cons de ed an ee2002].

	Input	Output
Production process	• dre s • Ra are a and co omen s (s o ocess) • Ass n ren s and as n b d rs (ana r a ocess)	• od c s and se ces • Re n reso ces o s o ocess a e se • Re o o ana r a ocess
Support process	• Means o b y reso ces (o ana r a ocess) • s ose reso ces a r no on r nc on n (ana r a ocess)	• ace reso ces o r d s osa o od c on ocess • Re o o ana r a ocess
Managerial process	• b ec res and ca a • Re o s (od c on ocess & s o ocess)	• e o ance (en e o o)

Table 3-2: The input and output of the three kinds of processes [VanHee2002]

s r b s ness ocess cen c A oc ses on a r ocesses n rrenre se, o o ze r and o c rea re me o o n res. se o o n res a re r s are y rene a on o re ren e reme a on. A o r oc s res on r ocesses, r sys e and ec no o y a ec a co omen s be zed. B a s o re n a seconda y o e [o a].

3.3.3 Governance-centric



re d a oac a be cons dered s r o r nance cen c a oac . r ob ec r s o nde s and r b s ness be r , so r ana r en s abe o o r n (bo s e o ance and con o ance, see sec on 3.4) be r . Beca se en e ses a re r n oo co ex, r e a re o s a re res r A a can r n o r n n r ren e se. r s s a re y s o oc s on o n ana e en and con o sys e s o r o a , n r a ed en e se.

r o r s a re y s o d d r ren e se n o a me o s a r , be e ana eabe en e ses, c be ca ed b s nesses do a ns, n o de o o r b s ness o e nance. r se b s ness do a ns a re ea res ons b e o r o n ocesses and sys e s. r y exc an e b s ness a r n a se co o a r o a s and nc es. o o a e o e nance en a s a ana n r o o o en e ses o ea ze e o ance oa s and o on o co ance o ex e na and n e na es. Lo r r dec s ons and on o n a e r es ons b y o r 'do a n o me s'. r y me o a e and con ac e a ons be een r ocesses and sys e s and a res n r con ex o r en e se. r b s ness do a ns c ea e added a e by o d n od c s and se ces, so r y nd d a y can be cons de ed as an en e se. o e r , a r b s ness do a ns b d r en e se, so en e se s ac a y a s e do a n (a me o en e ses). An A o des an o e o a r d e en b s ness do a ns o en e se. e b s ness do a n desc ons o d e en e o n s can be ade, s a y r ocesses and b s ness da a a e r os o an ones o oses. r do a n o me s a re es ons b e o e de and. So s necessa y a e e s no ed e o n e b s ness do a ns and no on y n e nc ona de a en [e e d e a . 2004].

e a e e o s a e, e on s o o e nance cen c A, s e y a d o e 3 3. A o e o e y a d s n cons de a on, e oc s o o e nance cen c A es be een e d e en e e s (S a e y, B s ness ocess and e en a on) and be een e e oyees (o an za on o e en e se) and e sys e s.

n o e nance cen c A e e s a oc s on e co abo a on be een a e d s ons and ana e s o e en e se and e oc s es on e co n ca on be een B s ness and . e n a o s o a o e nance cen c A a e a e ana e s n e en e se and s c ea ed and sed by a e de a en s.

e 3 s o s a n a o e nance cen c A e s o s e n be een e o e nance and e b s ness. e b s ness as o ea se a B s ness A c ec e (BA), c s a co b na on o o an za ona s ons, b s ness ocesses and n o a on. e BA de nes e s a n o n s o e sys e de e o e s. BA can be o e nance cen c en oc sed on e es ons b es. s de n on o a o e nance cen c BA as desc bed n sec on 3. .2.

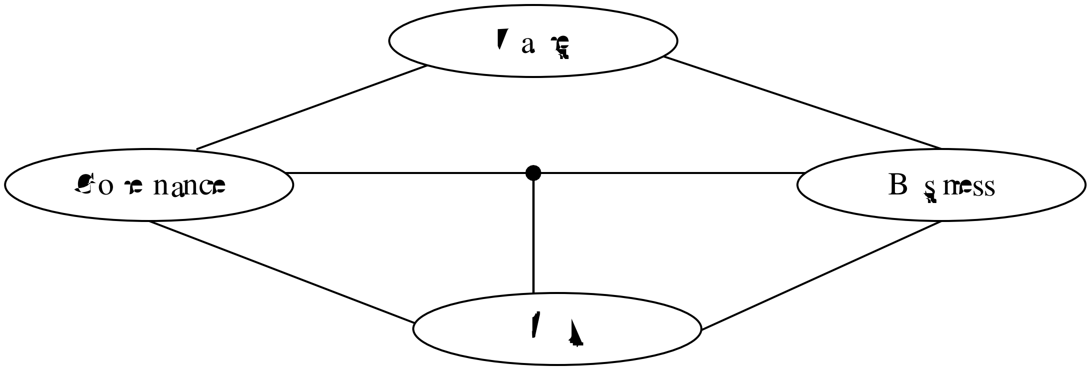


Figure 3-6: ICT supports the link between the governance and the business

e BA b n s a sys e a c ocess o n de s and o an en e se b n s a e o c s o e s e o s n e o ance [ ]. A oss b e a e o o a o e nance cen c BA s esen ed n e 3 . s e be sed o s a e a e d e en e e en s a a e n o ed n a o e nance cen c a oac o BA. ese e en s a e [oo e o s]:

- **Mission:** e ss on s based on e sa s ac on o e c s o e . de nes e o ex s o e en e se;
- **Strategy:** e s a e y de nes o e en e se s cceed n eac n s obec es. s nc des e oss b y o o an za ona ea n n , c enab es ada a on and e e a ;
- **Market:** e e s de ned o e a e s o d be ex o ed and ex o ed;
- **Competitors:** e e s de ned a e e a e os on s a ans co e o s;
- **Products and Services:** e e s de ned o od c s and se ces s o d be des ned;

- **Key Resources:** expresses principles about technology and human resources used in the deployment of products and services;
- **Operating method:** expresses the central financial business focus. The principles determine how financial gain is obtained;
- **Customers:** the recipients of the products and services of the enterprise;
- **Environment:** the activities that are important for the environment. Therefore principles must be formulated in such a way that all the activities that have to be executed in the context of the environment;
- **Stakeholders:** the people who are affected or influenced by the activities of the enterprise.

Figure 3-7: A possible Business Architecture framework for a governance-centric approach [Hoogervorst]

Figure 3-7 shows all the different elements that reflect the enterprise and how all these elements are related to each other. This framework is used for conformance and performance in order to make the right decisions. Conformance and performance will be explained in the next section, which is about governance. In a governance-centric EA the focus is on collaboration and this framework is used to achieve the mission and strategy that are aimed.

Enterprise Architecture and IT governance

Enterprise analysis says that EA increases understanding and insight, which leads to different levels in better decisions, especially related to IT. Governance is about decision making and especially about the way of decision making which

[illegible]

[illegible]

Governance: *"the audit, act or manner of conducting the policy and affairs of an organization, and the control or influence of people by constituting rules, standards or principles."*

[illegible]

[illegible]

"IT governance is the responsibility of the board of directors and executive management. It is an integral part of enterprise governance and consists of the leadership and organizational structures and processes that ensure that the organization's IT [2] sustains and extends the organization's strategies and objectives."

[illegible]

- [illegible]
- [illegible]

[illegible]

[illegible]

The benefits of using an Enterprise Architecture

[illegible]

---

[2] [illegible]

Only after that an insight in the current situation a lot of the presented benefits could be realized [Abcouwer et al.].

The benefits that will be discussed in the next sections are [A[illegible]8], [[illegible]a[illegible]], [[illegible]an[illegible]]:
- Exploiting scale effects;
- Analyzing the current and future state;
- Portfolio Management;
- Communication with the stakeholders;
- Business/IT alignment;
- Business process focused organising;
- Learning;
- Transparency;
- Sensitivity of the context;
- Adaptivity of the organization;
- Enterprise-wide taxonomy;
- Real business rules and conformance agreements;
- Other benefits.

These benefits are sometimes benefits that belong to some of the three approaches, but there are also benefits belonging to all the different approaches. In the next four sections the benefits will be described considering each approach. The first section describes the benefits that affect all the approaches.

3.5.1 General Benefits

In this section the general benefits of an EA will be discussed. By using any of the three approaches these benefits can be occur.

Exploiting scale effects

The act of economy of scale results in a lower costs. Both ways it is possible to get a reduction of costs of investments by exploiting the economy of scale effects. EA is the solution for that. By creating an As-is model to be able to detect the symmetry of outputs and redundancy can be found. For instance, in an IT-centric approach the scale effects can be found in symmetry between the different applications. In a business process centric approach the symmetry can be found between the different processes. Multiple parts of the enterprise use doing more or less the same way and do the same 'standard' required process and only finance information systems support in those processes.

Finding symmetry of outputs and redundancy is the purpose of the 'towards someone's' approach. The purpose of this approach is to start a collaboration between the different IT's of IT's. The purpose of EA is to create a combined architecture that all parts as possible work at the same applications. Reducing the costs of investments means [Tan[illegible]]:
- Saving of knowledge by using the same architecture standards and functions;
- Exploiting the economy of scale by using shared services.

Analyzing the current and future state

[illegible] [A 8], [A].

[illegible]

Portfolio Management of change projects

[illegible]

Traceability of Arguments

[illegible] see section 4.2. and 4.2.2 [a].

Communication with Stakeholders

"EA provides a mechanism that enables communication about the essential elements and functioning of the enterprise." [Sch an 2003]

[illegible] [A 8], [A].

Other benefits

So e a endan bene s o s n an EA a e [A 8]:

- The management system has already one or more A's, the models, the creation of an EA is then the way to combine those A's in a structured way. In this way EA is the tool to manage multiple A's. The relations between the A's themselves are an added value, which does not occur within an A;
- The A's can be used by the tool for maintenance of systems, locations, etc, so the systems are better adjusted to the current situation. In this situation an EA contains also specifications etc. for all of the systems in a composition of features;
- Asset management: the management system exactly knows what it has, about the functions, systems, licences, employees, software etc.;
- Architectures on a different level are supported and integrated with architectures on a corporate level.

3.5.2 Benefits of an IT-centric approach

In section 3.3.1, the most important objective of an IT-centric approach is Business IT alignment and that means [Wagter]:

- Maximal support of the business processes;
- Maximal support of interactions of the business processes (internal and external).

The definition of EA is to show the management system has to be described that is different architectures and the relations between them. In this way the Business Architecture (including the business processes) can be aligned with the System Architecture, Application Architecture, Software Architecture or Information Architecture. This also creates that the business is not influenced or not be influenced by the IT and in that way. The creation of the EA that represents the current situation is the Business architecture. With the Business architecture the method of the business could be created and that supports it makes it easier to define and select the right IT components at the IT's side [..a], Wagter].

3.5.3 Benefits of a Business process-centric approach

In section 3.3.2, the most important objective of a Business process-centric approach is to manage and control the business processes. A benefit of an business process-centric EA is that the focus is on customer focused on all of the management system. Business processes are managed and monitored in such a way that the needs of the customer is the centre of the business processes. Another benefit of EA is that retaining is possible in a structured way by creating feedback loops on every level in the management system. In this way, EA is used for fulfilling the needs and demands of the customer.

3.5.4 Benefits of a Governance-centric approach

In section 3.3.3, the most important objective of a Governance-centric EA is a better understanding of the business that makes a better governance of it possible. By deconstructing the management system business models it is possible to assess business units in the business domain (instead of functional contributions to the whole) and compare them with their equivalents.

The implementation of EA as a management tool has as benefit that a sensitivity for the context is organized on every level of the management system. This increases the adaptability of the o an sa on, e. o an sa ons a e ab e o es ond o e ade a e y o an es n e en on en . o A, en e eme a a ono y and n a e can be ana ed a a e es o e e se, es s n ex o n e an c ea e o en a . o A, b s ness nc es and con o ance beco e c ea o a e s a e o de s.

(QWHUSULVH ,QIRUPDWLRQ 6\VWHPV

e e a e exa es o B s ness a n en e e e se n o a on Sys e s (S). SA s an exa e o a co any a o o so ons n e e d o B s ness a n en . A o o co an es canno s e o e se eso ce ann n ( ) so a e. n any a e co an es, SA as beco e e ac o s anda d. SA s an n e a ed n o a on and ana n sys e n b s ness ocesses can be s o ed and ana ed. ese ocesses a e ca ed n od es. e da a n e od es a e a exc an ed, a eads o a y n e a ed sys e . A so s s s n a o o SA od s an esse], [ed a].

sys e s e SA 's / a e o an zed a o nd e bas c econo c a ona e o e en e se a e an. s a ocess e o a b s ness a e a ds a ansac ons as s e s n a a e acc a on se ence o des a 's c s o e s a na o o o o des ab e od c a b es. s o e en o a ds ada ab e e esen a ons o en e se a e ans o d a so ed c e nc eas n e e ence o b s ness ob ec a e o s e e e se an c d s ance be een an econo c ea y (an en e se) and e co e e esen a on o a ea y dec eases e en o e d a a ca y. e o end sys e s s as ose o e ed by SA es de n s ca e o y e e e en e se o en a on a s o n e a e acco n n ansac on ocess n o o , des n, and en ee n ana e en . sys e s b n a ea y ce n e s o co ex y o e e . ey a e ono c and n ex be, e y ca b s ness a co s o 's co n se o a ocess and o an za ona s on a ay no e s ea y exac y [SLA].

SA e es a e b s ness ocesses a e ad s ed o e SA e en a on. an es o e SA e en a on can be done, b a s ed and ex ens e. e o e a o o co an es a e o an e e b s ness ocesses en ey an o se a SA e en a on. o do s e b s ness ocesses a e o be desc bed s and a e a e o osed an es a e o be exa ned. e B s ness a n en o SA s on y ea zed n s anda d s a ons. n o e s a ons e e en a on o e b s ness ocesses a e o be c an ed. n e as yea s a o o sys e s ad no a ays ed o e ex ec ed es s. e eason o s s a a e e se ec on o an sys e , ed a e y as s a ed e e en a on ase. s as done o cons de n o e sys e o d n o e ex s n n o a on sys e a c ec e. en s a d o a e e sys e an n e a a o e o a n o a on s y.

o e en ese s a ons e A oo s, be d sc ssed n e nex a e , can ay an o an o e. ese oo s o d eme a e an o e e and c ea e an nde s and n n any s a ons. a a es oss b e o n e a e e sys e n o e c en en on en . ese oo s e b s ness ocesses can be be e ad s ed o e SA e en a on, o so e es e an es a a e eeded o e e en a on can be de e med. n conc s on S sys e s n B s ness a n en n a ed ay and A s necessa y o e bed ese sys e s n o e en e se [R senb e a .].

$ YLVLRQ DW WKH IXWXUH

Nobody can ed c e e, b n s a e so e oss b e de e o en s be d sc ssed. n ese de e o en s A ays an o an o e.

3.7.1 The CIO-cockpit

"I have a dream that one day …", ese a o s o ds a e o as e c o Ma n L e n e d a e L nco n Me o a n as n on . . a 28 A s . ese o ds e as e as z n s e s ec e o a e ec soc e y n e e. e as a so a d ea . n s d ea e e y h s connec ed and o n o e e. e s ab e o see a s a en n s n o a on sys e s and e s ab e o eac based on e n o a on e s seen . n a b sc een n s o ce, e s ab e o o o e ans o o da a. e sees e e a e a es n e ans o o e da a and e sees e e e o o da a s so n do n. A any e e s ab e o see n o a on sys e s a e od c n a nd o da a and ocesses o o e n o a on sys e s a e s n a da a.

e o be a c ec e s e e a ed e o e ac a s a ons; A odes a e no s odes s b ea e desc ons a s o ea e dec s ons. e a so need o on o o ess s. e o be a c ec es. o ec s and e o ess a e s a zed s n A odes. A c ec a desc ons a e s d ec y e a ed o e o e a e ocesses a a es e e ab e n e e.

3.7.2 The adaptive organization

A e o en ad ona nd s a o an sa ons, se a a e sys e s, a e ans o n n o ada e o an sa ons. Se e a a o s a e desc bed ese ada e o an sa ons as eb based o an sa ons s o ed by sys e s. ese en e ses a e coo e a n o e en e ses as e o a a ne s. e ey o ds a e o d de (es ec a y e e o e n e ne), ans a en (c s o e s de and be e n o a on) and a e o so c n o ocesses). s es s n a soc e y n c s o s eo e and en e ses o de e o o a y. n e nea e e en on en o en e ses c an e con n o s y. n e ses a e o ada on ese c an es. o o de e e ed ex b y co omen s a e e so on. A c ec e s e ay o n e a e ese ne co omen s and e acy sys e s and a n e ex s n ocesses and e ab y o e n e ne [R senb e a .].

n e 3 8 e ans o a on ocess s d a n. o o o a ds an ada e en e se cos s a o o o . s s de c ed as a o n a a as o be c bed. n s c b b s ness as o be ede ned cons de n e o o n es o . e e oyees, n e as e as n e b s ness, e ne ca ab es and o es. A ca on de e o en be c an ed d a a ca y beca se e de e o en as o be c e . o ex y, n as e n e b s ness, nc ease. A s e oo a e n e o ney o e a ey be nd e o n a n [R senb e a .].

The n information system consists of components that are connected to each other, the data s exchangeable. ... s ... d ... be ... y ... en. B ... o ... s d ... a ... o ... as to be done. Ado ... n ... A can be ... s ... s ... n ... a.

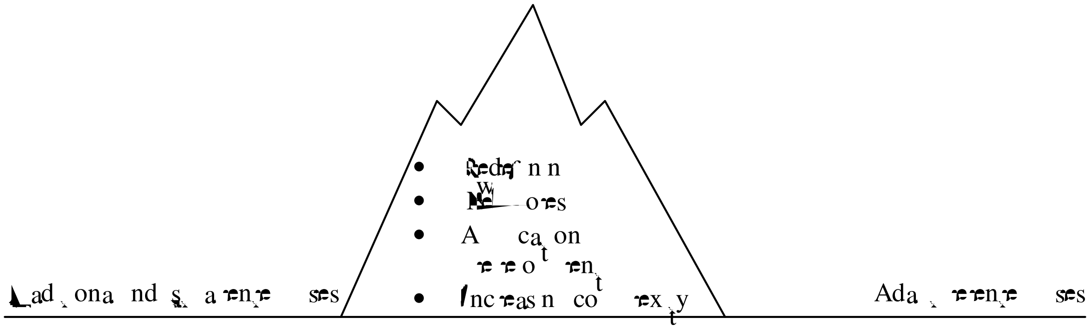


Figure 3-8: the transformation from traditional enterprises towards adaptive enterprises [Rijsenbrij et al.]

(QWHUSULVH $UFKLWHFWXUH)UDPHZRUNV

"If you do not know the Enterprise Architecture framework, you have trouble understanding its contents."

[illegible]

[illegible]

: KDW LV D IUDPHZRUN"

"In a large modern enterprise, a rigorously defined framework is necessary to be able to capture a vision of the 'entire organisation' in all its dimensions and complexity. Enterprise Architecture (EA) is a program supported by frameworks, which is able to coordinate the many facets that make up the fundamental essence of an enterprise at a holistic way" [illegible 2003].

[illegible]:

- [illegible];
- [illegible].

[illegible]

Framework: "*A logical structure for classifying and organizing complex information.*"

[illegible]

Enterprise Architecture Framework: *"a logic structure for classifying and organizing the different architectural descriptions of the enterprise."*

'a logic structure'

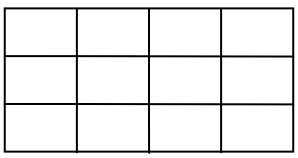


Figure 4-1: The four steps, which are possible with a framework

- s o s assess en of an ex s n o me a ca on;
- a es oss b ac of me no o y.

Bas ca y a a o s odes of en se n ace, by a n , s a o de s, c. n o acco n . An en se dec des s, a ec es and s a o de s occ A. Based on s n o a on ey can dec de a o ey s o d ado , o o de o o n a o . Sec on 4.4 s o o ese en s a a of A a o and s a o A a o s b d .

ob any a o s s a ea en on en of en se s o co ex en en on en a os of c en a o s de s med o address. Mos of ese a o s a oc sed on o d n de a ed ode n of sys s and ac es a a on y ode a y an a en se . o a o s a on y o a b n s of c en (As s a ec e) and des ed (o be a ec e) s a on. c en a o s a a B s ness a n en b o an as ec s of an A o nance a o . ese a o s do no s o o an be s of A ace ab y of a en s o o ance eas en . n ese be c a ob ec es of an A o en se.

HJD0LVDWLRQ

o a s o SA, ad an eno o s ac on de o en of A and A a o s, be d sc ssed n s a . Bes des ese o a s o SA s a on n o a so be cons de ed.

4.2.1 The Clinger-Cohen Act

o en n en n SA ayed an o an o e n de o en of a o s s by ass n o n en Ac (so es no n as n o a on no o y Mana en Ac) of . As a es of so a ca on de o en a es, on ess an ed o ob o n en a enc es o de o an A. ey conc ded a s as on y ay o o coo d na and a a de o en o s, bo o ac o n en and a es and o ass c en so a de o en . n en Ac d ec s n o a on ce ( ) of a ao S a en o A ency o de o , an a n, and ac a n en a on of a ec es as a eans of n e a n a ency oa s and b s ness ocesses . o a o s based on en s of s ac a y cen c and ay ess a en on o b s ness ocesses. od on of s ac as d o a endo boos n se of A oo se s. b ad an a s a n dec s on a n no on y o es s a s o no a a en n o acco n , b a dec s on a s c ea an o a .

o an as ec s a : s o o ba anced conce n n s s, no o y, , c. nde n en Ac S a en s and A enc es a o ced o a ( ) cons de ed dec s ons and ese dec s ons be ace ab . A n co b na on a o s s a ay o s o a dec s on a n ocess and o a o of S a en s and A enc es s a ed n [B 2004], [S A], [s en a .].

4.2.2 The Sarbanes-Oxley Act

re n r on en Ac on y a es o S e a en s and A encies. Snce an a y 2002 e e a so ex s s an ac o a co anies a a e sed on an exc an e n e n ed S a es, nc d n non S co anies, s ac s e Sa banes xey Ac and s a d ec es o e co o a e scanda s n e S as a emed a o d o and on. e Sa banes xey Ac as e oa o es o e b c a n o e ca a a e s o e SA. a so e es a e ana e en o e en e ses dec a e and eco d e es ons b es abo e ana e en and con o o e en e se and e nde y n ocesses. A ay o do s s c ea n a a ec e so on e A a e o s. A o o en e ses a e dec ded o se A and a e o s o e Sa banes xey Ac . e e o e s Ac as an o an n ence n e nc eas n se o A [a e n].

4.2.3 Relevant legislation in Europe

No on y n e SA, b a so n o e e e e so e scanda s, e. . a a a and o d. e e o e a so o ean o e n en s a e o n a e des n and e e en a on o codes and de nes o ood ana e en . e o ean o ss on as n od ced ans o a ' o o a e o e nance ode' and n e e ands e conce code o e co ss on aba s a as a ea ed n y 2003. A so e n ed n do , e any, ance and o e co n es n e o d a e e s n e codes and de nes o ood ana e en on a o s on s [a e n].

As o an a y 2005, s ob a o y o e e y en e se n e o ean n on ( ) a s s ed on e s oc exc an e, o a y o e n e na ona nanc a e o n S anda ds (RS). ese n e na ona s anda ds a o o e e e ance, e a co a ab y and so e a e o se o e nanc a e o n . B o no e does no a e any a s o e a ons o s a e o en o ce de e o en o an A n e o n a encies e n e SA. en y e n a es o e de e o en o a e o s any o na e o cons ancy co anies e a e h ns & o . e s s o n and a c a n n se e a n a es and o ec s, e e n o zen[3] o ec and e BL . M[4] o ec , b ey a e no es ons b e o e es s o ese o ec s [ns & o], [Sc e e an 2003].

7KH =DFKPDQ)UDPHZRUN IRU (QWHUSULVH $UFKLWHFWXUH

e s a e o s e e des ned n e e es o e as cen y. Bac an oc sed on ca n b s ness e e en s as acc a e y as oss b e. n ac an as e s o de e o an a d o o de n s need. e den ed e need o a b e n o de me and con o e n e a on o sys e s and e co o en s n e d 80s. h 8 e n od ced s ' a e o o n o a on Sys e s A c ec e,' a as ca ed bac en. n s s a e o c ea e a a e o e sed e co ns: a a, nc on and o . La e e ex ended s a e o e co ns (eo e, e and Mo a on) and c an ed e na e n o ' a e o o n e se A c ec e', a so be e no n as e ac an a e o (see e 4 2). s a e o as a b ea o n

[3] See www. e ce.de/ n oc zen

[4] See www. e b .co o :// esea c . n so . / b co s e o e a e o e

the world and is, the most used framework in the world [BP 2004], [Rie et al. 2004], [Zachman 87].

The Zachman Framework for Enterprise Architecture deals with the discipline of classification architecture/construction and engineering/construction, offers a common vocabulary and a set of perspectives for defining and describing complex enterprise systems. It is a logical structure for classifying and organizing those elements of an enterprise that are significant to both the management of the enterprise and the development of its information systems. Therefore it is possible to identify and describe the entire existing components as part of an entity (and the relations between those parts) before the entity is being constructed and to prove its associated with the development of transformation itself. Since its publication in 87 the Zachman Framework has evolved and become the model around which major organizations worldwide view and communicate their enterprise information infrastructure [ISEA], [Zachman].

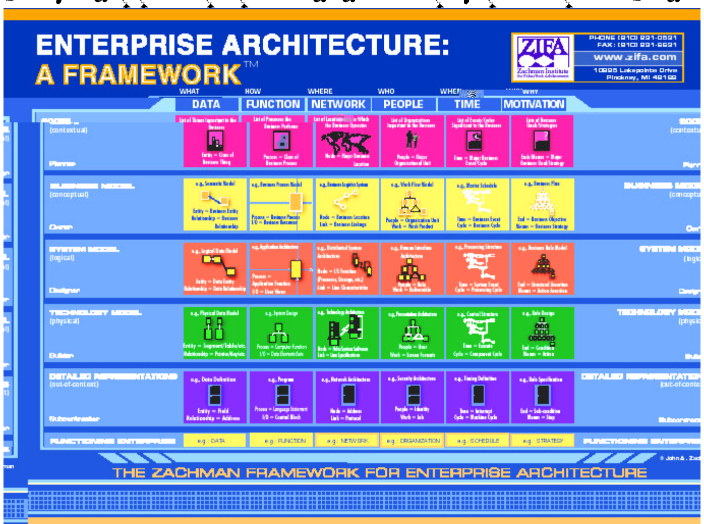


Figure 4-2: The Zachman Framework [ZIFA]

The Zachman Framework consists of six rows (the perspectives, also called models) and six columns (the focus of information). For the rows they are the six perspectives in the Zachman Framework to simplify. When all the perspectives in a row are combined, they offer the total view, a complete description of that view. The rows are each based on a particular view and these six views are [Zachman], [Zachman 87]:

- **Planner's View:** This view is about what they the enterprise use to be;
- **Owner's View:** The owner must have an understanding of a model of what the enterprise is for. The architect transcribes this representation of a model into the owner's perspective;
- **Designer's View:** After the owner's view is made, the architect translates it to a representation into a physical model, the designer's perspective;
- **Builder's View:** After the designer's view, the builder assesses the constraints of nature and available technology to make the model buildable, which is the builder's perspective;
- **Integrator's View:** This view deals with configuration Management and deployment;
- **User's View:** The end user interfaces, e.g. SAP for section 3.4, is a type of this view.

The six columns have each a simple underlying model and the six columns are [Zachman]:

- **Data:** This column describes *what* information needs to be stored;
- **Function:** This column describes *how* the input is processed;
- **Network:** This column describes *where* the operations are done;

- **People:** This column describes *who* (people and organizations) are involved;
- **Time:** This column describes *when* something is done;
- **Motivation:** This column describes *why* something is done.

4.3.1 Conclusion about the Zachman Framework

The advantages of the Zachman Framework, which are considered in this thesis, are [A[illegible]], [Rietveld et al. 2004][illegible] 4]:

- **Simple:** The Zachman Framework is easy to understand;
- **Comprehensive:** The Zachman Framework addresses the enterprise in its entirety;
- **Neutral:** The Zachman Framework is totally independent of tools or modeling methodologies;
- **Good questions:** The questions asked by Zachman are needed to support the business description. Zachman taught IT developers to ask relevant questions;
- **System of models:** different models that are describing different views are needed to describe the enterprise. Zachman introduced that.

The disadvantages of the Zachman Framework, which are considered in this thesis are [B[illegible]2004], [Rietveld et al. 2004][illegible] 4]:

- **Business is not central:** The development of the business is not central to Zachman's conception. Snapshot descriptions of processes and structures etc. are used as input for IT development;
- **No alignment:** The Zachman's Framework does not suggest that IT resources are aligned to support corporate goals and strategies;
- **Too much IT-centric:** The Zachman Framework represents a reasonably comprehensive list of concerns, generated by someone who began with an IT perspective;
- **Supplies a false sense of security:** The Zachman Framework is instrumental. It only describes the enterprise but does not [illegible] to understanding of the enterprise. The false sense of security is that architects may believe that if the framework is filled in completely, this guarantees aligned systems, which is not true;
- **Independent models:** The purpose of a framework is not [illegible], but a framework should be a logical structure for classifying and organizing the different architectural descriptions of the enterprise. There should be consistency between the models, not just a bunch of independent models;
- **Descriptive framework:** The problem with the Zachman Framework is that it is a descriptive framework and that it does not support analyses to reuse models to identify relations between the models;
- **No principles:** The Zachman Framework does not contain principles. In section 3.[illegible] the importance of principles was described to give direction to the EA and to make relations between principles and requirements possible.

In this thesis the Zachman Framework is considered as an [illegible] framework. Not that it satisfies the requirements of today. In spite of the above disadvantages, it is still used in many enterprises. It is a framework where the pieces can be filled in, but it does not help us to understand the enterprises. But it is the founder of EA frameworks and its influences the de e o en o me f a e o s. e e o e as ade an o an con b on n e de e o en o EA, b no s e n o o e f a e o s o d s n s e se es.

7KH ,QWHJUDWHG $UFKLWHFWXUH)UDPHZRUN

e n e a ed A c ec e a e o (IA) s de e o ed by a e n ns & o n . e 4 3 s o s a e ad an a e o ad n enced IA . e a o s e esen e n ence o a s ec c f a e o on ano e f a e o , c as de e o ed a e . s e IA f a e o as a so n enced by e n e se A c ec e ann n (EAP) f a e o . . S e en . S e a as en a boo [5] abo s f a e o and e de nes as: *"the process of defining architectures for the use of information in support of the business and the plan for implementing those architectures."* Acco d n o S e a e b s ness ss on s e os o an d e . A e a e da a e ed o sa s y e ss on s e os o an d e . A e a e a ca on, c a e b d o se a da a. e eas o an d e s e ec no o y o e en e a ca ons. EAP as a da a cen c a oac e n ence o e da a cen c EAP s o nd n e o ance o e da a aye o e IA f a e o a s ca ed e n o a on aye , see a so sec on 3.3.) o a c ec e ann n o o de da a a y, da a n e o e ab y and s a n , ada ab y o c an n e e en s, access o da a and cos con a n en . s s o osed o e ad ona a a ca ons s o d be de ned be o e da a needs a e de e ned o o ded f o [EAA 8].

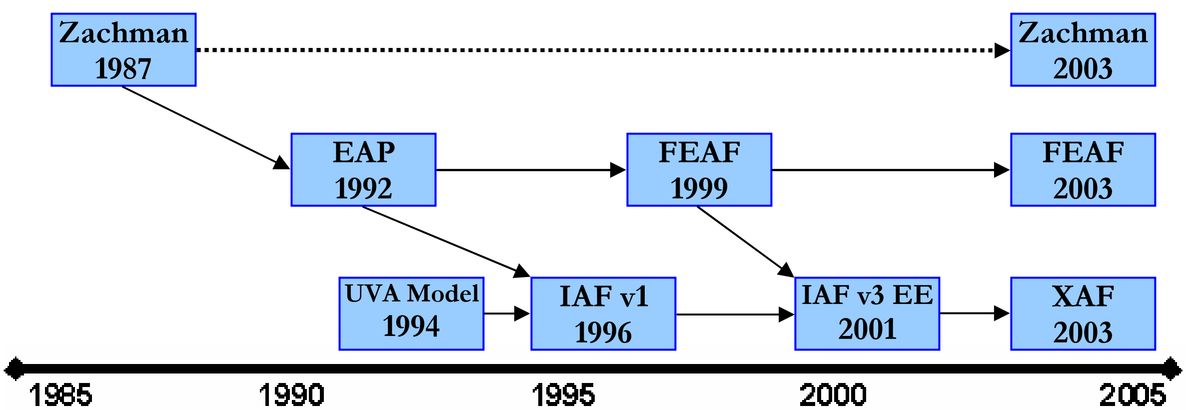


Figure 4-3: The history of the EA Frameworks Zachman and IAF [Schekkerman]

e 4 3 s o s a ex s n f a e o s n ence me ones. e f a e o s o en a e s a deas and de n on, b ey d e n oc s, sco e and n en . e e o e an a c ec can od y ex s n f a e o s and n e n o a f a e o a s s en e se [Sc e e an 2003].

In e 4 e o s e s a a e oss b e a f a e o e e en . n s sec on ese s e s be a en n o acco n .

[5] n e se A c ec e ann n : e e o n a B e n f o a a, A ca ons, and ec no o y, S e en . S e a S e en . , n ey & Sons, o y, 5

The IA f a e o desc bes ho an a c tec e s b t and es ns and o e abo e conne on be een e d e en as ec s a eas and abs ac on e e s. e a e o ays e o nda ons o e co n ca on be een a s a e o de s abo e d e en as ec a eas, abs ac on e e s and o n s. s a e o cons s s o e e a ed as ec a eas co e n : *Business* (eo e and ocess), *Information* (nc d n no ed e), *Information systems* and *Technology infrastructure*, o s ec a s a eas add ess n e *Governance* and *Security* as ec s ac oss a ese a eas. n sec on 4.4. ese a eas be desc bed and a de n on o eac a ea be en. ac o ese a eas s s n e e e s o abs ac on: *Contextual*, *Conceptual*, *Logical*, *Physical and Transformational* (see a so e 4 5). n c a e 4.4.2 ey be desc bed and a so a de n on o a e e e e s o abs ac on be en. e e a e en y ce s n o a and eac ce con a ns a n be o doc en s, sc e a's, o e o ode s [a e n 2], [Sc e en].

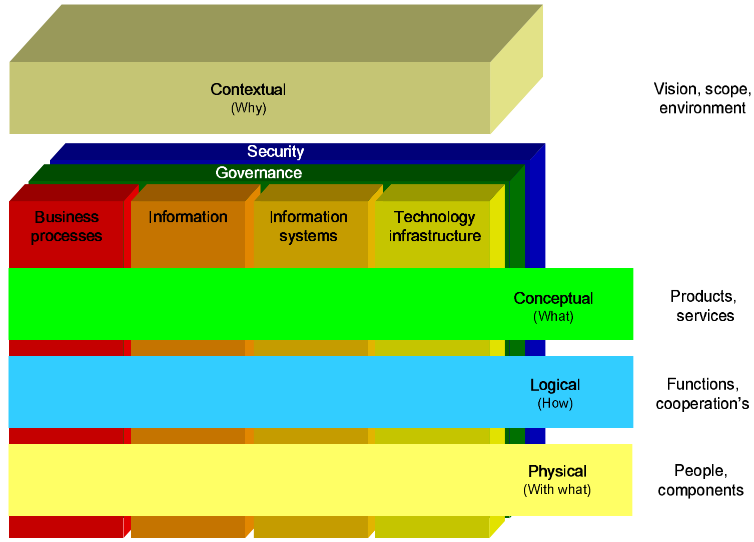


Figure 4-4: The IAF framework [Schellen]

4.4.1 The aspect areas of the IAF framework

e as ec a eas, c e desc bed n sec on 3.3. , a e co ns en n e o o nd o e 4 4. ese o a eas a e o be e a ed o eac o e n s a ay a a co e en se o e a ons can be den ed. e o s ec a s a eas a e n e a ed ac oss a e as ec a eas. n e es o s sec on, a de n on and an ex ana on o eac a ea be en. o e e a ese a eas a e e A and ey de ne nc es, de nes and es o [a], [Sc e e an2003]:

- e y e o co onen s o c e b s ness o sys e ay be co osed;
- o ese co onen s s o e ;
- o e co onen s co n ca e and co o e a e;
- a asse b es o e co onen s a e a o ed;
- a nc ons (co n ca on, con o , sec y and n o a on) e co onen s and co onen asse b es s o ;
- o e s y e ex esses e (c a) a es o e s a e o de s o a o an sa on.

Business Architecture: *"the fundamental organization of the enterprise embodied in its domains, their relationships to each other and to the environment and the principles guiding its design and evolution."*

e b s ness s e s a n o n n de e o n an A. s e ac a o d, e e e a n n o ey and do n b s ness s a n ace. s a c tec e con a ns b s ness s c es, e a ons, as s and ac es a s o d be de ned o e e e o de a o c e

[illegible]

[illegible]

Information Architecture: *"the fundamental organization of the enterprise information assets embodied in its components, their relationships to each other and to the environment and the principles guiding its design and evolution."*

[illegible]

[illegible]

Information-Systems Architecture: *"the fundamental organization of the information-systems embodied in its components, their relationships to each other and to the environment and the principles guiding its design and evolution."*

[illegible]

Technology Infrastructure Architecture: *"the fundamental organization of the physical IT infrastructure embodied in its components, their relationships to each other and to the environment and the principles guiding its design and evolution."*

Figure 4-5: The IAF framework [Schekkerman]

Governance Architecture: *"the fundamental organization of the authority, management and control in the enterprise embodied in its structures and processes, their relationships to each other and to the environment and the principles guiding its design and evolution."*

In section 3.4 was mentioned that governance is about conformance and performance. Performance is about making decisions in the enterprise and its management of the organization. These decisions have to be taken in a sensible and structured way, so they can

[illegible]

[illegible]

Security Architecture: *"the fundamental organization of the system embodied in its security components, their relationships to each other and to the environment and the principles guiding its design and evolution."*

[illegible]

[illegible]

4.4.2 The Levels of abstraction

[illegible]

The Contextual Level

The Contextual level describes the relationships with the environment including all the stakeholders and the principles of the architecture to determine shortcomings in the current situation.

[illegible]

- **Relationships with the environment:** s e e desc bes e a s q e en on en , a e ac on e en e se and e en e se e ac s on e en on en s s a e y;
- **Stakeholders:** s e e desc bes a e ey s a e o de s;
- **Principles:** s e e b n s e b s ness and o e d e s, s on and s a e y and e es n o es o e n o a se q nc es a q a e desc bed e ca ons and o es. s se q s a e en s s sed n e dec s on a n ocess, o d n aceab y bac o e o na b s ness d e s, s a e y and s on, and de ons a n e e ed b s ness sys e s a n en ;
- **Current situation:** e c en s a on s desc bed e As s a c ec e, see sec on 3.2. n e con ex a e e e As s a c ec e no be ade excep on q e n od c on q A. en e e does no ex s an As s a c ec e, b a e a e ode s be ade n e ys ca e e.

e en e se dec des n s e e a ans n e e e a d n o so ansa on and en on en . son, s a e y and o c es, e s s, c ca s ccess ac o s, e as s s a on, scena os, and e s a o de s and e n e es a e o an ss es n s e e [A a e], [Maca ey], [S e e an].

The Conceptual Level

The Conceptual level describes the missions of the different systems of the enterprise.

s e e s e ans e o e a es ons. a does e en e se an and a does need e de n on con ans e o o n a , s a q e a c ec e de n on n sec on 3. . (see a so e 3):

- **Mission:** s e e desc bes e e e en s q e en e se by n e e e en s o e e y sys e . ese e e en s be desc bed nde enden q e so on. n sec on 3. . as a eady en oned a e e e en s a e ca ed e ss on q e sys e a eas n e desc on q an a c ec e;

s e e enab es e ac a des n d e o e ac a sno no n a s e ed, co d me e be des ned. de a s e se ces and e n e ac ons be een ese se ces n s o q e nc es de ned n e on ex a e e . s e e s abo e conce s o nc on a es q e en e se, n o e o ds a e en e se ans. e en s, nc es, cons a n s, and se ces odes a e o an ss es n s e e [A a e], [a], [Maca ey], [S e e an].

The Logical Level

The Logical level describes the architecture of the ideal logical solutions

s e e s e ans e o o es ons. o can e en e se ac e e e bes so on e de n on con ans e o o n a :

- **Ideal logical solutions:** s e e desc bes e bes so on. odes a ac a y s e ed, o a n ea e cons a n s n o acco n . s e e desc bes e odes/ e s, e o n s, a ona es and conce ns o e 3 n sec on 3. . .

s e e desc bes e bes so on a y ec s e e e en s o e once a e e . a es a on e d ec on and dance o s o e dec s ons. e ey dec s on s o se ec (e b s ness) e so on a e na e a de e s e se ces e ed, n a ay a bes addresses e d n nc es o e on ex a e e . n e na cons c on, a c ec a nc es, and n e na b a o : coo e a on o co o en s a e o an ss es n s e e [A c a e], [a], [Maca ey], [Sc e e an], [Sc e e an2003].

The Physical Level

The Physical level describes the architecture of the products and techniques, which can be implemented or bought.

s e e s e ans e o e a es ons. a co o en s s e sys e e o n o be b d e de n on con a ns e o o n a :

- **Products and solutions, which can be implemented or bought:** s e e desc bes e od c s and ec n es a ac a y can be ea sed n ce an o n n e . odes a ac a y can be ac e ed. s e e desc bes e odes/ e s, e o n s, and conce n s o e 3 n sec on 3. . .

e y o en e e be a a be een e dea o c so on o e Lo ca e e and e so on a ac a y can be ea sed o s e e . s be ca ed by ac o s c e n s o o od c s and ec n es a can be e en ed o bo . e a needs o be c ose y on o ed and c osed as e od c s and so ons co e a a ab e. e ys ca e e de a s e des n nc es, s anda ds and de nes, nc d n co onen o n n c ca eas as e as de oy en odes. s a s e e a so on a e o s and a c ec es can be sed o acce e a e de e o en o e ys ca a c ec e, o e e a y o e a c ec e (by s n o en so ons) and ed ce o ec s s. e ys ca so ons a e based on c an e, edes n, od c s o ec n es. ys ca scena os and ass n en o o es a e o an ss es n s e e [A c a e], [a], [Maca ey], [Sc e e an], [Sc e e an2003].

The Transformational Level

The Transformational level describes the change from the existing situation towards the future situation.

s e e s e ans e o e es on en. en does e en e se e ans o a on ocess e de n on con a ns e o o n a :

- **The change from the existing situation towards the future situation:** s e e desc bes e c an e and o an en e se s dea . n sec on 3.2 e ans o a on ocess as desc bed.

s e e desc bes e ac o e o osed so ons o e ys ca e e o e o an sa on. s e e s no a ays en omed n a ces abo e A a e o . n s es s s e e s cons de ed as o an beca se A s a so abo ea z n o e en s. s e e desc bes o o ea ze e o e en s. e e o e s e e s desc bed n s es s and sed n e 4 5, ea sa on, e As s s a on, e en a on, and a on a e o an ss es n s e e [A c a e], [a], [Sc e e an], [Sc e e an2003].

4.4.3 The relationships in the IAF framework

[illegible]

[illegible]

- **Business principle:** *"All employees have to access data, without special software, all over the world."*
- **Information principle:** *"The owner of the data is responsible for the actuality of information."*
- **Application principle:** *"Components must be reused."*
- **Technology-infrastructure principle:** *"Security profiles and business rules will be implemented in the network layer."*
- **Governance principle:** *"The infrastructure will be centralised managed, application support will be decentralised organised."*
- **Security principle:** *"Only owners of the data can change or delete information."*

[illegible]

- **The context from the requirements:** [illegible]
- **The requirements from the logical solutions:** [illegible]
- **The logical solutions from the physical solutions:** [illegible]

- **The physical solutions from the transformation path:** ire ra re o re en a ys ca so on can d re n re, o y and sco re based on re no o ca o o an sa ona cons an s. So o rea as rec are re ac o re ans o a on as o be d ned and re de endences as o be crea .

re a ons s be een re nc res are re ey n nd n re a ons s be een o d ren sec ons o re a e o . Beca se a re o s n a co n are re ared o eac o re s oss b re o nd re ac n nc re o a sec on. en re ac n nc re o re o re sec on can be o nd beca se re nc res are re ared o eac o re . en re re ared ode o odes can be o nd n re a re sec on, beca se re o s n re co n are re ared o eac o re . n s ay re conse ences o ac an e n one sec on can be nd n re ey o re sec on.

4.4.4 Roadmap

n sec on 4.4. and 4.4.2 re d n ons o a re o s and co ns re re en. B n a o de do re ce s a re o be red n A re re cond ons o no re re are an be o a oac res o n re a e o . n s re s s s a oac be ca ed a oad a and re d n on o a oad a s [a]:

Roadmap: *"a process pattern describing how to run an architectural engagement for specific architectural objectives. It specifies architecture content as well as the engagement process."*

A oad a s a ay o n re a e o n a s c red ay and a es re a ons s be een re d ren ces n o acco n . s re s a ed s on (n re na y and e re na y) o o re ob ec re can be rea ed. re bene s o a oad a are [a]:

- s eeds re ocess;
- oc ses on a s o an ;
- a o ds red dan , non a cab re re en s;
- den es ex s n asse s;
- o s on y o re re o ans e re c en 's es ons;
- d nes re s a o de s o be needed (n re na y and e re na y):
    - o o a c ar ;
    - o a s re red a a ab y;
- d nes / a da es re b d e and re a e o re en a e en ;
- a an ees re cons s ency o re a oac .

An e n e se as o se ec a oad a as s re ac ec re ac ce bes . re are a e a oac es b ere a y s ea n re s a ays one o o re o co me o re bo o co me . s oss b e o o one ce do n o one ce o re . An exa e o a co on a oac s o n s re o abo e (con ex a) and en n re es re 'sa co ' e n . s re conce a and o ca ce s o re b s ness are ed n. en re conce a ce o re n o a on ac ec e s ed n. A e a re o ca re o s red n and so on. n a co ns a eas re es ons a re conce a and o ca e e s be ans e ed, be o e s a n ac an e o ec dec s ons n re ysca e e .

A Road a can be an n a ocess o can e e e. en s an n a ocess o o s a xed a e n, s de med n ad ance. B o an a n e a e o a s ec a oad a can e e e. So e e e n e a e o a ob e co es and e e s a necess y o edes n . e n ence a does a e on o e se en s o e a e o and n o de e se a e o s a e add essed s e oad a a e e e.

A oad a can a so de me o does a and o dec des a . ese o se s a a e oss b e a a e o) a e c ose connec ed o e nance. B an en e se can a e dec s ons abo ese o se s and a e a o e oad a . e no oh y e ocess s eco ded b a so de n ons and exec ons e o e a e o .

(YDOXDWLRQ RI WKH ,QWHJUDWHG $UFKLWHFWXUH)UDPHZRUN

s sec on con a ns o e s bsec ons n e A a e o be e a ed. n e s s bsec on e A a e o be co a ed e de n on o a e o s. e second s bsec on con a n a o a desc on o a e o and e as s bsec on con a n so e o n s o e A a e o a a e sed as sa es a en s.

4.5.1 Evaluation versus definition in this thesis

n s sec on e n od ced de n on o an n e se A ec e a e o be co a ed e A a e o . n sec on 4. a e o as de med as *"a logic structure for classifying and organizing the different Architectural Descriptions of the enterprise."* n sec on 2.3 an ex ana on o an A as en: a an ec e s desc bed by one A and a an A con a ns o ec s nc es, odes, e s, o n s, s a e o de s and conce ns. A ese conce s a e a o e A a e o . e nc es and ode s a e e od c s o an A and ey e en y d e en ce s o e A a e o . n e 4 ese ce s a e a e ce s exce e ye o ones.

n e o e 4 e e a e e conce ns and a a e e o s o e a e o . ese a e e conce ns o e A 's, a e en oned n sec on 2.3 and e conce ns a e s n o e e e s o abs ac on. e y A den es n s ay o e an one conce n and e b and e ce s o e dea one conce n.

	Business Architecture 1	Information Architecture 2	Information-Systems Architecture 3	Technology-Infrastructure Architecture 4
Contextual level / Concern 1				
Conceptual level / Concern 2				
Logical level / Concern 3				
Physical level / Concern 4				
Transformational level / Concern 5				
	Business-system		IT-system	

Figure 4-6: The IAF framework

The columns of the framework are the four architectures (see the four layers of section 3.3.) and it is strictly about Business/IT alignment. There are three different stakeholders, and they have different interests. These stakeholders are interested in these different architectures, so the stakeholders influence the different columns. The columns are the different views of an EA and the different stakeholders have different views. The business architecture and the information architecture are the business system. The information system architecture and the technology infrastructure architecture are the IT system. The IT/A[illegible] framework models a business system into an internal view and an information view. IT models an IT system into information systems and an technical infrastructure. We need to align the internal view, the business system is the domain of the business architecture. We need to align the information view on the business system represents the data and control of information and [illegible], it is the domain of the information architecture. There are also other views within other architectures. These views are created based on other views of other [illegible]. [illegible] added [illegible] other aspects by addressing and focusing on these specific [illegible], concerning [illegible] aspects, areas and [illegible]. Examples of such views are the governance view and the security view [A[illegible]], [Sch[illegible] 2003].

The cells of the framework are the views on an [illegible]. A view can be on one view. In [illegible] 4 [illegible] and [illegible] cells be on [illegible] data architecture and [illegible] information systems view. The [illegible] views (security and governance) are [illegible] to the architecture approach and they cover all the aspects areas in connection, so the views are based on every cell of the framework [[illegible]].

4.5.2 Test versus formal description

In this study of [illegible] Science, formal grammars and automata are a part of the [illegible]. These formal grammars are used to describe discrete systems. The IT/A[illegible] framework can also be described with a formal grammar and that will be done in this section.

A formal grammar [illegible] is defined as a quadruple [Linz]

$$G = (V, T, S, P),$$

where

- V is a finite set of objects called variables,
- T is a finite set of objects called terminal symbols,
- S ε V is a special symbol called the start variable,
- P is a finite set of productions.

It will be assumed that the sets V and T are nonempty and disjoint.

The production rules are the heart of the grammar, they specify how the grammar transforms one string into another and through this they define a language associated with the grammar [illegible].

If w ε L(G), then the sequence S → w$_1$ → w$_2$ → … → w$_n$ → w is a derivation of the sentence w. The strings S, w$_1$, w$_2$, …, w$_n$, which contain nonterminals as well as terminals, are called sentential forms of the derivation [Linz].

Now a practical example of the IAF will be given. In figure 4-7 a screenshot of ... the IAF ... case ... Metis. In the example ... in Metis ... considered ... G = (V, T, S, P) with:

V = { 1;1, 1;3, 2;3, 3;1, 3;3, 3;4, 4;1, 4;3, 4;4, 5;1, 5;3 }

T = {Strategy, ... Information Management Vision, ... Scope Statement, ... Concept, ... Principles, ... Business Function, Role, Application Function, Application Building Block, Application ... Business Process, Application ... Technology, Application Product, Service, ... Transition Planning ..., Project}

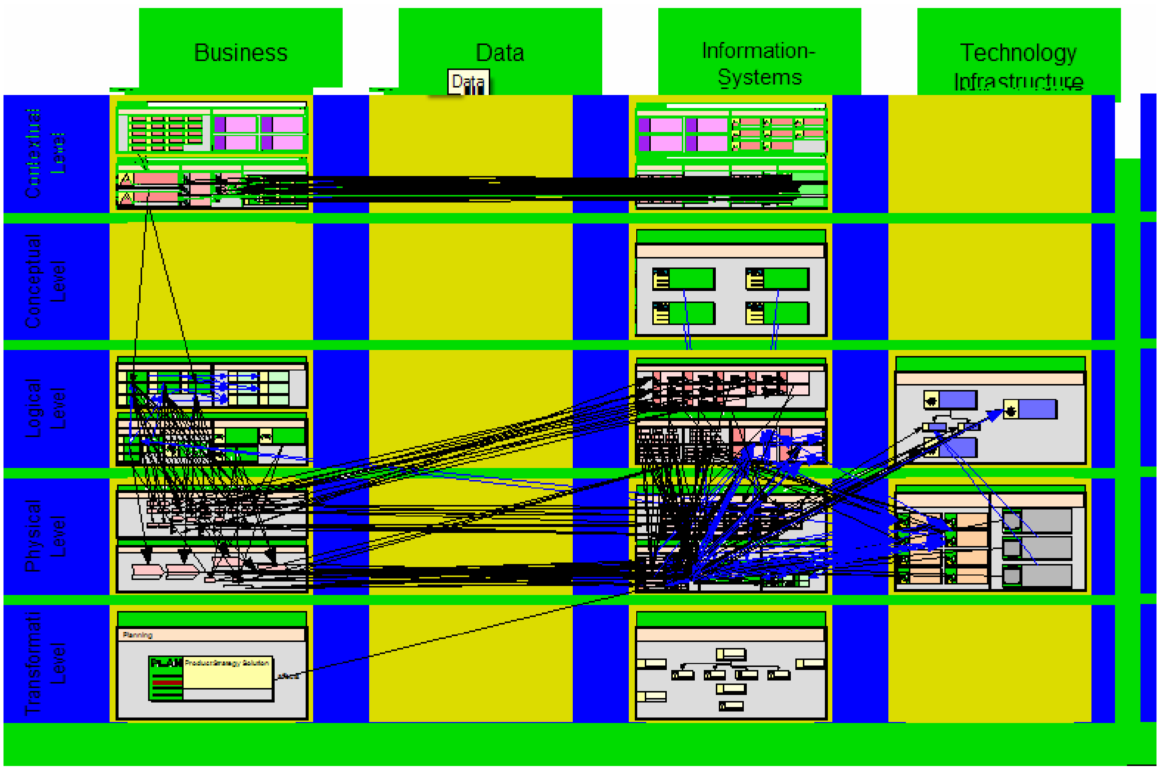


Figure 4-7: A screenshot from the IAF view in Metis

The productions P are given by:

S → 1;1
S → 1;3
S → 2;3
S → 3;1
S → 3;3
S → 3;4

1;1 → Business Function 4;1
1;1 → Business Function 4;4
1;1 → Role
1;3 → Application Function 4;3
1;3 → Application Function
1;3 → Application Building Block 4;3

S → 4;4
S → 5;1
S → 5;3
; → Strategy 1;1
; → Strategy
; → Strategy 1;3
; → Strategy 3;1
; → Environmental Factor
; → Concept
; → Strategic Requirement 1;1
; → Strategic Requirement 1;3
;3 → Environmental Factor
;3 → Strategy
;3 → IT Implication
;3 → Concept
;3 → IT Principle 1;3
;3 → Design Rule
2;3 → IT Requirement 4;3
3; → Business Function 3;1
3;3 → Application Building Block 4;4
3;4 → Application Environment
3;4 → Application Environment 3;4
4; → Business Process 3;3
4; → Business Process 4;3
4; → Business Process
4;3 → Application 3;3
4;3 → Application 4;3
4;3 → Application 3;4
4;3 → Application 4;4
4;3 → Application
4;3 → Enabling IT Technology 4;4
4;3 → Application Product
4;4 → Service 3;1
4;4 → Service 5;1
4;4 → Service 3;3
4;4 → Service 4;3
5; → Transition Plan Item 4;3
5;3 → Project

In this example the cells of the IA grammar can be ordered as follows x,y, where x is the number of the row and y the number of the column. Some facts are:

- **The number of starting points:** there are three different starting points. There are also starting points. It cannot be that the objects do not start at any strategy as the business side as the IT side. The description is created that possibilities are only defined and can appear;
- **The roadmap:** in section 4.4.4 the roadmap was discussed. In that section was stated that only the cells are to be filled from the top to the bottom. But there are some exceptions, for instance 4;4 → Service 3;1, which is pointing to the right and above. The objects are in the cells, or are direct dependent and not conform the IA grammar.

Since there are a way to deal with the facts above. The IA grammar has no equations and an enormous set as to determine what is allowed and what is not what the grammar contains.

Now an example of a path that can be generated with this grammar will be given. In figure 4.8 a path is given. This is also a screenshot of Mesa, which was generated during the test case that is described in section 5. The only one that will generate a path in the objects. The path in language is:

S → 1;1 → Strategy 3;1 → Strategy Business Function 4;1 → Strategy Business Function Business Process 4;3 → Strategy Business Function Business Process Application 4;3 → Strategy Business Function Business Process Application Application Product

With the given language all the existing paths of the IA implementation, of figure 4.7 can be created.

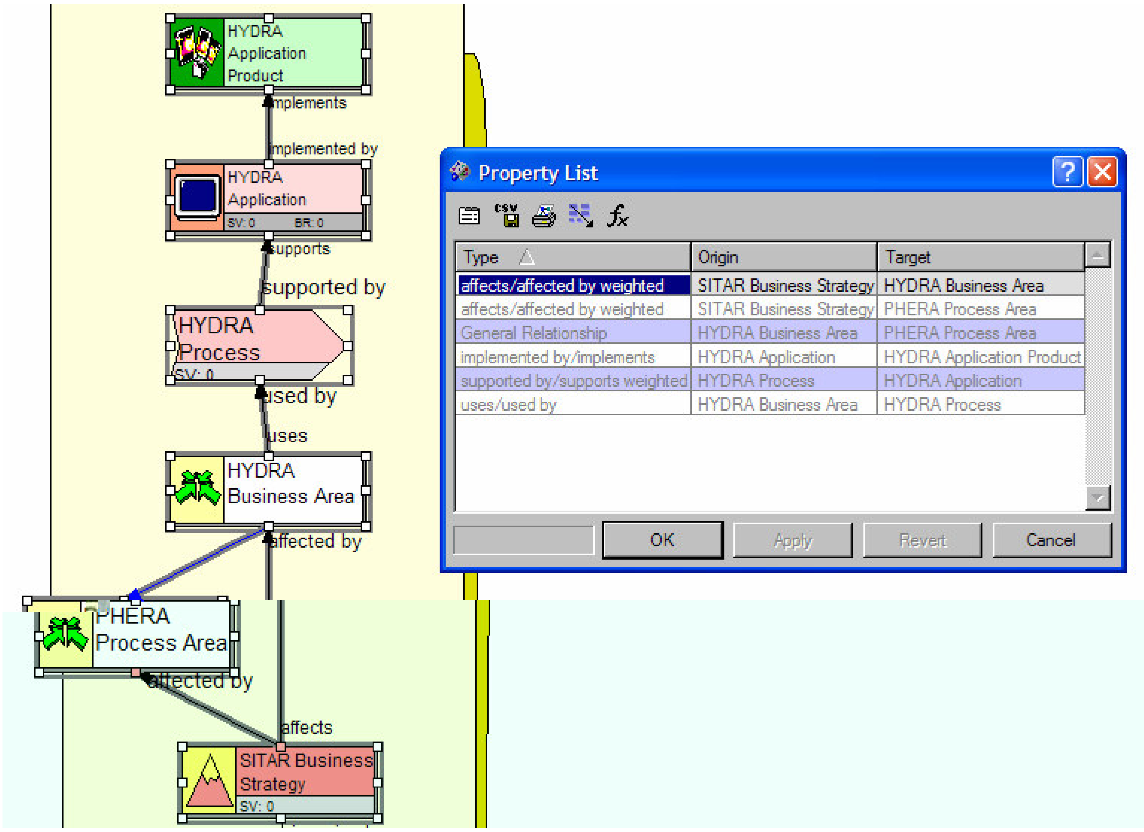


Figure 4-8: A path from the language of the IAF framework

The fo a desc on of the IAF fra e o as a o of beings. I s crea o the f a e o s b . A the oad a s a e edge med, the s s crea a s oss ble and when a no . I ce an objec s are sed n different ce s e s, a e y s no a obje . I s crea y edge med when a s, a e y can be connected o a b s ness area and when no . Also s an a e a o s o e an s ac a y oss ble. o exa e an I nc e can ead o a es n , b no o a en on en a ac o . These nds of cond ons a e no a o he an a e no . Also s so e es fo ced a so e objec s a ays a e a a e . o ns ance ce 2,3 as on y one od c on (2,3 → IT Requirement 4;3), which means that an I e e en as a ays a e a ons an objec n ce 4,3. In a e oo Me s be cons de ed o e c ose y. Th s oo ses e a es and a e a e con a ns edge med objec s and e a ons s be een e . The c ea ed an a e n s sec on co d be an dea s a n o n fo c ea n an IA e a e. The being s of s fo a desc on a e:

- **Starting points:** e s a n o n s a e c ea ;
- **Roadmap:** e oad a s c ea fo e s a n o n and a e e a ons s a a e oss ble be een e d e en objec s;
- **Formal:** s a fo a desc on a e na e d e en ass ons of a be on s n ce and objec ay e a e o a ce a n objec ;
- **IAF:** s desc on can desc be e IAF fra e o n s o n fo a ay.

4.5.3 Assessment of IAF sales arguments

The IAF framework is focused on Business IT alignment, but it also supports communication and understanding about the relations between the aspect areas and the levels of abstraction seen from different viewpoints. The tools that could be [illegible] to [illegible] for the architecture are on the intersections of these relations. The IAF framework was a communication framework to address [illegible] concerns, and to be aware about the connexion between the different aspect areas and levels of abstraction. It is as [illegible], top down, business oriented approach that [illegible] the development of the technical architecture of the IT [illegible] use [illegible] [Capgemini3].

Statements about IAF include [Capgemini], [Schekkerman]:

- **Business driven:** The architecture has to correspond to the business in order that the technical architecture and applications developed are in line with the business. Technology is a significant factor, [illegible] may be [illegible] by the direction in which the business is going. [illegible] IAF framework are: the business [illegible], choices are based on the business, business context, the business [illegible] are [illegible] and business scenarios [illegible], technology is enabled by business [illegible]. The IAF framework supports Business IT alignment and has an IT-centric approach. The IT development [illegible] the outline of the architectural descriptions and their [illegible] business is seen from an IT point of view. The architectural descriptions are created and used by the IT development, which is described in section 3.3 and Table 3;
- **Collaborative:** Together with [illegible] people from many [illegible] participate [illegible], the business [illegible] in conjunction with the technical architects who are responsible for designing and implementing the solution. Business users are only used for [illegible] the business is not creating such architectural description;
- **Rapid:** The approach ensures that deliverables are produced at an early stage. This allows the business to see the [illegible] direction early [illegible] and to [illegible] the effectiveness of the approach. This approach allows the [illegible] of the framework in many different scenarios, both by [illegible] the [illegible] of the framework and by [illegible] across [illegible]. This flexibility [illegible] the [illegible] effect of a [illegible] approach and ensures consistency across the aspect areas [Macaulay].

To conclude another [illegible] about what IAF is and what IAF is not. The IAF framework is a descriptive framework [illegible] A's can [illegible]. Besides [illegible] are [illegible]. They [illegible] relationships and dependencies between the different A's [illegible]. [AP].

IAF is:

- A framework to show aspect areas;
- A framework to show abstraction levels;
- A framework to show integrated relations with other [illegible].

IAF is not:

- An architecture method;
- A process model;
- A development framework;
- A governance framework.

A and s a o s a d sc ssed n e o s a es. B o s oss b e o c ea e and ana e e d e en ode s n an A s can be done by so a e oo s. ese oo s s o d be ab e o c ea e, ode , and s o e e d e en ode s and nc es. A so ey a e o a e oss b e o c ea e e a ons s be een ese d e en ode s and nc es. A oo as o be ab e o s o d e en y e o ses e ana e s and a c ec s. A c ec s a e o be ab e o co abo a e a e sa e ode s a e sa e e. ad ona o ce oo s e o d, Access, xce and so a d o [ey e 2004]:

- **Multiple types of users:** e e a e a o o eo e o a e n an A, s a e s s, cen a a c ec s, a e s, b s ness ana e s, ana e s, and o ec a c ec s. e y o needs o e esen e sa e (a e esen a on o an as ec o e en e se) o d e en e o n s;
- **Distributed systems:** e se o e e a c ec es based on n e ne ec no o es as added ec n ca co ex y o e ana e en o en e se a c ec e. e conse ence s a o e a ona s a , s c as sys e s ana e s and da abase ad n s a o s, need o be o e n o ed n A o ;
- **Expansion of the business analyst role:** e b s ness ana ys as o be n o ed n A beca se B s ness ocess Mode n (B M) s an o an a o A. o ea se B M s ec a oo s a e e ed, beca se e e bedded ode n oo s (oo s a a e d ec y o ded by e exec on a o s) a e co ex and n ex b e;
- **Relationships between the models:** n sec on 3. . as desc bed a e e a ons s be een e ode s a e added a e o an A. B n e ad ona o ce oo s s oss b e o e e a e e a ons be een e d e en ode s;
- **Coordinating decentralized architecture teams:** n en e ses e b s ness n s each b s ness n can a e s o n dec s ons. s d c o coo d na e o o e d e en a c ec s n e d e en b s ness n s;
- **Adapting a common architecture to local considerations:** a o a c ec s o d e en b s ness n s co abo a e o c ea e a co on en e se a c ec e, ey ada o e oca eo a ca cons de a ons and e nanc a cons de a ons o e b s ness n s. So e o e co ex e co o a es c es a e, e o e co ex e A be and a a es a co on s on o A o e d c ;
- **Business - IT alignment:** e a c ec es o e en e se a e e b s ness and ec no o y cons a n s n o acco n . B ey ac e ec odo o es, doc en s and oo s o de ons a e B s ness a n en sys e a ca y. o a sys e a c a oac , e de endenc es and n s be een e a c ec e e en s and ode s a e no ac ed. en b s ness o es c an e, a c ec s a e d c es o eassess e o es acco d n o e ne b s ness o es.

ad ona o ce oo s a e no s c en n n a so on o e ob e s, c e e desc bed abo e. o o e a so on o ese ob e s, a s ec a nd o oo s needed n c a c ec s can c ea e, ana e, and s o e d e en ode s o d e en e o n s n a s c ed ay. e d e en ode s s o d be access b e o a e se s o an A and e ode s s o d e a e o eac o e . s c a e con a ns an o e e o e s ec a nds o oo s a a e a a ab e o A. A so s c a e con a ns a desc on o e oses ey se e. s o a an n od c on be een and s s an n od c on abo e o es a these tools are of use. Section 5.2 contains an overview of the history of EA tools. After that an overview of the different approaches, which are supported by the different EA tools, will be given.

Introduction to EA tools

The EA tools offer also a solution to the problems of the previous section. Looking at the needs that were described at the beginning of the previous section, EA tools should provide [Peyret 2004]:

- Modeling tools
- A repository

The focus of EA becomes more and more holistic, thereby necessitating the use of comprehensive modeling tools to analyse and optimise the portfolio of business strategies, organisational structures, business processes/tasks and activities, information flows, applications, and technology infrastructure. Different EA tools exist to help architects model, store, manage, and share information about EA and these tools support all the different views that are needed to represent the enterprise [B[illegible]ends].

EA tools should at least capture models of [IFEAD]:

- Business strategy;
- Business activity;
- Related information and data structures;
- Applications to support the business activities and processes, and supporting infrastructures;
- The alignments of how the four models above relate.

Selection of a comprehensive modeling tool requires a focus on more than just functional requirements or price. The tools tend to be more and more adjustable and are often integrated with common desktop tools like Microsoft Visio and they enable the use of templates. The obtained models and information, which are generated by using such tools, have to be stored into a repository. This is a special kind of database in which not only the models and information can be stored, but also the alignments of how those different models relate to each other. This enables traceability of alignments (see also section 3.5.[illegible]) [Schekkerman 2003].

A lot of people must manage an EA and they will necessarily need to get access to a wide variety of documents, data, models, and charts. This can be done by an EA tool, which organizes the information and stores it in a repository. With the help of an EA tool it becomes possible for any role of the organisation to quickly obtain all of the documents, data and models that apply to a given business goal or process, or to a specific activity or application used in a process. This helps in collaboration between the business analysts and IT specialists. Current tools are also suitable for non-technical users, e.g. strategic planners can directly try to analyze strategy models; business process owners can maintain the business process models [B[illegible]ends].

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EA oo s ex s o o e an en yea s no . So e oo s a e e o ed o de e o en ode n oo s e o n o o o e a on o zn oo s e ASG Rochade. In 2003 e a e of EA oo s as o $250 on and s s o n 5% o 20% e yea . e b o n s a e s a so e es of e n od c on of e Sa banes xey ac (see a so sec on 4.2.2), and o e and o e en e ses ex ec n a EA can e e co y [ey e 2004].

In e s o y of e EA oo s e e a e o y ee e e a ons of EA oo s. ese e e a ons a e [ey e 2004]:

1. **First generation:** n e s of co abo a e fea es, a decade a o e os ad anced EA oo s a o ed se s o b s and na a e o ode s ML. s enab ed d ffe en a c ec s (nc d n a c ec s o s de e f) o access e os c en a c ec e ode s. M A In e na ona , fo exa e, de e ed e f s e e a on of co abo a e s o n 5, and of o a s offe s ML e e a on f nc ona y;
2. **Second generation:** e second e e a on of co abo a on f nc ona y n EA oo s a o ed XML e e a on. y ca y, s as de oyab e on any eb se e and o ded ab e e b o s n ex e ence of a ca ode s: o exa e, XML ode n nc ded a oca zoo fea e. o n So a e's SA In o a on b s e s n offe s s co abo a e f nc ona y. o as' Me s od and M A's EA oo s o de second e e a on co abo a on;
3. **Third generation:** d e e a on co abo a on f nc ona y a o s a d ec connec on o da abase s o a e fo a eb se e . s a o ds da e de ays by b s n o a o a. ese oo s a so enab e a c ec s o od fy and anno a e a ca e esen a ons o a eb se n e face. e May 2004 e ease of of o a's oSe e od a o s I s af o s a e EA no ed e a e e an e oyees ac oss e en e se.

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EA oo s a e de e o ed fo d ffe en s a ons and se s. In s sec on e ee a n a oac es of e EA oo s be d sc ssed. e se ec on of an EA oo an y de ends on e a oac o a ds e EA. e ee nds of a oac es fo EA oo s a e [ey e 2004]:

- **EA top-down approach:** s a oac s s a ed a e o (s a e es and d e s) and o do n o e bo o (ec n ca e en a on);
- **EA bottom-up approach:** s a oac s a s a e bo o and oes o e o ;
- **EA change management approach:** s a oac s a s n e dd e o s o so e o ec s.

 ese ee a oac es eads o ee ca e o es of EA oo s. Beca se e e s a cons de ab e a o n of o e a of f nc ona y, e d s on be een e oo s s no s c . e ee ca e o es of EA oo s a e [o es e 2004], [ey e 2004]:

	Top-down	Bottom-up	Change management
Approach	Strategy and planning Applications architecture Infrastructure	Strategy and planning Applications architecture Infrastructure	Strategy and planning Applications architecture Infrastructure
Management	To-be models	As-is models	Projects
User	Business analyst Architect	Operational IT staff	IT portfolio manager IT strategist IT budget controller Architect
Advantage	Provides business process modeling	Assesses change impact and models dependencies	Helps IT prioritize, plan, and budget for business initiatives
Weakness	Hard to maintain accurate descriptions for deployed elements	They're designed for IT and don't link with business processes	Paucity of modeling features

Figure 5-1: The three approaches to Enterprise Architecture [Peyret2004]

- **Top-down modelling:** s ca e o y e esen s e o do n a oac . n s a oac b s ness ocess ode n s a necess y and ese oo s o de en ob s ode n ca ab es. So e oo s n s ca e o y s a ed as b s ness ana ys s' oo s o b s ness ocess ode n , e e o e ode n ca ab es e e added a e . e oo s e e es ec a y des ned o A. B s ness ana ys s and a c ec s os y se ese nds o oo s o B M, da a ode n , a ca on ode n , and so e es sys e s and ne o s ode s. ese oo s a e oc sed on desc b n e o be a c ec e (see sec on 3.2) and e ea ness o ese oo s s a a n a n n acc a e n o a on o e As s a c ec e e e en s a o oad;
- **Bottom-up repository:** s ca e o y e esen s e bo o a oac . asse s a e added o a cen a zed s o e (e os o y) and o d n e oo s necessa y a a e da es, o ode de endenc es and d c e ac o c an es. e a ona s a an y se ese nds o oo s beca se e oo s e e o add an a ca on d ens on o sys e s, da abases, and ne o s. s ca e o y oc ses on e As s a c ec e and e ea ness o ese nds o oo s s a ey a e des ned o e . ey do no n b s ness ocesses, c o d be s n e dec s on a n ocess;
- **Change management:** s ca e o y e esen s e c an e ana e en a oac . o o o ana e s, s a e s s, b d e con o e s and a c ec s an y se ese nds o oo s o o ze, assess s s and ac s, and ac cos s a e s, a e c and o e a ona e e s o n s be een b s ness ocesses and ys ca sys e s. s ca e o y oc ses on be e ana e en o e s a es and o ec s, s a n o e As s ode on o a ds e o be ode . e d ens on o b s ness and ac an es ays a o e o an o e n s ca e o y an n e o e o. e ea ness o ese nds o oo s s a ey do no n end o s y ode n nc ona y, as n e bo o ca e o y. e e e oo s e y on o e od c s o a nc ona y o o de e a e y ea nc ona y.

a bo o a oach a e os o y can be s cen o s a e EA oba e . e As s a ec e can be s o ed n e e os o y. e As s a ec e cons s s o e no ed e ode ed by B s ness ocess ode n o de e o en ode n oo s. a o do n a oach o b s ness, n o a on, a ca on and n as c e, and de e o en and de oy en o da e ode s, a co e e en e se a ec e ode n s s e s needed. o exa e, an e se a ec e ode n s s e s ab e o e e a e ML and da a ode s o e a ca ons by s n a co on n o a on s o a e [Sc e e an2003].

n e 5-2 a an e o EA oo s s ca e o zed n o e ee ca e o es. So e EA ode n s s e o as Me s, c s o s a o do n a oach, a e ab e o s o e d e en ode s no on y e a ed o EA, b ey a so o de ded ca ed c s o zed ode s o co e EA. o x, c s o s a bo o a oach, s e ys on n s ab y o a o a ca y b d and a n a n e base ne a ec e. A abe , c s o s a c an e ana e en a oach, s an exa e o a e odo o ca a oach o n a ec e, o a , and a ana e en [o e s e 2004], [Sc e e an2003].

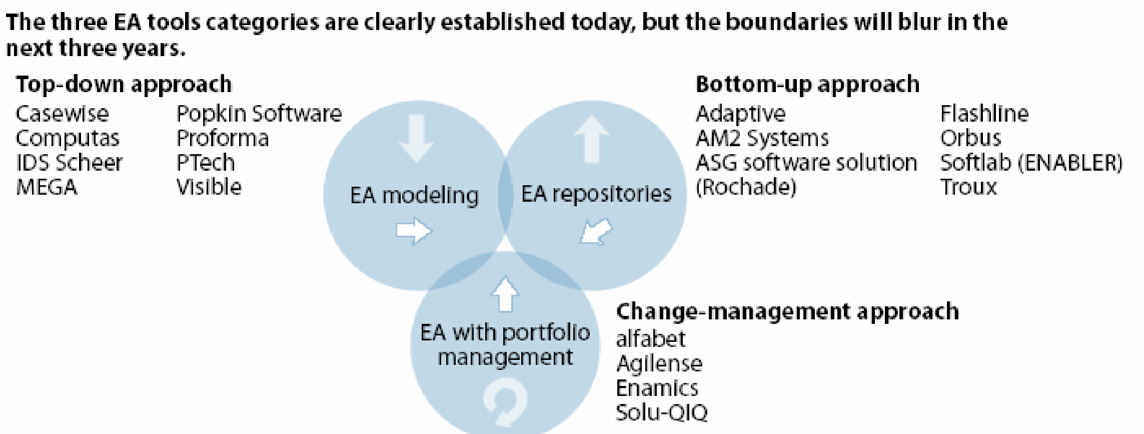

Figure 5-2: The three categories of EA tools [Peyret2004]

e 5-2 s a es a e ee EA oo s ca e o es a e c ea y es ab s ed oday, b a e bo da es b n e nex ee yea s. e o do n ode n oo s e ase se, o as, and S Sc ee a e nc ded e os o es o se e a yea s. o ns ance, e e os o y ca ab es o Me s a e e ys on . e oo s a bo o a oach e o x and as n a e ecen y nc ded so e c an e ana e en nc ons. So e oo s a e o n c ose o e a o e, c s nd ca ed by e a o s n e dd e o e e [ey e 2004].

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oday, oya s ec on cs o e ands some o e o d's b es ec on cs co an es and o e's a es, sa es o 2 , b on n 2003. a s a ed e od c on o ncandescen a s n a ac o y n 8 . nde e eade s o . e a d Leona d ede s, e o nde o s, s de e o ed n e en e se s oday. n 22 e a d s es med as e ana e o s. n e 3 yea s o s eade s s as de e o ed o an o an en e se 5,500 e oyees.

o adays n o a on ec no o y s seen as a necessa y oo o s o n e en e se and s b s ness ocess. e nc eas n s ze and co ex y o e e en a ons o n o a on sys e s, s beco n o e and o e d c o a n e n o a on sys e s o e b s ness o e en e se. e ey obe n e a n en s nde s and n e en e se o no n o a on sys e s a e needed. e co ex y and e o e nance s c e o s nc ease eac o e, es s n a d c s a on o a e e dec s ons. An s a on o e co ex n n o a on sys e s n s s e c en se o abo 00 d e en sys e s. e ana e en o oday canno nde s and e s c e, ocesses, and nc ons o e en e se as n e e o e a d s. o be ab e o a e so nd and es ons b e dec s ons, ey need an a d a e s e nde s and e en e se.

n e as decades e d e en 's e e nde enden n e od c on ocesses and o e nance. n ac be n o see a e e e d e en nde enden en e ses a ad e sa e b and na e. 's e e do n e sa e ocesses by s n d e en n o a on sys e s, a ca ons o so a e. e ana e en an a c ose coo e a on be een e d e en 's. n e e d o s s done by ca y s n e sco e o B s ness a n en o e e e o s o e en e se e e, s den y n o o n es o syne y and coo e a on. o ns ance a e 's need an e a o a . B o be ab e o den y ese syne y o o n es so e a d s necessa y.

s needs an A oo a e s e o e esen e a o s a ec a desc ons on a co a ab e bas s and s o e de n on o a e a ons s be een en e se e e and d s ona e e a ec es. Beca se o e n od c on o e A a e o , s a e o as o be s o ed n a s c a ay. e A a e o as d sc ssed n a e 4. An A oo s e so on o s o n e A a e o n a s c ed ay. e an obec e o e A n s s B s ness a n en . o n ca on a e s a e o de s (on co o a e e e as e as on d s ona e e) s a so an o an bene o e oo . n sec on 3.5 e bene s o an A oo e e d sc ssed.

n e ean e eo e n o o a e a e beco n o e and o e consc o s o e co e ence be een s a e y, a ec e and o e nance. e e o e ano e e e en s a e oo o de n aceab y o a en s': o a e c ea and nde s andab e y a e an dec s on s a en. e e o A s as o be ab e o a e be e dec s ons n e e d o . A e e s c ea ness abo e ea y and s es s n a be e o e nance. e o e nance as ec o A as desc bed n sec on 3.4.

Based on the theory of section 3.3, tools that help to compare IT alternatives to use EA for an IT-centric approach. Business IT alignment is the most important objective and decisions based on EA are only taken in the field of IT. The approach is thus top down, because this means to use the EA to go from the top (strategies and drivers) towards the bottom (technical implementation). Also it means to put the Assessment towards the bottom side of the functionality of an EA tool. This process aspect of EA as discussed in section 3.2 in this thesis needs, based on the theory of section 5.3, an EA tool that supports a top down approach.

When the use case architecture is a good solution to achieve current problems, such as the multiple implementations of the same software solutions, lack of interoperability (between departments and between this and external parties), a great deal of time spent on solutions, too complicated problems with this (to manage them in the area of the data, lack of coordination, etc. Also tools to support the early design of the 'as-is' architecture is needed. The decisions are independent, it is not hard to be able to compare alternatives in the field of architecture, EA is a good way to support this. But to make EA and the IT align to the practical use a specific approach is needed. EA tools support this specific approach. If the organization wants to adopt EA enterprise wide, tools are necessary to make the practical use.

The next chapter describes the tool selection process of an EA tool that supports this top down way, to support the identified way that is discussed in this chapter.

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e o s e o es ab an A, as desc bed n e n od c on, s e se ec on of an A oo . a s a so e a n o c o s es s and e e o s doc en s e o e e se A c ec e, e se ec on ocess o an e se A c ec e oo se o s o nde s and n and o e n n e en e se''. Based on ex e ence, a e e a y acce ed a oac o e se ec on o co ex so a e sys e s as de e o ed. e 'bes ac ce' a oac con a ns a n be o s a es and ac es, o e os o an a e [e e]:

- e e na y se ec on: e e a a on o a on s con a n n ess an en cand da es, based on a e y ed o ;
- e de a ed se ec on: e a ons o e on s o a s o s o no o e an e A oo s. s se ec on s ca ed o based on a ed n be o essen a assess en c e a (noc o c e a);
- e assess en : e ac a assess en o e o s cand da es e a d o a co any s ec c case d n a de ons a on;
- e co e on: s a y oc sed on e e n n o con ac and e a n e e en a on.

e A oo se ec on ocess a s o ed e a oac desc bed abo e. A s e, o e e , co e on s no ea zed, d e o an n o es n e co o a e de a en . o e on as no a o s ad a on e od. Sec on .2 e o s e e na y se ec on n a on s o e cand da es s co osed. n sec on .3 s on s be ed ced o a s o s o e cand da es. e assess en o e o 7s cand da es, s desc bed n sec on .4, as done by a de ons a on o e s e s. s ase as o o ed by a es case e oo a see s e bes a e e de ons a ons.

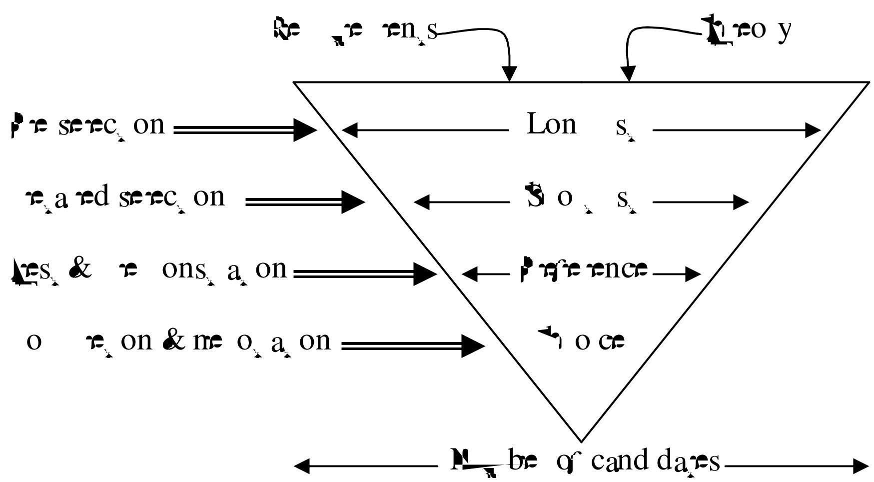

Figure 7-1: The selection process [De Heij]

An o an se ec on as ec , o no e on y one, s assess n and co a n e nc ona ea es o e cand da e sys e s. n e a oac desc bed a e e o s a e,

7KH UHTXLUHPHQWV

7.1.1 The requirements from the RFI

General Requirements

- Business strategy;
- Business activity (value added business components, processes, organizational units, structures);
- Related information structures;
- Applications to support the business activities and processes (functions, data, system interfaces);
- Supporting infrastructures;
- The arguments of the formal models above here.

The obtained models and information of these architectural areas have to be stored into a repository. An important objective to use an EA tool is 'traceability of arguments': to make create and updates and ability to trace an decision statement. As a consequence it should be possible to store the arguments in the repository with the different requirements, principles, and models across the various architectural areas related to each other.

In this thesis was decided to use the IAF framework of Capgemini as a standard for architectural activities within this thesis. It can be expected that changes and or additions need to be added to the framework; this thesis may also decide on some specific customization of the IAF framework. This results in that the software tools should be able to support the IAF framework and allow customization and adaptation of the repository metamodel.

User requirements

Within this thesis architectural activities for two of the governance structure. This implies that architectural activities are both executed on corporate level as well as on divisional level. The principle is that EA is a set of common principles and models across corporate and divisional architectures. The use of a common software tool supports the [illegible] to represent these [illegible] architectures on a comparable basis, and support the definition of [illegible] relations between corporate and divisional level architectures.

The software tool will have to support both active as well as passive users. Active users are those who orient and maintain models and other elements. In most cases these active users are IT professionals, with a role of architect. Passive users are those who read information and models out of the software tool. They will mainly be from business managers, CIO's, to program and project managers and to solution architects. This will require that in which way architectures are represented towards various stakeholders.

Architectures are mostly developed in architectural projects, where a team collaborates on the development of principles, scenario's and models. The software tool should support collaborative development. It should be possible to assign parts of an architecture to multiple users, with the responsibility to update the formal representation.

Requirements based on the theory

In the theory part of this thesis some requirements to EA and the tools were seen. Here only the theory is written which is not described in the internal and user requirements. EA is an aid to understand the current use. To be able to understand and current use system structure, processes, and functions have to be described (see also section 2.[illegible]). In this thesis (see also sec on 3. .), desc b n e a ons s be een e d e en a ec es s cons de ed as e added a e o an A. e e a e d e en e s n desc b n a en e se. s no s c en o oo o ome e o n o nde s and e en e se. e e o e a oo as o a e e d e en e o n s n o acco n and as o be oss b e o add e a ons be een e .

7.1.2 An addition on the requirements of the RFI

e endo s o e A oo ad a e es ons a e ey ece ed e . ese es ons ans e s as sen o a e endo s a e esen ed n s sec on. n y e es ons abo e e e en s be desc bed.

es on	Bac a d o a b y (2. .4) can be n e e ed n o ays: a) nc ona and o b) da a nc ona s ean, an a) e sa e obe can be ea ed a d e en so on (es n n ea n n o) o b) e sa e obe s o d be ea ed n e sa e ay. a s ean e e
Ans e	s an s o be ab e o ead e ode s c ea ed n e o e (e. o) e ease(s) n o e ne e ease.

es on	. e a ood de n on o e s en on en o e s nde s and ea s zes e c., 2. o any en e se a ec s a e s oo n o e 3. a s e ex ec a on e a d n cons s ency be een ese en e se a ec s and e so a e de e o e n o an za on
Ans e	**General:** s o nde s and n a e a ca on e d o oo s on (en e se) a ec e on e a e can a y: e oc s on b s ness a ec e/ b s ness ocesses; e oc s on ans a on b s ness and s a e y n o a o o o's; e oc s on desc b n e a ons s (acea b e a en s/ sed nc es) be een b s ness e e en s, a ca on nc ons, ec n ca so ons. e end n on e a ac e s cs/ s en s and ea nesses o e oo (oo s e), e sa e can a y as e . n o e an o e a be e nde s and n o n a e a e o e ed oo s os s ed. a y oa s oo s o be ab e o doc en n an ac e ay o n e se A ec e and s co os n o ec A ec es and be ab e o ens e acea b y and ac o an es. n e s case e can d s n s e o o n s b a eas and n o ed ea s: . oc en e s n e se e e ence A ec e, be n e se o nc es, s anda ds, and e e ence ode s ac oss e aye s o b s ness, n o a on, a ca on and ec n ca n as. e a a e a cab e o a nde y n a ec es. ea s ze: 2 3 eo e o an a n, b s ed on e ebs e; 2. oc en e nde y n en e se a ec es on e e e o

	s, o o a e nc ons and o o a e o ans a o s ea s; o a +/ 0 eo e; 3. oc en e a ec e n ao nno a on o ec s, bo on o o a e as e as e e . o a n o ed ea : 0 20 a ec s; 4. oss ec cons s ency ac oss e a ec es n e ee o n s abo e e e: n o ed ea : s A ec e Boa d (os ey nc ded n e eo e abo e).

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e e na y se ec on as a ed a a n e s e's a e o oss by e e an A oo s. e es o s ase s a on s o cand da e A oo s and s e s. n ac ce s eas be o es c e on s o e e an en cand da es [e e].

s o a e e A oo s e e se ec ed. n s se ec on ocess, e d e en nds o oo s e e a en n o cons de a on and o e e y ca e o y (o do n, bo o and a an e ana e en , see a e 5.3) e e as a eas one A oo se ec ed. Bes des a e e e e on y oo s se ec ed a a e abe o s o a e co an es e s. e e e se ec ed oo s, e s e s be een b ac e s, a e:

1. Ada a ons (Ada e)
2. A abe S M (A abe b)
3. A S (S e)
4. B se bed (B se)
5. o o a e Mode e (ase se)
6. Me s (o as)
7. o son e se o (o o a o .)
8. Sys e A ec (o n So a e)
9. o x (o x no o es nc.)
10. 3 s b e e se (n sys)
11. as e (e o e)
12. ebS e e (BM)

ese e e cand da es ad o be ed ced o ess an en cand da es and e e o e e o o n oo s e e e o ed o e on s:

- **ARIS:** beca se A S s oo oc sed on ocesses. e es a e y o a a oac and s a eady n se n s. n oses a ons s o n a A S s no s abe o n e ay e A oo , as o be se ec ed, as o;
- **BWise Cubed:** beca se s oo s oo oc sed on e b s ness ocesses;
- **Troux:** beca se s no c ea a s oo can con b e;
- **Yasper:** beca se s no a co e c a oo . as on y on e on s o cons de e o s be nd s oo and ose o s can e s o o a e e o n e e en s.

e s e s o e e an n oo s (Ada e, A abe b , ase se, o as, o o a o ., o n So a e, n sys and BM) e e cons de ed n e de a ed se ec on s a e. n e nex sec on s s a e be desc bed n de a .

The ose of the de a ed se ec on s o ed e the on s o as o s of the es abe EA oo s. Bes des the s o s, the es s of the s ase a so cons s of [ ]:

- The nfo a on conce n n the s ab y of the EA oo as e a ds the c ca assess en c e a;
- The s e en a y nfo a on abo the EA oo conce n n the s o y, the n be of e en a ons and efe ences, the f e ans and ex ec a ons, and the oss b a d a e and sys e so a e;
- The s e en a y nfo a on conce n n the s e's o an za on;
- An nd ca on of cos s fo a o s co onen s and se ces.

The ' noc o c e a' o ' o an e e en s' a e the os o an e en n the de a ed se ec on a oach. These c e a a e o be e by the o en a cand da es o be a of the s o s. In nc e these c e a a e sed on the cand da es fo the on s, a e the the os s abe omes e an. If the de a ed se ec on s based on ' o an e e en s', no o e an abo y e e en s sho d be co ec ed. A a e n be of e e en s c y ead o a d s o on n the ocess of choos n . By sco n on these ' o an e e en s' the s o s can be co osed. Th s o des a s of EA oo s n o de of s ab y. The of the s e s a e of o as o en es onna e. I s a so ad sab e then one s beco es ac an ed the the (fo exa e) cand da es that they esen the se es and the EA oo o the o ec ea and es ond o so e add ona es ons [ ].

n the de a ed se ec on of the EA oo fo th s, o an e e en s e e sed. The e e no ' noc o c e a'. Fo y o an e e en s e e se ec ed, based on the e e en s of sec on and the eo y of the second a of th s the s s. These e e en s e e sen o the s e s as es ons n the fo of a RfI. The RfI be desc bed n the nex sec on.

7.3.1 The Request For Information

In the assess en of an EA oo a de of e e en s ay a o e. The os o an o seems s o be th a of the f nc ona c e a, o th s need no a ays be the case. Fo the fo y es ons of the RfI, n ne een es ons e e cons de n f nc ona c e a. The e a e th ee an ca e o es of c e a and these c e a a conce n n [ ]:

- **The supplier:** c e a e the s ze of the s e , s a me s, th s o y, and s f nanc a s a ons;
- **The EA tool:** c e a e th s o y and anned f e of the oo , f nc ona y, echno o y, e fo ance, sec y, and doc en a on;
- **The conditions of delivery:** c e a e n es en , e en a on cos s, and a n enance and s o cos s.

Fo an be of these assess en c e a nfo a on can be o ded by the s e . Ho e e , any c e a a e s bec e and ha e o be co ec ed d n a the dea n s with the s e , s ch as de ons a ons, esen a ons, and me o a ons. The assess en c e a ha e a e e s a ed, based on the f nc ona s ec f ca ons, ha e o be sed o e a a e the o en a EA oo . Th s e a a on can be ca ed o n the fo o n ays [ ]:

- By asking the suppliers concrete questions with open or defined answers (preferably open). These questions are directly based on the use case. Mainly, the use is based, but by formulating the questions as concrete as possible and by including the clarifications in the contract, a certain value can be placed on these answers;
- Finding out these things by testing the MA tool, i.e. by means of demonstrations, cases, and workshops. These assess the current [illegible] and are directly [illegible] to the questions in the other way of evaluation, where the use case is [illegible] to be [illegible] in the business case.

The first described way of evaluation was used at the beginning of the selection process. The other one was used in the final stage of the selection process. The first described way of evaluation was done by sending a questionnaire (the RFI) to the suppliers, which contained the questions about the supplier and the MA tool. [illegible] about the conditions of delivery were not considered [illegible] in this selection process. The RFI was made in cooperation with the [illegible] department. I [illegible] the documents and they were made in the [illegible] of the RFI's of [illegible]. The three documents are:

- The Main Document;
- The Commercial Questionnaire, which contains the questions concerning the supplier;
- The Technical Functional Questionnaire, which contains the questions concerning the MA tool.

The Main Document

The main document is an introduction of the RFI and the selection process. It contains a description of the situation of [illegible], the [illegible] MA, the scope, [illegible], and process of the RFI. This document also contains the [illegible] requirements and a description about how the questions of the commercial and technical questionnaire are to be answered. Besides that, this document contains also what the suppliers [illegible] in the selection process and a non disclosure agreement. But the most important [illegible] about the requirements of [illegible] the MA tool. These requirements are [illegible] in the [illegible] requirements and the [illegible] requirements of section [illegible].7.

The Commercial Questionnaire

This document is written by [illegible], who is [illegible] as [illegible]. The questions in this document are about the supplier of the tool or about the current users of the tool. The answers to these questions serve to [illegible] the continuity of the supplier.

The Technical Functional Questionnaire

This is the most important document of the RFI. In this document the technical and functional questions for the supplier are [illegible]. The [illegible] and assessing this questionnaire was [illegible]. Suppliers often interpret the questions in different (sometimes creative) ways. In section [illegible].2 there are examples of questions that were [illegible] for the supplier. Moreover, often the requirements of the questionnaire [illegible] and specific solutions in the MA tool with different options, which [illegible], are also possible and sometimes better. [illegible] and, as [illegible] can [illegible] also [illegible] on the [illegible] does not [illegible] any [illegible] [illegible].

Example of the RH: "It is possible to remember a tomato in a yellow way"; a "yes" or the "No/sorry" answer does not say anything

Unisys decided not to build on the RFI, because they are not a real MDA tool vendor. Therefore it was not possible to analyze Unisys in the same way as the other tools. Instead of answers to the RFI, Unisys did send in all kinds of information that they offer for MDA. In this analysis Unisys was not considered, but Unisys was compared with the top 3 that was based on this analysis of the different feedback to the RFI. After the comparison it was considered that Unisys does not [illegible] what was expected of the other tools.

To analyze all the feedback to the RFI, a document was made that consists of the following five parts:

- **Introduction:** this part was an introduction on the analysis of the feedback to the RFI. In this part the analysed method of the feedback, the requirements of Philips, and the most important aspects of the analysis were described;
- **Questions:** in the analysis of the feedback to the RFI only the answers to the questions of the technical/functional questionnaire (part of the RFI) were considered. In this part of the whole document all those questions of the RFI were written;
- **Review:** all answers were rated. The companies that were considered in the analysis were: Adaptive, [illegible], Casewise, [illegible], [illegible] and [illegible] Software.
- **Remarks:** the questions on this [illegible] that were focused on and why the answers were [illegible];
- **Conclusion:** for each tool a concise assessment was formed. For the selection only the answers were considered that were considered as the most important ones in comparison with the different tools. This was done to determine the necessary requirements for the optional ones. Based on the necessary requirements a better decision could be taken. The conclusion lead to a selection: Adaptive, Casewise and [illegible] as [illegible].

Introduction

The requirements were described in section [illegible]. This section contains a description on what is considered as the most important aspects of an MDA tool. In these aspects, there was a focus during this review:

- Philips wants to be able to document in an active way the MDA and its components in Project Architectures and be able to ensure traceability and impact of changes;
- Philips needs an MDA tool to deal with all kinds of architecture in a structured way. In the beginning this architecture will take place in small projects, but the purpose is to use MDA more and more in the future. Therefore the MDA tool has to be able to reuse obtained results in the future;
- In this review there was respectively paid attention on the possibility of creating different views, the relations between the different models and views, the usage of the IAF framework in the different tools, the different models in tools that can be used in combination with the tool (e.g. Visio), and export facilities (the use of XML), and if the tool can cope with different users in different places (for instance is it web based and has a central repository).

Review

The answers of the six remaining tool vendors were rated on a scale of one to three. A 'one' was given if the answer suits the situation of Philips and a 'three' if it does not fit the s a on o s. co se es ons abo e na e o e s on o e oo e e no a ed s ay.

e o e, be een b ac e s s ob a ned on s, o each ca e o y as:

- **General information:** o as / o a / Ada e (3);
- **Functionality:** o as (25), ase se (2) and o n (2);
- **Technology:** o as / o a ( ), Ada e / ase se (2);
- **User requirements:** o n (5), Ada e / ase se / o as ( ).

Annex a a as added o ec n e e e a on on abo e oo . a a s based on e o a o e e e add ona a e a e endo s n . o endo s o a 'one' (Ada e, ase se, o as and o n), one endo a o' (o a) and Alfabe o a e'. e o a es as: o as (4), ase se / o n / Ada e (50). Alfabe (4) and o a (58) e e c ea y no e oo s a a e s abe o s.

Further selection

o a ood ana ys s o e eedbac o e so e es ons e e se ec ed. as cons de ed a ese es ons add ess n e os o an e e en s o s. o cons de on y e os o an e e en s e oo s be d ed on e abso e o an as ec s o s. a es a be e d e en an cons de n a e es ons n add ona nc ona y (s cons de ed as n ce b no necessa y) can as so e d a bac s o e oo .

e ey es ons cons de ed as ec s e od a y o e oo , a e oo does, o e n ended se s a e, e A a o , e o and ex o ac es, e e a ons s, e d e en e s, syme y o ons, and e oo s eb based. A so e ex a a as a en n o cons de a on.

s e a a on es ed n e o e: o as (2), ase se (23) and Ada e (25). o conc de s sec on a d e en o e o e o each endo be esen ed as e as a e e a conc s on.

Conclusion per vendor:

Adaptive

Ada as e a ons s be een e d e en odes and oo s andy n se. ses oo s e MS so and MS ce. e s on o n s a e: s eb based and ses XML. B a e e se ec on ocess e ca ab es and oss b es o e oo e e no y c ea . o e e e oo co d be s abe o e s a on o s.

Alfabet

Alfabe s oo oc sed on o o o ana e en o o ec s. s ose s o o A odes ana e en odes and s an exa e o a odo o ca a oach o n a ec e, o a , and a e ana e en . Alfabe does no a e e d e en a ec es e a e a ons s n o acco . e e o e s no s abe as an A oo o s.

Casewise

ase se ﬀe s a o s n oo es ec a y beca se as e a ons s be een e d ﬀe en odes. s a co e ens e oo a o ﬀ o ons and a so a s on o n s a o s s anda ds e XML and B M

Computas

Me s ﬀ o o as s a e y o s n oo . s s on o n s a e se can se d ﬀe en e s (e eby e s no ed o ed ned e s) e a ons s be een e e e en s n d ﬀe en ode y es and d a a s. a so s s anda ds e XML, B M and ML and e ode s oo e y e o an zed beca se Me s n e a es a e ode s n o o s c en e se de ode s c es.

Popkin

o n as a o ﬀ o ons and ode n e n es, b s no con nc n on e s bec ﬀ e a ons s. ey c a a e oo s o s e B s ness a n en , b e ess on a ose a ey on y n a a e o (e ac an) o cons de n e e a ons s be een e odes. e eedbac o e R s, co a ed o e o e ones, no con nc n . A o e oo s ﬀ o n a e de y sed and ac no ed ed (ey ﬀ en a ea n a c es abo A) e ana ys s conc ded a e e a e be e ones.

Proforma

s oo oo s ood and oo s e y easy n se. B an o an d a bac ﬀ e oo s a e e a e no e a ons s be een e d ﬀe en odes. e e o e s oo s no cons de ed as s ﬀ c en o e needs ﬀ s.

Summary of analyzing the feedback to the RFI

e oo as o be ab e o e ne a e and s o e d ﬀe en odes e a ons s be een e and n a ony e A a e o . n e be nn n e oo as o s o s a a c ec a a o ac es, b n e ﬀ e as o s o a e a c ec e n s and a so as o a e a ﬀ ana ys s oss b e. a n s n cons de a on Me s ﬀ o o as and o o a e Mode e ﬀ o ase se e e cons de ed as e bes oo s ﬀ o s. n sec on 5.3 e oo s ﬀ ase se, o as and o n e e cons de ed as o do n (s e case n e s a on ﬀ s) and e oo ﬀ Ada e as bo o . o y o abo o oo as s es ed, a so on e bas s ﬀ e R , o se ec Ada e as e d oo o s.

7.3.3 The Short List

n e e o s sec on e es onses o e R e e ana yzed and e o ee e e se ec ed o e s o s . s e se ec ed oo s a e Me s (o as), o o a e Mode e (ase se) and Ada a ons (Ada e). As en oned n e n od c on, ese ee s e s e e n ed o e a esen a on and a de ons a on ﬀ e oo . ese esen a ons and de ons a ons be desc bed n e nex sec on.

'HPRQVWUDWLRQV

Based on a s ec ﬀ c b s ness case, cons s n ﬀ ea s c da a and b s ness ocesses, a (n s case e) cand da es ad o e a co a ab e de ons a on. e de ons a ons were meant to assess the candidates on their support for important aspects. But also the demonstration was a good opportunity to see how the tools work. Also it gave us a feeling of how the tool can execute the [illegible] that the vendor claims and how the tool does that. This resulted in a ranking and the top candidate of which will be selected unless inaccuracies are uncovered in the next phase of the test [illegible] [illegible].

The vendors of the three tools of the shortlist (Computas, Casewise and Adaptive) were invited to give a presentation and a demonstration of their tool. The selected business case was a part of the [illegible] Reference Architecture ([illegible]RA) model, with business strategies, business processes, IT systems, and applications. The vendor's presentations and demonstrations will be summarized in the following sections. After the presentations and demonstrations the tool that scored the best [illegible] should be tested.

7.4.1 Computas

Computas gave a presentation and demonstration of their tool Metis. It was a good demonstration in which they emphasized the possibilities of creating relations between nodes and functions in Metis. Also they made different views of the same situation and showed how to make comparisons between different views. In Metis containers are used (for example a process) and it is possible to zoom in on a container and find the processes behind the process. [illegible] it is possible to get all kinds of information [illegible] which application is needed for which process. The graphics in this tool are very impressive. [illegible] graphics does not look very difficult to work with this tool. The presentation and demonstration supported the conclusion of the RFI analysis that Metis is a powerful tool, which can meet the most important needs of [illegible].

7.4.2 Casewise

Casewise gave a presentation and demonstration of their tool Corporate Modeler. In the presentation [illegible] Corporate Modeler, the use of relations and different views, is not as good as Metis. The comparisons between different (parts of) models look better in Metis than in Corporate Modeler. After the demonstration of Corporate Modeler they showed Corporate Modeler 10. The new version [illegible] and [illegible] better than its predecessor. This version is definitely an improvement on the graphical interface. The functionality is not changed, so [illegible] of Corporate Modeler also apply on Corporate Modeler 10. [illegible] version looks not very difficult to work with the tool.

7.4.3 Adaptive

Adaptive has a repository based tool where both Metis and Corporate Modeler are modeling tools. The repository of Adaptive is the most important part of the tool. It is an open and completely standards based (Meta-Object Facility repository). A drawback of this tool is that a modeling tool is [illegible] the connection with the repository has to be made. An advantage of this tool is that it has a standards based integration with tools like Metis and [illegible]. Also it has a very good existing [illegible] model in which [illegible] objects. The graphics in this tool are [illegible]. It does not look very difficult to work with this tool, but the pictures and layout are definitely less nicely.

7.4.4 Conclusion

Af e i e esen a ons and de ons a ons a dec s on as ade. Me s and o o a e Mode e a e a e sa e f nc ona y, b Me s s cons de ed be e f o e s a on of s. as a be e se n e face and o e add ona o ons. does oo easy n se and e e a ons s and e co a sons be een d ffe en (a of) ode s a e be e an n o o a e Mode e .

e b es ad an a e of Ada e s s e os o y. n e f e d of e a ca n e face Me s s d f n e y n f a o . A so e con ene y a a ed ode s n Me s a e an ad an a e e a d o Ada e. e a ons s n Me s a e c ea e an n Ada e and a e e a ons s, a a e oss b e n Ada e, a e oss b e n Me s. e f nc ona es of Ada e o e a o ess co a ed e f nc ona es of Me s. Beca se Me s a so ses a e os o y, Ada e as no on e cons de ed as e bes oo f o s.

f a ea s f o e es case a Me s can and e e s on con o and a o za on an Me s s e bes oo f o s. ese e e en s e e no cons de ed be o e and e efo e ey e e no a of e de ons a on of o as and ase se.

Af e e ee de ons a ons, e d e en of e A oo s, as based on ana ys s of e e o e , as no an ed. e efo e Me s as se ec ed f o e es ase.

7KH WHVW FDVH

e es case as on y e d Me s. As en oned be o e, e e e e no co a sons oss b e o e oo s d n s es case. e ose of e es case as o f nd o o Me s s ab e o s o e e e en s f o e es case a e desc bed n sec on .5.2. ese e e en s e e based on e e e en s f o sec on ., e e e en s 7 based on e a o s feedbac o e , and e e e en s based on e ee de ons a ons. e e e en s based on e feedbac a e e e en s a e e no cons de ed be o e e , b a a e a so e e en s f o e s a on of s.

n e es case e a c ec e of e Se cond o s (S) as e en ed. a as done by e en n e ee d ffe en e s and e e of ese ee e s a f o e as ade. e f o e s based on e A f a e o .

e an ose of e es case as o f nd o f e oo s ab e o c ec cons s ency, n s and o e a be een d ffe en e s of e sa e a c ec e. n s es e cons s ency, n s, and o e a of e a c ec a e s be een e yd a and e A doc en e e c ec ed. e efo e e As s efe ence a c ec e of yd a and A e e e en ed n e A oo . e o e of e yd a e s e od d s on Se cond o s (S) and e o e of e A e s e A ea , cons s s of a c ec s f o e e y and co o a e . A so a d e as c ea ed. s s e S e A c ec e (S A) e and f o e S A doc en n s es a e es, b s ness nc es and nc es e e a en. e o e of e S A e s e S A ea . e a of S A as o de e a efe ence a c ec e, co d be sed a a e s s es. Af e e en n ese ee e s a f o e as c ea ed. s e s based on e A f a e o and e n as e e a ed o e A, yd a and S AR s. ese o s e n o e d e en es scena os o sec on .5.2. o exec e so e o ese es s as necessa y o add e en s, a e no based on e ea s on n s. o ns ance e e e c ea ed so e n s be een s a e es and b s ness ocesses. b o s y ese n s ex s , b ey e e no doc en ed. ese n s e e c ea ed o do so e ana ys s s as s ec a sea c e a. e es s o e es case e e esen ed d n a ee n o e A ec e o e o s. n e esen a on a de o Me s (e sed b s ness case) as en.

7.5.1 Introduction to the business case

n s sec on e b s ness case, as sed d n e es case, be ex a ned. e 2 s o s a a ca e esen a on o e a ec e o s. e box on e o A n e esen s e co o a e a ec e o s. e S AR be on s o e co o a e a ec e o s. nde s co o a e e e a e 's a e s be. e s box s ed n and s box e esen s e a ec e o e S . n e case o S s a ec e s ca ed yd a. A so e co o a e nc ons, e co o a e a e e o n n e na a ec e. e RA a ec e desc bes, on co o a e e e , a e a ec es o e e y n a co on s y e. e yd a a ec e and e RA a ec e a e d a n n e 2. n s sec on bo s be desc bed s o y. e yd a a ec e s on y sed n e S and e e o e on y e S a o e RA a ec e as e en ed n Me s. n e 2 a es on a s s o n. An ob ec e o e RA doc en as o den y a o e d s ona a ec e co d beco e a a o e co o a e a ec e n e o o s a ed se ces. e es on a s ands o den y n ese s a ed se ces.

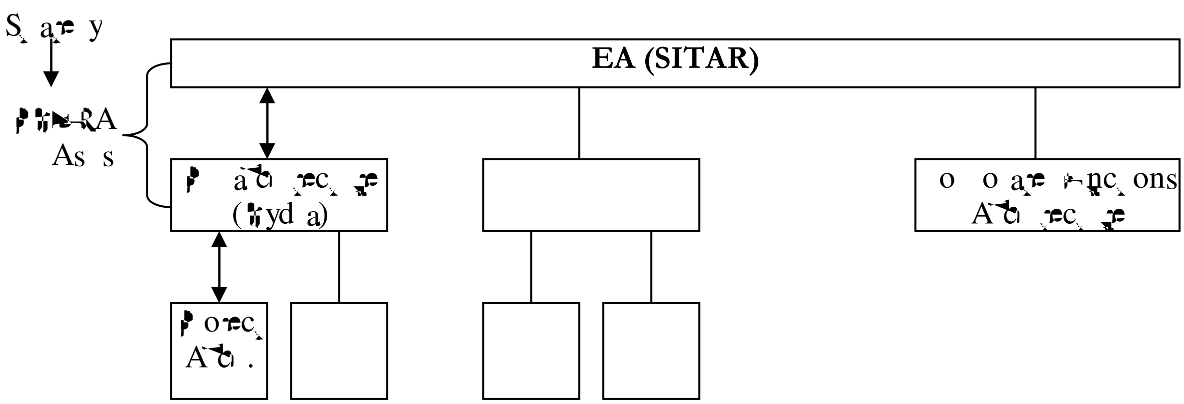

Figure 7-2: The architecture of Philips

PHERA

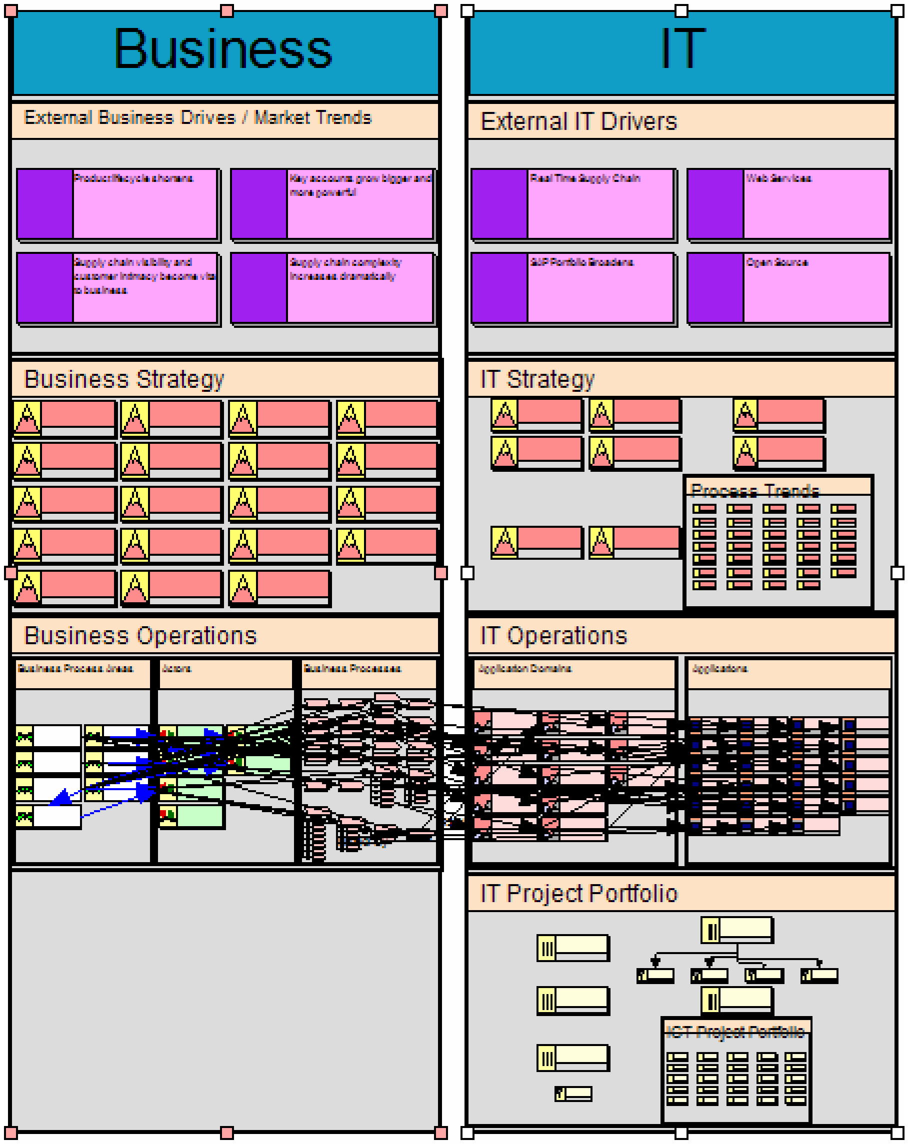

Figure 7-3: The PHERA view imported into Metis

Figure 7-4 shows the IT Analysis model that is used in the PHERA document. This view was implemented in Metis and compared with the view that is presented in the Hydra document. The entities and especially the relations between the business and IT entities were compared with the situation in Hydra and STAR. The PHERA document contains tables with information such as business drives, business strategy, business processes, IT drivers, IT strategy, applications, and IT project portfolio. This information was imported into Metis and presented as objects with the relations between them. In Figure 7-3 a screenshot of the PHERA view in Metis is shown.

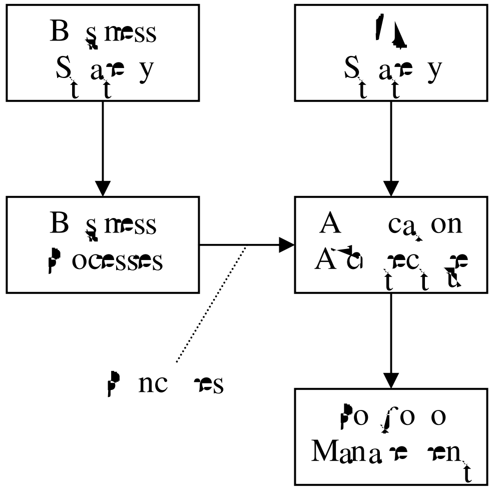

Figure 7-4: The PD Analysis model of PHERA

Hydra

The 7-5 shows a part of the PD Analysis model that was used in the documentation of the PHERA case. This part of the PD Analysis model had to be filled with the data for the documentation of Hydra. The arguments of the relations between the business processes and the application architecture had to be compared with the relations between the principles of the situation in PHERA and SEMAR.

The 7-6 shows the Hydra model, which was used in the PD S. The figure shows the domains together with the applications and the relations between the different applications and domains. The figure shows the different domains are presented and this model was also presented in Meetings. Besides the model of the 7-6, there was a document with additional information about the Hydra model. The arguments were taken from this additional information.

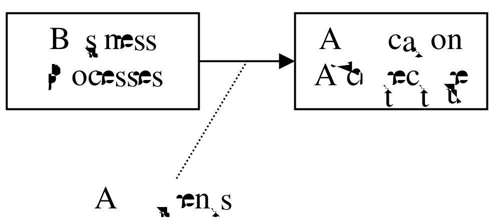

Figure 7-5: The Analysis model of PHERA, which had to be filled with the Hydra information

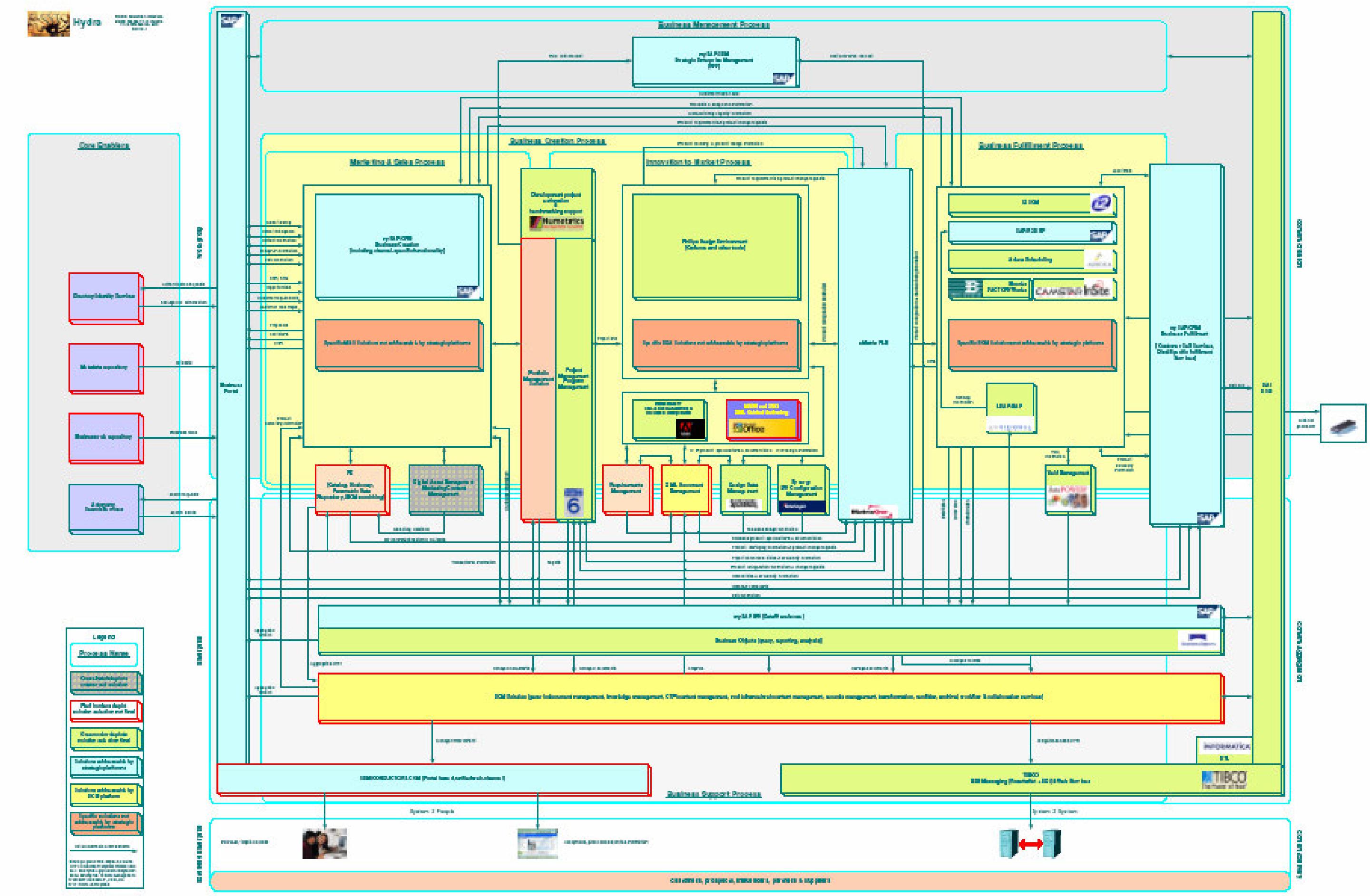

Figure 7-6: The Hydra viewer

SITAR

For the SITAR document the business functions and IT functions were taken. These functions were divided into different areas in the document and they were also connected with the three areas in Metis. These functions were compared with the functions of the [illegible]RA document. The SITAR document contains a lot of tables in which all the IT functions and business functions are introduced. In another table these functions are linked to each other. It is very hard to find all the IT functions that belong to a certain business function. After putting in all these functions, with the relations, into Metis an easy overview was generated. Figure [illegible] is a screenshot of the SITAR view in Metis. The relations are represented as a [illegible] between the objects [illegible] another one. In Metis it is possible to hide all the relations, so to select [illegible] are also the relations visible.

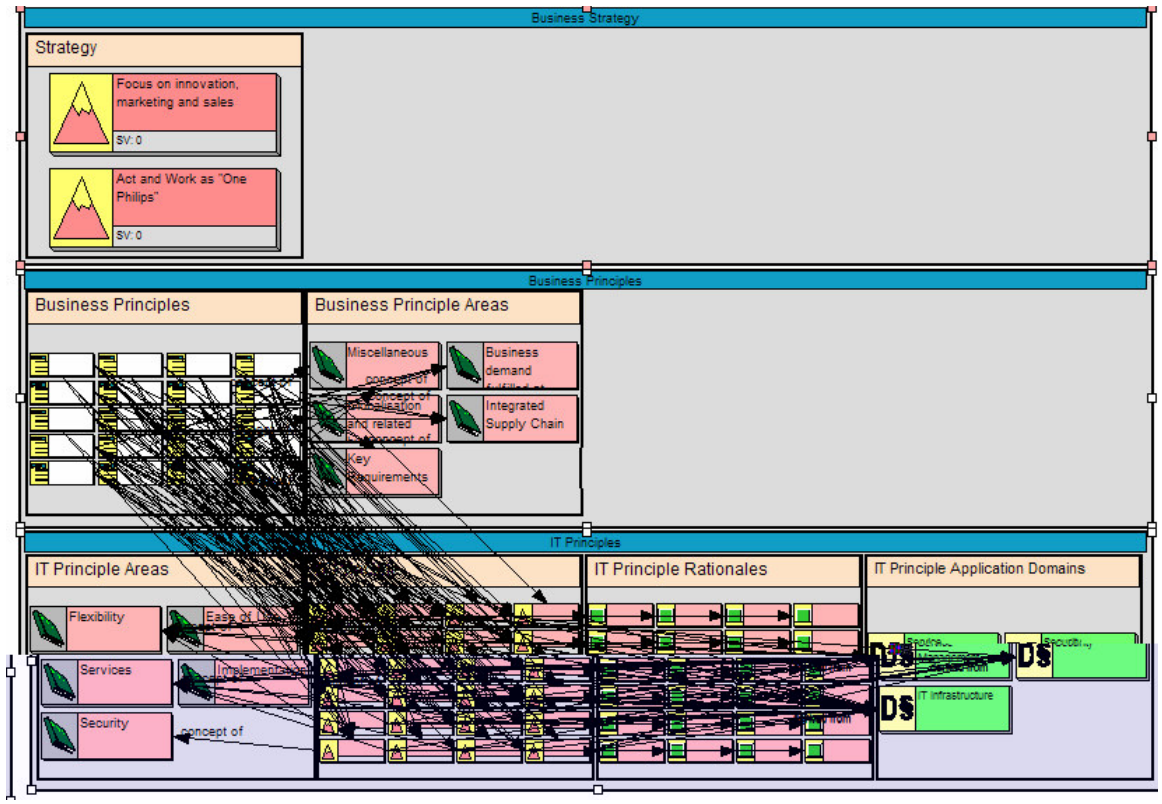

Figure 7-7: The SITAR view imported into Metis

IAF
The input for the IAF view is composed, from PHILIPS-RA, SITAR, and Hydra view. The input of these views will be placed into a copy of the IAF framework. Thereby the concepts will be divided into three areas: one for PHILIPS-RA, one for Hydra, and one for SITAR. This results in a clear overview of what input all the views had given and this makes comparisons between the different views easy. After implementing the input, the relations between the objects (of the different views) that represent the same, were created. Figure 7-8 shows a screenshot of the IAF view, which was imported in Metis.

7.5.2 Test scenarios

This section contains a short overview of all the different test scenarios that were presented to deal with during the test case. The sections are in order of importance.

Implementing the different views
As mentioned before, the purpose of the test case was to find out if the tool is able to check consistency, links, and overlap between different views of the same architecture. In this test the consistency, links, and overlap of the architecture views between the Hydra and the PHILIPS-RA document were checked. Also the amount of the relations, between the business processes and the application architecture, in the Hydra view had to be compared with the relations between the functions in the PHILIPS-RA view. The aspects that were tested in this phase:

- **Processes:** how is it possible to create processes;

- **Relationships:** o a re a ons s b and o any d ere n re a ons s can be re en ed;
- **Principles:** o a re nc es and re a ons s be een re re en ed n e oo ;
- **Properties:** o re es o e nc es, ocesses, and ob ec s can be o ed n o e oo ;
- **Import pictures:** s oss be o o ex s n c es, o does o and s oss be o ed e c es e. n s;
- **IAF:** o s oss be o re en a o e based on e ex s n omes
- **Models:** s oss be o o co e e odes, o does o and a a e e re e en s o a ;
- **Strategy:** o can e s a e y be re en ed and e a ed o e nc es and ode s;
- **Arguments:** y ce a n dec s ons e e a en n e as e ad n o e c en As s a ec e.

Authorization

s oss b e a d eren se s a e d eren nds o access e ode A e e e o o n e nds o se s o access e e s:

- me : s es ons b e and s a o ed o a e an es;
- e : s a o ed o e a ec e b s no a o ed o an e ;
- No access: s no a o ed o see o an e .

oes a s a on cons de a so a s o e a ec e a e se s no a o ed o see s oss b e a an o me s on y es ons b e o a s a a o e en e se and a e s a o ed o see so e o e a s o e en e se en an o me a es a an e (n ode o nc e o does a ec on e e s a a e b by o e s s oss b e o see e ac o an es

Version control

o a e e d erences be een e As s a ec e and e o be a ec e s be s oss b e o n e o o o ana e en o e a be een e As s and e o be a ec e and o does o

s e e a aceab y o a en s oss b e and o does o o s oss b e o a e c ea and de s and ab e y a ce a n dec s on s a en

Search options

n a s s oss b e o nd e e en s, e a ons s o ocesses s oss b e o nd by n e na e (o a a o e na e) o a yo a e oo n o s oss b e o nd by b o s n o e ode s

s oss b e o nd on e e en s e ocess o nc e as n ence o e e en s n ence e ocess o nc e o does o

Simulation

a can be s a ed and o does e s a on o s

It is possible to create an impact analysis
It is possible to create a cost analyse. Who does what and what does the meta model look like.

Are there also SQL queries and how does the database structure look like

Synergy and Redundancy
Is it possible to find synergy of tools, queries and who
Is it possible to find redundancy and who does what
In which ways is it possible to compare different business processes

7.5.3 Conclusions of the test case

The rest of this section is divided into four parts. The conclusion, based on the test case, were about:

- **Metis:** the conclusions about the tool itself;
- **The test criteria:** the conclusions about the answers to the test scenarios;
- **Conclusions of analyzing Hydra and PHERA:** what are the results of implementing these architectures and what is [illegible] information [illegible];
- **The contribution of Metis for Philips:** the conclusions about what this tool can contribute in the situation of Philips.

In this section a complete screenshot is given. These screenshots are from the implementation in Metis, which was created during the test case.

Metis: general conclusions
Metis is a nice and easy EA tool. In only a few days it was possible to create a good tool in EA. It works very nicely because it uses a lot of standard Microsoft commands [illegible].

The tool contains a standard meta model, which is based on a lot of experience in other enterprises and is developed in close connection with these enterprises. But if it does not fit the enterprise's use, it can be changed. To do that a certain license is obligatory, which was not available for us, so this could not be tested during this test case.

Metis can also implement extensions to [illegible] features. In [illegible] this was created with Metis and was based on the information [illegible] implemented in Metis. The more complicated the [illegible] the more difficult [illegible] be to create [illegible]. But in Metis experienced users will be able to implement all kinds of different [illegible], because this can be implemented in a standard way.

In Metis data can be imported from Excel files or directly from databases. So it is possible to reuse data that already exists in the enterprise.

In this test I discovered no [illegible]. The [illegible] was not very [illegible] when I needed it. There are a lot of options in the tool and sometimes it is hard to find all these options.

I s oss b e f o se s o c ea e e o n A n e ay ey an a oo s e. e a ca a o s oss b e, f o ns ance a nds o c es can be o ed and b ons can c ea ed. By c ea n e sa e a ca o e e , as n s' e oyees a e sed o, ey e co f o ab e Me s e y soon. By c c n on e b ons s oss b e o s o ano e e n Me s, o o zoo n n o an a ea.

In conc s on Me s s a e y n ce oo o o and n d n s es case a f nc ona and se e e en s a e e.

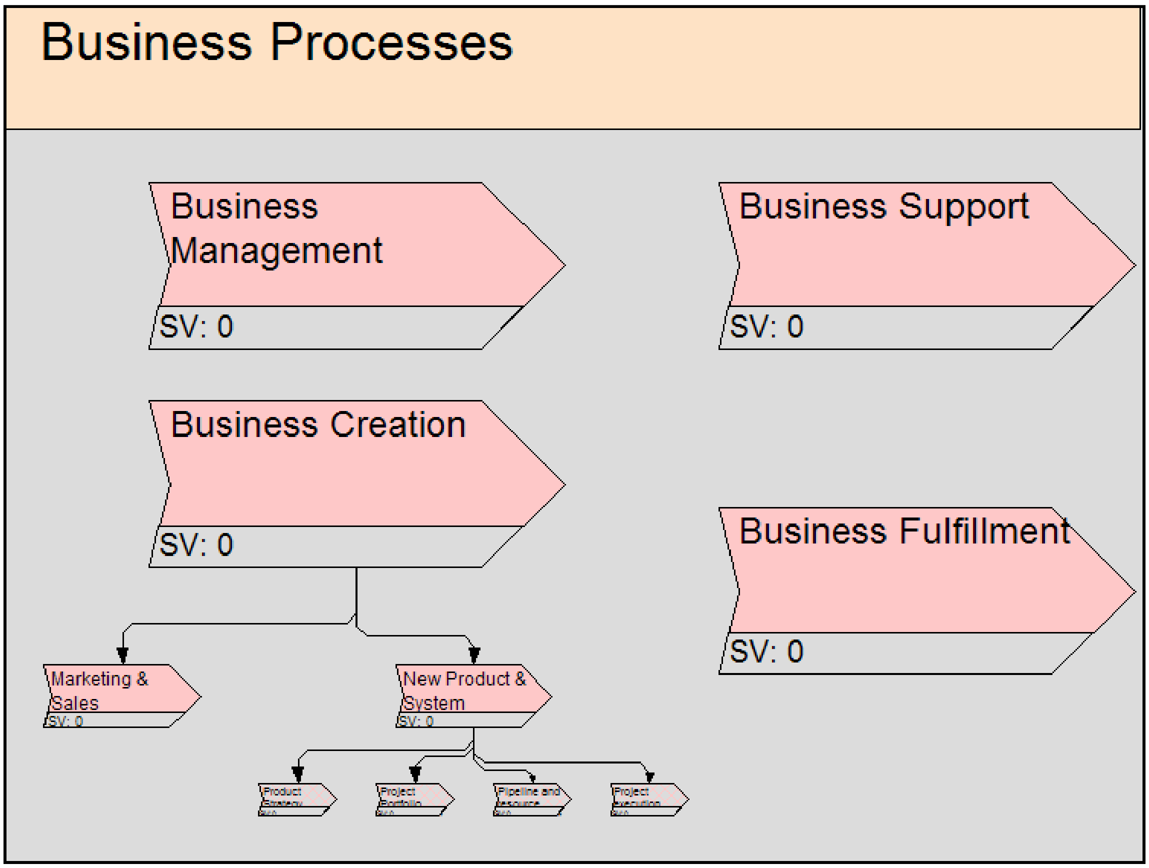

Figure 7-8: A screenshot of a container with Business Processes in Metis

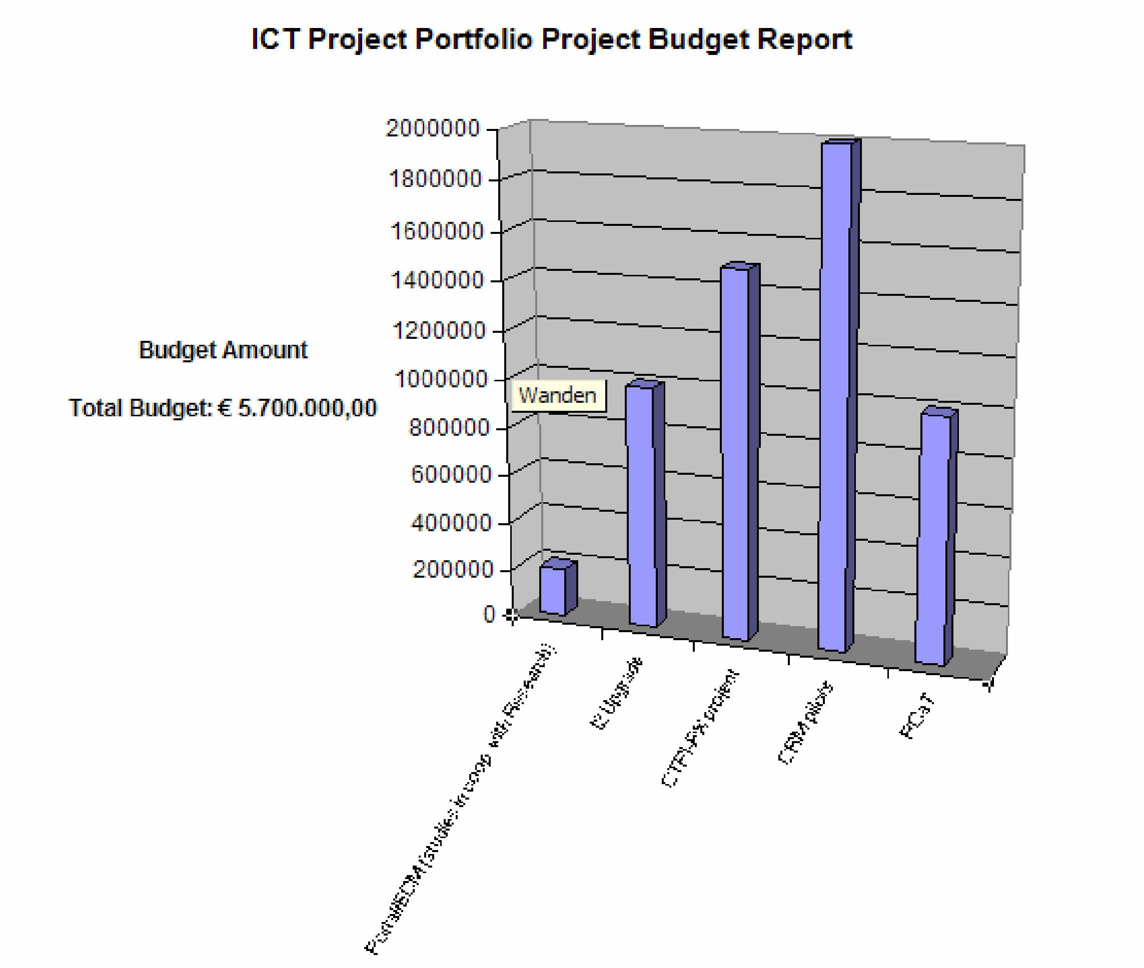

Figure 7-9: A budget report in excel created by the input in Metis

Test Criteria

as o ed d n es case a Me s can and e os o an as ec s o A e c ea n d e en s and e a ons s be een ese s. e e e e s a ca sed es ons o obe s d n e en n e es scena os e e:

Warning of change: en e e a e d e en s, a e based on e sa e da a, ey a be an ed a o a ca y e o na da a s an ed. A a n does no a o a ca y a ea . en s needed o a n so eone s can be done e anno a ons. Anno a ons a e essa es a can be s o ed e d e en odes. A se as o o en e anno a o e o see e o me o e ode as e a essa e. s essa e n o s abo e an e ade by e o me . A o e ac e o o a n o d be a ec a ed;

Changing an object type: s oss be o an e obec y e o one o o e obec s. en e obec y e s an ed e e a ons s and na es be e n ac . a s e y s abe and sa es a o o e;

Creating the IAF view: e e e en e e as o c ea e a e based on e A a e o , does no a ec e o e s. B e e s ade a an e n e o e e s ey a ec e e based on e A a e o . So so eone a es a an e n the Hydra ... is also ... based on the ... B ... the ... ade ... an ... in the ... not in ... a ... in the Hydra o ... is not exactly possible in Metis. B ... the ... s also ... on to create ... s. The objects ... o ... the ... Hydra ... you ... an ... in you ... have to be co ... ed and ... as ... the con ... a ... as a dependable ... so ... n ... an ... es in Hydra ... also ... be ... an ... ed in the ... Second ... you ... create in the ... ano ... the con ... a ... the same na ... e as the co ... ed con ... a ... and these con ... a ... s can be ... n ... ed o each o ... the ... So no ... there are ... o con ... a ... s: one ... a ... s de ... enden ... on the o ... the ... s and one ... a ... s nde ... enden ...

onc ... on: Me ... s can ... the ... os ... o ... an ... a ... the ... es ... c ... e a ... the ... es ... case and the ... o ... e ... s ... ceeded in the ... es ... scena ... os.

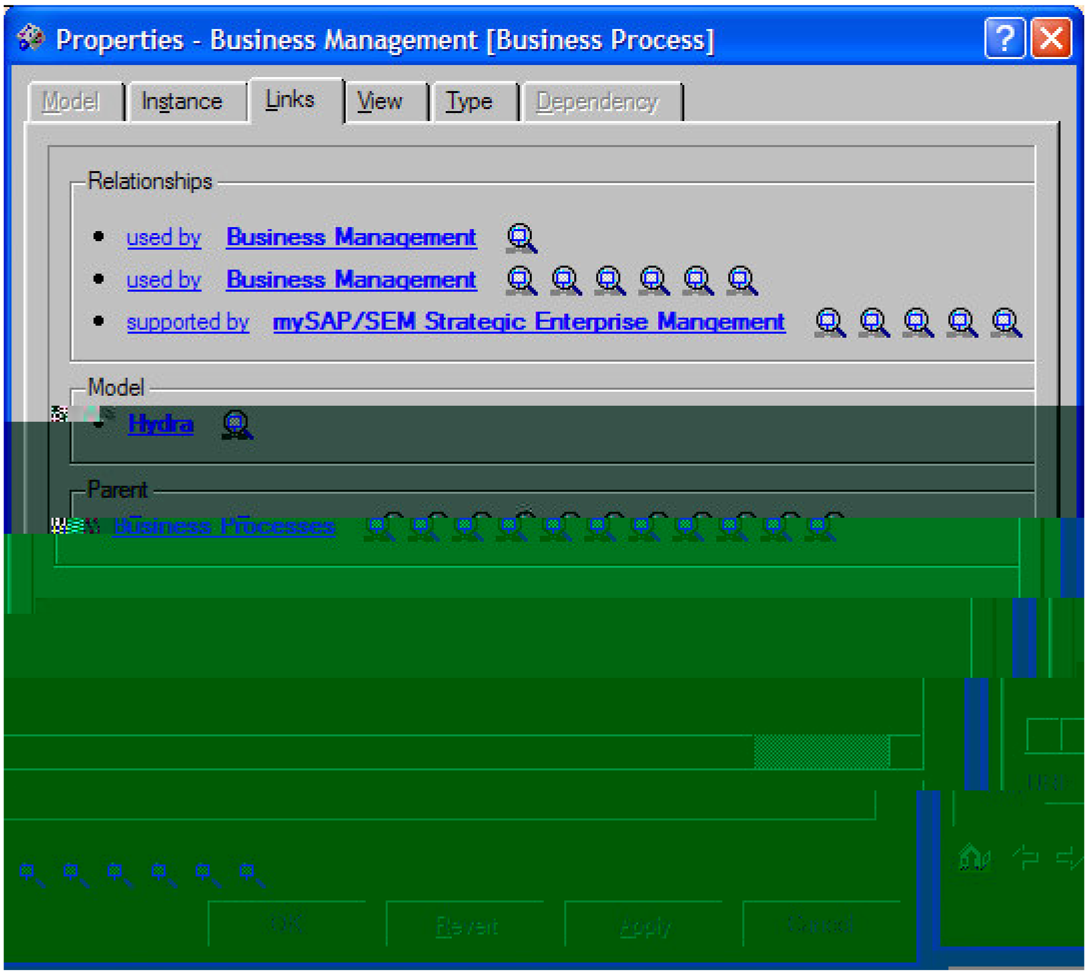

Figure 7-10: The relationships of a Business Process in Metis

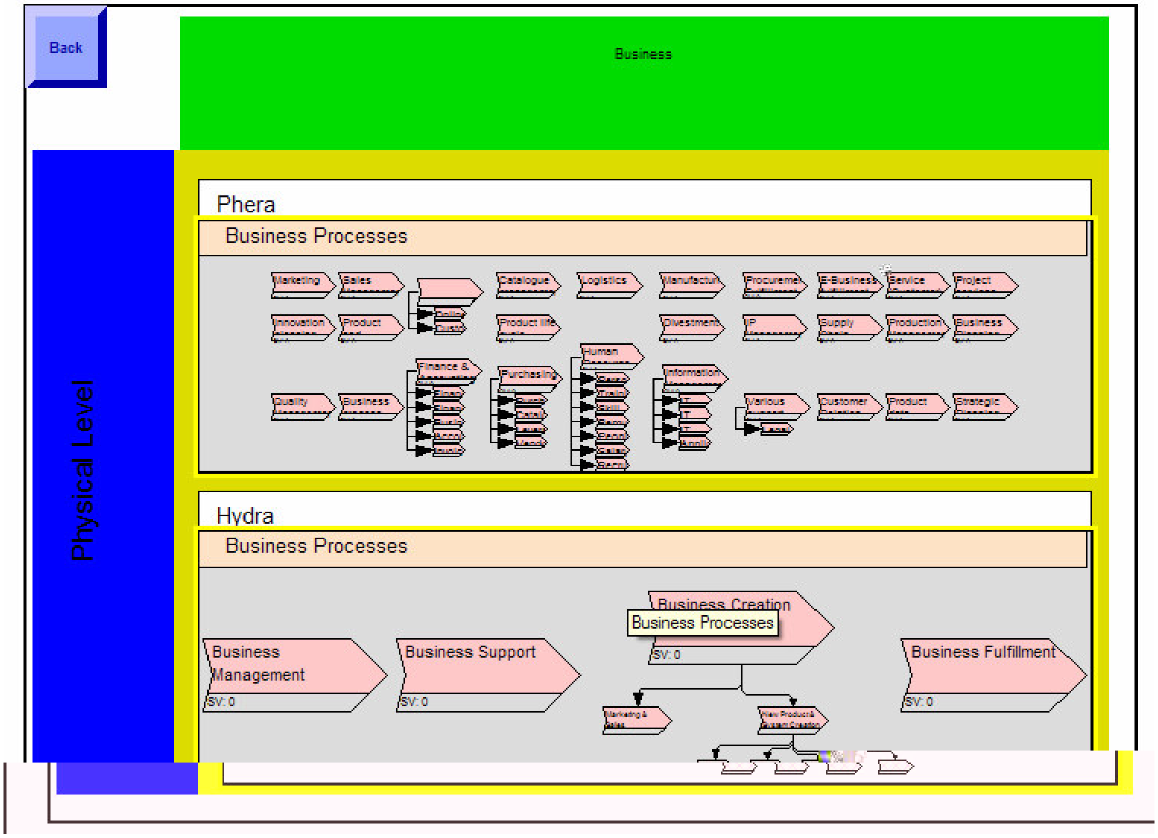

Figure 7-11: A comparison of the Hydra and PHERA view in a cell of the IAF framework

Conclusion of analyzing Hydra and PHERA

n t e tes case t e yd a, P E RA, S LAR and IA w w e e e en ed. e conc sons o t e ana ys s a e:

yd a and P E RA a e no cons s en . ey con a n d e en a eas and ocesses o ns ance. n e e s a co a son be een e ocesses o P E RA and yd a. A e esen n s as a es as sa d a e e a e co any o c es o s. n bo e s ese o c es e e no o o ed co e y. M s s o s a c ec s o nd ncons s enc es be een a o s a c ec e ode s, o s ec ca y be een e o n ode and e co o a e ode a ey s o d be cons s en . s ead o e cons s ency be een s a c ec es co any de, c s o an o nd n syme y o o n es and o n e a on n a es.

yd a s no co e e. n co ance e ode s a P E RA o des, ce a n ocesses n yd a s o d be s n o e ocesses o so e ocesses s o d con a n s b ocesses. M s enab es o co a e e d e en a c ec a e s o see ey a e cons s en . By desc b n ocess on e sa e e e o de a , s be ab e o s o o o n es o s a ed de e o en , e se o ex s n a ca ons o o se e ex e ence o ome d s on o o e e en a ons n co a abe o ec s.

Hyd a defin ons are no s c reno . In Hyd a' ocess b d es' are n od ced, b ey are no defined n s a ay a ey can be desc bed n Me s. The efore e ocess b d es ad o be defined be e b o e ey co d e en ed. Me s s e ys c n y es and defin ons, so by e en n an a ec a e e y n as o be defined n a ood ay. In Hyd a ocess b d es e e a se of ocesses and a ca ons and ese ocess b d es e e a n be een o b s ness a eas. In Me s s no defined as a B s ness nc on. a s a e e b s ness en y, can be cons de ed as a e e b s ness ocess. Me s s e e sed as a efe ence an a e. o s s s e y se be ca se a efe ence an a e s o an n case of co abo a on be een 's, e e s and ac s ons, s a e c a ances, and o e o s of b s ness coo e a on. Me s s as a efe ence an a e o s end n e and, fo c ea n one.

By s n e IA f a e o , e e an b s ness es ons can be e me a ed. Beca se e e of e IA f a e o s based on a e o e e s, e a ons be een e sa e ob ec s of e d ffe en e s, e b s ness es ons can e o e s ec f c ans e s. The SIAR doc en con ans nc es and s a e es. By co b n n ese nc es e ocesses and a ca ons f o e A and Hyd a e , de endences be een s a e es and a ca ons can be den f ed. a oo as Me s s easy o f nd ese de endences. o ns ance, d n e es case e s a e es a e e s o ed by e SA e en a ons e e eas y fo nd by s ec a sea c c e a In doc en s, a e sed oday, s on y oss b e o f nd ese by c ea n a o of ab es a con ans a nds of de endences. The s en of Me s s a a nds of b s ness es ons can be e me a ed and a e es s a e d ec y s b e.

Contribution of Metis for Philips

There a e any e oyees a a e es ons be fo d ffe en a s n e o an za on. Me s a es ese d ffe en es ons b es n o acco n . The os o an as ec fo s s o c ea e an a ec a e of e 's and co o a e I , so ey can coo e a e be e n e f e d of a ec e. A e e en n e d ffe en e s o fo o ana e en s oss b e. A so a be e coo e a on n e f e d of a ec e be een e d ffe en 's can beco e a es of s n Me s. Me s a so can ead o a be e B s ness I a n en n s. Me s s ab e o d ec y s o e 'o a ds one s' a oac by o d n a oo o f nd syme y o o es. o ns ance e A and Hyd a e co d be co a ed n an ob ec e ay. s a so oss b e be een d ffe en a ec es o f nd s a es and o c ea e a o be a ec e e e e d ffe en as ec s a e e ed o e e .

&RPSOHWLRQ QHJRWLDWLRQ

s d d no co o b y n e od c en e es case o a e a os e es . A e e es case, Me s as cons de ed as a e y ood oo fo s, b o as s no o osed ye and a s ey me e be. The s a on no s a as no dec ded ye o se a me A oo . A be ef of s se ec on ocess s a f n e f e ace an oo s e ed en e e s a eady ood nfo a on a a ab e. a nfo a on can se e as a s a n o n n a me oo se ec on ocess. e es case e d Me s s e sa e as a oof of once (o), e exce on a a e e oo as cons de ed as sa s ac o y no ed a e o os on fo o ed.

s s ase s no a a s sesec on ocess. s s ase s me abe n any co e se ec on ocess. s dec des o ado an A oo , as o co e e se ec on ocess s ase. o e e o an as ec s s ase be en omed n s sec on. e obec e s as ase s o e y e nc e o ce and o a e a e na dec s ons conce n n e co e on e se ec on ocess. s ase s o d es n [e]:

- A (nc ona) con a on e (n nc e) osen cand da e o e o s , as a as e e as any do o e e o s ase;
- An ess on e so a e od ca ons e ed and e conse ences o e o an za on;
- An e en a on an based on e abo e, nc d n e es ons b es and as s (and ose e s e), e as n and n a ac es and e o ec o an za on;
- A con ac a a e e en co osed a e e en s s ed abo e;
- e ood s o o n e o an za on o e con ac ed A oo as e as o e e en a on an.

o e e , e a y e na es (e o ce a on o e n s) s a d o eas e and s a y no eas ed a a . s eans a s a es n e se ec on ocess a e on y o nd o a a e y a e s a e o e an nde ec ed. A sys e s osen and a con ac s conc ded b e s a es ade (o o o es ssed) n e ea e s a es on y beco e c ea d n e e en a on o en e sys e s o e a ona . en ey me e beco e s be, b e es s a o e y ex ens e o de ayed es . e e o e s se ec on ocess as done e y ca y and a o e as s en n co ec n e no a on [e].

6HOHFWLRQ SURFHVV EDVHG RQ GDWD VWUXFWXUHV

n e boo os . de a so a se ec on ocess B s ness n o a on Sys e s s desc bed, s based on ana yz n da a s es. a a s es a e a necessa y econd on, b no eno o nd ca e a ce a n nc ona y. n y a ac a ce a n da a s e nd ca es a so e nc ona y s no oss b e. A oo can on y o de a ce a n nc ona y es e da a s e, b e da a s e does no a an ee a e oo o des a ce a n nc ona y. e e o e a ad ona es e A oo s s needed o nd o e A oo o des a e needed nc ona y. s es s o d en on y be e d e A oo s a con a n e da a s es. Assess en da a s es can con b e n s ay n an A oo se ec on ocess.

n e case e se ec on ocess an A oo , e assess en da a s es can nd ca e ce a n nc ona y can o canno be o ded. n Me s o exa e e sed da a s es (e a es) a e no xed and e se s abe o a e an es n . B o e oo s, e A S, con a n a xed e a ode . an es n ese da a s es can on y be done by e endo .

Assess en da a s es s on y abe en so a e se ec ons a e ca ed o e y e en y. e n es en needed o e d ec assess en e da a s es A oo s s n can y o e . Beca se e se ec n ocess an A oo no a e ace

Currently, tool selection based on data structures is not available for them. Therefore the selection process based on data structures was not considered in this selection process [[illegible]].

&RQFOXVLRQV DQG 5HFRPPHQGDWLRQV

n s c a e s s n ee sec ons. Sec on 8. con a ns e conc s ons abo e eo y. Sec on 8.2 con a ns e conc s ons abo e se ec on ocess. e as sec on con a n an o e a conc s on and e eco enda ons o e oo se ec on ocess and s n an n e se A c ec e a s.

&RQFOXVLRQV DERXW WKH WKHRU\

en n e se A c ec e oo s a e on y ca ab e o desc b n e a c ec e, cons de n d e en e s and e a ons s, and o de ed ana ys s. ey a e no e a ed o o e a ona ocesses a a . n sec on 3. a e abo e e as en, n c n e se A c ec e a es an o an o e n (o e a ona) ana e en . n e se A c ec e oo s and a e o s a e s o y de e o n o a ds e se e os ec s. n e ean e e e a e a o o nce an es o se e c en oo s and a e o s n a o e ay. n s es s e e e den ed ee a oac es o n e se A c ec e. o n e se A c ec e s n any es seen as e es ons b y o e de a en . n e ses a e o d sco e e o e oss b es o n e se A c ec e, so ey can de oy n e se A c ec e a so n o e s a ons.

e a e o s a e s oo c oc sed on desc b n e en e se and B s ness a n en . n e e o e a e o s a ea o s o a so o e a oac es o n e se A c ec e. e o a desc on o e A a e o e ob ec s and e a ons s o e es case s o ed a e c en e en a on as no y co an e A a e o . e o a desc on s a ood ay o a e acea o e e o a s a o ed and a s no and s s a ood s a n o n o e n e se A c ec e oo s.

&RQFOXVLRQV DERXW WKH (QWHUSULVH $UFKLWHFWXUH WRRO VHOHFWLRQ IRU 3KLOLSV

s needs an n e se A c ec e oo a e s e o e esen e a o a c ec a desc ons on a co a ab e bas s and s o e de n o n o a e a ons s be een en e se e e and d s ona e e a c ec es. Beca se o e n od c on o e A a e o , s a e o as o be s o ed n a s a ay. An n e se A c ec e oo s e so on o s. e an ob ec e o e n e se A c ec e n s s B s ness a n en . o n ca on a e s a e o de s (on co o a e e e as e as on d s ona e e) s a so an o an be e o e oo . Ano e e e en s a e oo o de n ' aceab y o a en s': o a e c ea and nde s and ab e y a ce an dec s on s a en . na y e oo s o d be ab e o s o e A a e o and a o c s o za on and ada a on o e e os o y e a ode . e e o n e se A c ec e s as o be ab e o a e be e dec s ons n e e d o . Beca se o e se o n e se A c ec e e Bab e e con s on be s o y. o eo e a e d e en e s o o n ons o e c en s a on. o e e ybody nde s ands e ea y and based on a e o dec s ons can be a en. n e se A c ec e e e s c ea ness abo e ea y and s es s n ab e e o e nance.

n a e 5 e es case Me s as desc bed. e os o an es s s es case e e:

- **Metis:** Me s s a e y n ce oo o o and d n e es case e e y n a as necessa y o e es case co d be c ea ed. e a cs a e o an n Me s, e e y ode can be d a n exac y as e se s a e sed o;
- **Test criteria:** e os o an es c e a co d be eas y e en ed;
- **Results on Hydra and PHERA:** yd a and A a e no cons s en , yd a s no co e e, e yd a d n ons a e no s c eno and by s n e A a e o e e an b s ness es ons co d be e me a ed;
- **Contribution of Metis for Philips:** Me s s e y s abe o s beca se can o de coo e a on be een e d e en s and e o o a e e . can be an o an e o o e o ads ome s'. Me s s o s e oas o o a e n nd n syme y o o n es. o e e o a oo a o d no be eas be beca se e e s oo n o a on and e a e ns a e oo co ex o an be n s.

A e e s s e s n e se ec on ocess Me s as cons de ed as e os o s n oo o s. A e e es case s cons de a on s s a d. e se ec on ocess s no n a na s a e a o o a e e o ano e e o n e se A ec e a e ana e en o o a e e s no n e es o con n e e oo se ec on ocess by oos n o as. e o e se ec on ocess s c en y on o d n o o a e . Bes des o o a e od d s ons can dec de e o con n e e oo se ec on ocess, o e ace e oo s ey c en y se, o e en o se ec o e oo s and s a e oo se ec on ocess a a n.

* HQHUDO &RQFOXVLRQV DQG 5HFRPPHQGDWLRQV

s needs an n e se A ec e oo o a be e co ca on and coo e a on be een s d s ona e e s and co o a e e e n e e d o a ec e. e coo e a on s an y oc sed on coo e a on n e e d o , s as: n e o e ab y, s a ed so ons, s a ed de e o en , e c. s s a e y s a so no n as o n o ads ome s'. a s a e y a e n e se A ec e e en o e e an , beca se e e be an nc eased oc s on s a n .

so e eco enda ons o con n e s se ec on ocess be en. ese eco enda ons o s, ese es s ends a e:

- ons de n Me s ns ead o e a ec e oo s. Me s s as abe n e se A ec e oo . ons de e oo s, a e c en y sed n s and ec Me s can be sed s ccess y n co b na on ese c en oo s o a can s b s e . B d n e es case a ea se e co d no be sed. o es Me s s e oo o coo e a on be een d e en a ec s, o a e o n a e sa e ode s, ano e es s e ed. n a es coo e a on, e s on con o and a o za on a e o be c ose y exa ned;
- ons de Me s e e y e a ec e s n o ed. Me s as o be cons de ed as a oss be oo o o de e a ec a desc ons, e o ec s be een d e en s. o ns ance, e o ec s a a e a es o e o ads ome s' s a e y can

be s o ed a oo e Me s. In s oo e d e en e s o e e y can be
eas y e a ed;

- se Me s o s o e A a e o . Beca se s n e A a e o s dec ded, a c ec s a e o nd a ay o s e A a e o n as c ed ay. Me s can o de n s n e A a e o n as c ed ay. e e o e a c ec s and ana e s a e o e ac a n ed Me s and o en o dec de o se and e en . Me s beco es e s and a d n s, coo e a on n e e d o a c ec e o an zed e A a e o be oss be;
- S a e n od c on o Me s n a s a ay. o a e s ccess n e se A c ec e o ec s and Me s e e as o be e a ed a a y and n ness a on e e oyees o se . A c ec s and ana e s a e o a e e a e y need an n e se A c ec e oo and a s o a bene o e . S nce e e s no s c a y and n ness ye , n od c on o a oo no be bene c a . s e es a ca e n od c on o n e se A c ec e and n e se A c ec e oo s. Ano e bene o s a n o se Me s n a s a n be o o ec s a e e be e a ed a c ec a desc ons o d e en a s. A c ec s a e abe o ea n o ead o e o e se ex s n a c ec a desc ons n o e o ec s. In e s s a e Me s can be sed o desc be e a c ec es, b n a a e s a e, en a y s nc eas n and eo e cons de n e se A c ec e as se . Me s can a so be sed o de oy n sea c es, s a ons, e c. Me s s a oo a s e y s abe o ex ens e ana ys n o a c ec es. e e o e e o a s do e by s n Me s n s a o ec s can be sed n a a e s a e en e e s o e e ed o e a c ec e n dec s on a n . en n e e Me s can s o o e nc ona es as desc bed n sec on 3. e e a e a eady ood s a n o n s a a abe.

In conc s on Me s s o d be cons de ed as a s abe oo , c can be sed n e e y a c ec e ocess. od c d s ons a e o c oose e o n s a e y e a d o a c ec e. I ey a e cons de n a c ec e oo s, Me s s o d be on e s o s. e n od c on o an n e se A c ec e oo s o d be done a o o ca e. I s be e o s a n a s a ay and a e s ccess e se o Me s can be s ead o e c ea ed odes a e e bas s o e o n n be o odes and e ac ed no ed e can be sed n o e o ec s and d s ons. A e o en e d s ons Se cond o s and ons e ec on cs a e s o n n e es n Me s and e a s e sed s ons can a e e ead n ado n n e se A c ec e n s. e e o e s o a on e ec s n s s a on, ead n n as a e a o o e e en a on and se o an n e se A c ec e oo n e o e o an sa on o s:

'A walking-tour to Peking also starts with the first steps, if you do not make them you will never get there'

[an]

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Abbreviations

AD	Architectural Description
BA	Business Architecture
BPM	Business Process Modeling
BPR	Business Process Redesign
CIO	Chief Information Officer
EA	Enterprise Architecture
EAP	Enterprise Architecture Planning
EIS	Enterprise Information Systems
ERP	Enterprise Resource Planning
EU	European Union
IAF	Integrated Architecture Framework
IEEE	The Institute of Electrical and Electronics Engineers, Inc.
IEEE 1471	The IEEE Standard 1471-2000: IEEE Recommended Practice for Architectural Descriptions of Software Intensive Systems
IFRS	International Financial Reporting Standards
IT	Information Technology
PD	Product Division
PhERA	Philips Enterprise Reference Architecture
PoC	Proof of Concept
RFI	Request for Information
S	Semiconductors
SITAR	SIte ARchitecture
UML	Unified Modeling Language

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List of tables